

Required Readings

Required Readings for this program are open access, which means you should be able to access them at no cost. The link provided in the citation will take you directly to the reading or to a page where you can download it.

1. Lee, Daniel D., and H. Sebastian Seung. "Learning the Parts of Objects by Non-negative Matrix Factorization." *Letters to Nature*, vol. 401, no. 21, October 1999.

<https://www.cs.columbia.edu/~blei/fogm/2020F/readings/LeeSeung1999.pdf>.

- **Required pages:** All
- **Estimated reading time:** 25 minutes

2. Technische Universität Dortmund. "Non-negative Matrix Factorization." <https://dm.cs.tu-dortmund.de/en/mlbits/matrix-factorization-non-negative/>.

- **Required pages:** The entire content including YouTube video
- **Estimated reading time:** 35 minutes

Lesson 1 - Non-Negative Matrix Factorization

December 25, 2025

MODULE 5 | LESSON 1

##Non-Negative Matrix Factorization

Reading Time	50 minutes
Prior Knowledge	Basic understanding of linear algebra and statistics: Matrices and vectors; Eigenvalues and eigenvectors; Basic statistics; Familiarity with Python programming: Basic Python syntax, data manipulation and numerical operations; Fundamental financial concepts: Asset returns and correlation; Portfolio diversification; Sentiment analysis
Keywords	Non-negative Matrix Factorization (NMF), Dimensionality Reduction, Feature Extraction, Parts-based Representation, Sparsity, Interpretability, Sparse NMF, Constrained NMF, Semi-NMF, Convex NMF, Online NMF, Sentiment Analysis, Principal Component Analysis (PCA), Singular Value Decomposition (SVD), Independent Component Analysis (ICA), Factor Loadings, Factor Scores

In this lesson, we explore non-negative matrix factorization (NMF), a technique to decompose a non-negative matrix into two smaller non-negative matrices. It's often used for dimensionality reduction, feature extraction, and topic modeling. This lesson offers a comprehensive overview of NMF, its variations, and its applications, with a focus on financial engineering, particularly portfolio diversification.

```
[1]: !pip install pip --upgrade
      !pip install yfinance --upgrade
```

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Requirement already satisfied: pip in /usr/local/lib/python3.11/site-packages
(22.3.1)
```

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Collecting pip
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Installing collected packages: pip
  Attempting uninstall: pip
    Found existing installation: pip 22.3.1
    Uninstalling pip-22.3.1:
      Successfully uninstalled pip-22.3.1
Successfully installed pip-25.3
WARNING: Running pip as the 'root' user can result in broken permissions
and conflicting behaviour with the system package manager. It is recommended to
use a virtual environment instead: https://pip.pypa.io/warnings/venv

Requirement already satisfied: yfinance in /usr/local/lib/python3.11/site-
packages (0.2.55)
Collecting yfinance
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bytes)
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Collecting certifi>=2024.2.2 (from curl_cffi<0.14,>=0.7->yfinance)

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 curl_cffi-0.13.0-cp39-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (8.3 MB)

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Installing collected packages: websockets, protobuf, certifi, curl_cffi, yfinance

Attempting uninstall: certifi[90m

1/5 [protobuf]

Found existing installation: certifi 2022.12.7

1/5 [protobuf]

Uninstalling certifi-2022.12.7:

1/5 [protobuf]

Successfully uninstalled certifi-2022.12.7

1/5 [protobuf]

Attempting uninstall: yfinance

3/5 [curl_cffi]

Found existing installation: yfinance 0.2.55

3/5 [curl_cffi]

Uninstalling yfinance-0.2.55:[90m

3/5 [curl_cffi]

Successfully uninstalled yfinance-0.2.55m

3/5 [curl_cffi]

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[yfinance]4/5 [yfinance]
Successfully installed certifi-2025.11.12 curl_cffi-0.13.0
protobuf-6.33.2 websockets-15.0.1 yfinance-1.0
WARNING: Running pip as the 'root' user can result in broken permissions
and conflicting behaviour with the system package manager, possibly rendering
your system unusable. It is recommended to use a virtual environment instead:
https://pip.pypa.io/warnings/venv. Use the --root-user-action option if you know
what you are doing and want to suppress this warning.
```

```
[2]: # Loading libraries
import pandas as pd
import yfinance as yf

from sklearn.decomposition import NMF
```

##1. What is Non-Negative Matrix Factorization?

Non-negative matrix factorization (NMF) is a dimensionality reduction technique that decomposes a non-negative matrix into two smaller non-negative matrices. It's often used in data analysis, machine learning, and recommender systems. Please review the required readings and come back for the remainder of this lesson.

The mathematical definition of non-negative matrix factorization (NMF) is as follows:

Given a non-negative matrix V with dimensions $m \times n$, NMF seeks to find two non-negative matrices:

- W with dimensions $m \times k$ (m rows, k columns, where k is the reduced dimensionality);
- H with dimensions $k \times n$;

Such that:

$$V \approx W * H$$

Where: - $*$ denotes matrix multiplication - The approximation $\approx$ is typically measured using a cost function, such as the Frobenius norm: $\|V - W * H\|^2$

Constraints: All elements of W and H must be non-negative real numbers, i.e., $W \geq 0, H \geq 0$

In simpler terms, we want to find two matrices (W and H) that, when multiplied together, closely resemble our original data matrix (V). The elements of W and H must be non-negative (greater than or equal to zero).

Factorization rank in NMF, often denoted as k , represents the number of factors or components we are trying to extract from data. It essentially determines the dimensionality of the reduced representation.

Iterative Update Rules: There are several ways in which W and H may be found. NMF algorithms typically employ iterative update rules to refine the matrices W and H until convergence. These rules aim to minimize the difference between the original matrix V and the product $W * H$. One common update rule is the multiplicative update rule, which is derived from minimizing the Frobenius norm:

$$W_{ia} \leftarrow W_{ia} * \frac{(V * H^T)_{ia}}{(W * H * H^T)_{ia}}$$

$$H_{aj} \leftarrow H_{aj} * \frac{(W^T * V)_{aj}}{(W^T * W * H)_{aj}}$$

Where:

- W_{ia} represents the element in the i -th row and a -th column of matrix W .
- H_{aj} represents the element in the a -th row and j -th column of matrix H .
- V , W , and H are the matrices as defined in the NMF problem.
- H^T and W^T denote the transpose of matrices H and W , respectively.
- $*$ and division represent element-wise multiplication and division, respectively.

The iterative process continues until a convergence criterion is met. This can be based on:

- Difference in cost function: The algorithm stops when the change in the cost function (e.g., Frobenius norm) between consecutive iterations falls below a predefined threshold.
- Maximum number of iterations: The algorithm terminates after a fixed number of iterations, even if the cost function hasn't fully converged.

Mathematically, convergence can be expressed as:

$$\|V - W^{(t+1)} * H^{(t+1)}\|^2 - \|V - W^{(t)} * H^{(t)}\|^2 < \varepsilon$$

Where:

- $W^{(t)}$ and $H^{(t)}$ represent the matrices W and H at iteration t .
- ε is a small positive value representing the convergence threshold.

In simpler terms, the algorithm stops when the difference in the approximation error between consecutive iterations becomes sufficiently small.

0.1 2. Applications of Non-negative Matrix Factorization (NMF) in Financial Engineering

NMF has gained popularity in financial engineering due to its ability to extract meaningful and interpretable features from financial data. Here are some notable applications:

Portfolio diversification: NMF can be used to identify groups of assets that exhibit similar behavior, allowing for the construction of diversified portfolios with reduced risk. By decomposing the correlation matrix of asset returns, NMF can reveal underlying factors that drive asset co-movements and help investors allocate their investments across different asset classes.

Risk management: NMF can be used to identify and quantify risk factors in financial markets. By decomposing historical market data, NMF can uncover hidden risk factors that may not be apparent using traditional methods. This information can be used to develop risk management strategies and improve portfolio hedging.

Asset pricing: NMF can be applied to asset pricing models to identify factors that explain asset returns. By decomposing the cross-section of asset returns, NMF can reveal underlying factors that drive asset prices and help investors understand the determinants of asset performance.

Sentiment analysis: NMF can be used to analyze textual data, such as news articles or social media posts, to extract sentiment and gauge market sentiment toward specific assets or the overall market. This information can be used to inform investment decisions and predict market movements.

Fraud detection: NMF can be used to detect anomalies and patterns in financial transactions that may indicate fraudulent activity. By decomposing transaction data, NMF can identify unusual patterns and highlight potential areas of concern for further investigation.

High-frequency trading: NMF can be used to analyze high-frequency financial data and identify patterns that can be exploited for trading strategies. By decomposing price and volume data, NMF can reveal hidden relationships and predict short-term price movements.

Algorithmic trading: NMF can be integrated into algorithmic trading systems to automate trading decisions based on extracted features and patterns in financial data. This can help improve trading efficiency and profitability.

These are just a few examples of how NMF is being applied in financial engineering. Its ability to extract meaningful and interpretable features from complex financial data makes it a valuable tool for various tasks, including portfolio management, risk management, asset pricing, and trading. As financial markets become increasingly complex and data-driven, the applications of NMF in financial engineering are likely to continue to expand.

0.2 3. Toy Example

Let's illustrate how the NMF procedure functions with this small toy example. Let's consider the following 4x5 data matrix V :

$$V = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 6 & 7 & 8 & 9 & 10 \\ 11 & 12 & 13 & 14 & 15 \\ 16 & 17 & 18 & 19 & 20 \end{pmatrix}$$

We want to decompose this matrix into two smaller matrices, W (4x2) and H (2x5), such that $V \approx W * H$. We initialize $W^{(0)}$ (feature matrix at $t = 0$) and $H^{(0)}$ (coefficient matrix at $t = 0$) with random non-negative values:

$$W^{(0)} = \begin{pmatrix} 0.2 & 0.5 \\ 0.8 & 0.3 \\ 0.6 & 0.9 \\ 0.4 & 0.7 \end{pmatrix}, \quad H^{(0)} = \begin{pmatrix} 0.7 & 0.1 & 0.4 & 0.6 & 0.2 \\ 0.3 & 0.6 & 0.2 & 0.4 & 0.8 \end{pmatrix}$$

Now, we iteratively update $W^{(t)}$ and $H^{(t)}$ using the multiplicative update rules to minimize the difference between V and $W^{(t)} * H^{(t)}$. After a few iterations, we might obtain the following updated matrices:

$$W = \begin{pmatrix} 0.9618 & 0. \\ 1.9262 & 1.0167 \\ 2.8902 & 2.0347 \\ 3.8542 & 3.0527 \end{pmatrix}, \quad H = \begin{pmatrix} 1.0413 & 2.0823 & 3.1232 & 4.1642 & 5.19 \\ 3.9269 & 2.9399 & 1.9529 & 0.966 & 0. \end{pmatrix}$$

The final values of Feature matrix W and Coefficient matrix H represent the decomposed factors of the original matrix V . Please note that the specific values in the matrices may differ slightly depending on the actual calculations. By multiplying W and H , we get an approximation of V :

$$W * H = \begin{pmatrix} 1.0015 & 2.0028 & 3.004 & 4.0052 & 4.9919 \\ 5.9983 & 6.9999 & 8.0017 & 9.0034 & 9.9972 \\ 10.9996 & 12. & 13.0004 & 14.0008 & 15.0002 \\ 16.0009 & 17. & 17.9992 & 18.9983 & 20.0032 \end{pmatrix}$$

This resulting matrix is an approximation of the original matrix V . The difference between V and $W * H$ represents the reconstruction error.

$$V - W * H = \begin{pmatrix} -0.0015 & -0.0028 & -0.004 & -0.0052 & 0.0081 \\ 0.0017 & 0.0001 & -0.0017 & -0.0034 & 0.0028 \\ 0.0004 & 0. & -0.0004 & -0.0008 & -0.0002 \\ -0.0009 & -0. & 0.0008 & 0.0017 & -0.0032 \end{pmatrix}$$

The goal of NMF is to find W and H that minimize this reconstruction error while keeping all elements of W and H non-negative.

This very simple numerical example demonstrates how NMF can decompose a larger data matrix into smaller matrices, capturing its underlying structure and relationships.

0.3 4. Basics of Non-Negative Matrix Factorization

In a nutshell, NMF is about finding a simpler representation of data by breaking it down into two parts that capture the essential features and their importance. It has applications in various fields, including image processing, text analysis, and bioinformatics. Implementation involves the following steps:

- **Initialization:** Starting with a non-negative matrix, often denoted as V , that represents data. This is our matrix that contains information about users, products, or any other kind of data.
- **Decomposition:** The core of NMF is to find two non-negative matrices, typically called W and H , such that the product of W and H closely approximates the original matrix V i.e. $V \approx WH$. In simpler terms, we're trying to break down the original data into two parts that, when combined, resemble the original.
- **Iteration:** NMF algorithms employ iterative processes to refine W and H , minimizing the difference between the product WH and the original matrix V . These iterations involve updating W and H repeatedly until a satisfactory level of accuracy is reached. This process is often based on gradient descent or multiplicative updates.

- **Convergence:** The iteration process continues until a convergence criterion is met. This usually means that the difference between WH and V is small enough or that the algorithm has run for a predetermined number of steps.
- **Interpretation:** After convergence, the resulting matrices W and H provide a lower-dimensional representation of the original data. W can be seen as containing the basis vectors or features, while H represents the coefficients or weights of these features for each data point. Each column of W represents a feature, and the corresponding row of H shows how strongly this feature is expressed in each data point.

0.4 5. Properties of Non-negative Matrix Factorization (NMF)

Some of the key properties of non-negative matrix factorization (NMF) are:

Non-negativity: The most fundamental property of NMF is that it produces non-negative matrices W and H . This means that all elements of these matrices are greater than or equal to zero. This property is crucial for interpretability, as it allows us to understand the resulting factors as additive combinations of original features.

Dimensionality Reduction: NMF reduces the dimensionality of the data by decomposing it into two lower-rank matrices. This can be useful for simplifying the data, removing noise, and identifying latent features.

Parts-based Representation: NMF often leads to a parts-based representation of the data. This means that the resulting factors (columns of W) can be interpreted as representing individual parts or components of the original data. For example, in image analysis, NMF might identify factors corresponding to edges, textures, or objects.

Sparsity: NMF often produces sparse matrices, meaning that many elements of W and H are zero. This can be beneficial for interpretability, as it highlights the most important features and reduces the complexity of the model.

Interpretability: Due to its non-negativity and parts-based representation, NMF is often considered more interpretable than other dimensionality reduction techniques like Principal Component Analysis (PCA). The resulting factors can be more easily related to the original features, making it easier to understand the underlying structure of the data.

Flexibility: NMF can be applied to a wide variety of data types, including text, images, audio, and biological data. It can also be adapted to different applications by using different cost functions and constraints.

Computational Efficiency: NMF algorithms are generally computationally efficient, especially for sparse data. This makes them suitable for large-scale datasets.

Non-uniqueness: The NMF decomposition is not unique, meaning that there can be multiple solutions for W and H that give a good approximation of V . This can be addressed by using regularization techniques or by imposing additional constraints.

These are some of the key properties of NMF. They contribute to its popularity and effectiveness in various data analysis and machine learning tasks.

0.5 6. Challenges and limitations of Non-negative Matrix Factorization (NMF)

Non-negative Matrix Factorization (NMF) has some challenges and limitations:

Non-Uniqueness of Solutions: The NMF decomposition is inherently non-unique, meaning that there can be multiple pairs of factor matrices (W and H) that, when multiplied together, approximate the original data matrix (V) equally well. This arises from the fact that there are often many ways to represent the same data using different combinations of non-negative factors. > - Implications: This non-uniqueness can make it challenging to interpret the results of NMF, as different solutions may lead to different interpretations of the underlying factors. It can also make it difficult to compare results across different runs of the algorithm or when using different initialization strategies. > - Mitigation: Techniques like regularization, which adds constraints or penalties to the objective function, can help reduce the non-uniqueness issue by encouraging solutions with specific properties, such as sparsity or orthogonality.

Initialization Sensitivity: The performance of NMF algorithms can be sensitive to the initial values of the factor matrices (W and H). Different initialization strategies can lead to the algorithm converging to different local minima, resulting in different solutions. > - Implications: This sensitivity to initialization can make the results of NMF less reproducible and potentially lead to suboptimal solutions. > - Mitigation: Exploring different initialization methods, such as random initialization, NNDSVD (Non-negative Double Singular Value Decomposition), or using prior knowledge about the data, can help mitigate this issue and potentially improve the quality of the solutions.

Determining the Optimal Number of Factors: Choosing the right number of factors (or components) is a crucial step in NMF. Too few factors may not capture all the important information in the data, leading to a loss of information and reduced accuracy. Too many factors can lead to overfitting, where the model captures noise or irrelevant patterns in the data, reducing its ability to generalize to new data. > - Implications: Selecting an inappropriate number of factors can significantly impact the performance and interpretability of the NMF model. > - Mitigation: Model selection techniques, such as cross-validation, silhouette analysis, or using domain expertise to assess the interpretability of the factors, can help guide the choice of the optimal number of factors.

Non-negative Data Requirement: NMF is fundamentally designed to work with non-negative data. The algorithm assumes that the input data matrix (V) and the resulting factor matrices (W and H) have only non-negative elements. This is because the non-negativity constraint is essential for ensuring the interpretability of the factors as additive combinations of original features. > - Implications: This limitation restricts the applicability of NMF to datasets where negative values are not meaningful or where they can be transformed into a non-negative representation. > - Mitigation: For data with negative values, techniques like shifting (adding a constant to all elements), scaling (multiplying by a positive constant), or using alternative matrix factorization methods that can handle mixed-sign data might be considered.

Interpretability Challenges: While NMF is often considered more interpretable than other dimensionality reduction techniques like PCA, interpreting the resulting factors can still be subjective and require domain expertise. The factors represent latent features or patterns in the data, and their meaning may not always be immediately obvious. > - Implications: The interpretation of NMF factors requires careful analysis and consideration of the context of the data and the specific application. > - Mitigation: Techniques like visualizing the factor loadings, examining the top contributing features for each factor, and using domain knowledge to relate the factors to real-world concepts can aid in interpretation.

Computational Cost: NMF algorithms can be computationally expensive, especially for large

datasets or a high number of factors. The iterative nature of the algorithm and the need to update the factor matrices repeatedly can lead to significant computation time. > - Implications: The computational cost of NMF can be a limiting factor for large-scale applications or when real-time analysis is required. > - Mitigation: Using efficient algorithms, such as those based on sparse matrix operations, can help reduce the computational burden. Additionally, techniques like online NMF, which updates the factorization incrementally as new data arrives, can be more computationally tractable for streaming data.

0.6 7. Comparing NMF with Other Matrix Decomposition Techniques

Principal component analysis (PCA) is a dimensionality reduction technique that transforms high-dimensional data into a lower-dimensional space by identifying the directions (principal components) along which the data varies the most. These principal components are orthogonal and capture the maximum variance in the data, allowing for data simplification, noise reduction, and feature extraction. PCA is often used for data visualization, noise reduction, and as a preprocessing step for machine learning. It's a versatile technique applicable to various data types, but the resulting components can be less interpretable. In summary, PCA finds the most important patterns in the data and represents them using a smaller number of variables (principal components) that are orthogonal. This simplifies the data while preserving its essential information.

Singular value decomposition (SVD) is a more general-purpose matrix factorization technique that decomposes data into three matrices, capturing latent factors and their relationships. SVD is a matrix factorization technique that decomposes a matrix into three constituent matrices: U , Σ and V^T . These matrices capture the latent factors and relationships within the data, allowing for dimensionality reduction, noise reduction, and data compression. It's widely used in various applications, including recommendation systems, image compression, and noise reduction. However, like PCA, SVD components can be less interpretable. In summary, SVD breaks down a matrix into simpler components that reveal its underlying structure and relationships.

Independent component analysis (ICA) is a computational technique that aims to separate a multivariate signal into additive subcomponents that are statistically independent and non-Gaussian. In summary, ICA aims to uncover the hidden sources or factors that contribute to a mixed signal. It's like "unmixing" a cocktail to identify its individual ingredients.

Non-negative matrix factorization (NMF) specializes in decomposing non-negative data into two non-negative matrices, resulting in a parts-based representation. NMF decomposes non-negative data into two non-negative matrices, W and H . This decomposition results in a parts-based representation, where the components (columns of W) represent individual parts or features of the original data, and their weights (rows of H) indicate their importance in each data point. This makes NMF particularly useful for tasks where interpretability and a parts-based decomposition are desired, such as image processing, text analysis, and topic modeling. NMF often produces sparse results, further enhancing interpretability. In essence, NMF breaks down data into additive, non-negative components that represent its essential parts or features.

PCA, SVD, ICA, and NMF are all techniques used for dimensionality reduction and feature extraction, but they differ in their approaches and properties. Key differences are:

- **Constraints:** PCA and SVD enforce orthogonality on their components, ICA enforces independence, and NMF enforces non-negativity.
- **Interpretation:** PCA and SVD components can be difficult to interpret while ICA compo-

nents represent underlying sources and NMF components are often more easily related to the original features.

- **Sparsity:** PCA and SVD typically produce dense results while ICA can be sparse or dense and NMF often leads to sparse representations.
- **Data types:** PCA, SVD, and ICA can be applied to any type of data, while NMF is limited to non-negative data.

The choice between PCA, SVD, ICA, and NMF depends on the specific application and the desired properties. If capturing the maximum variance is the primary goal, PCA might be more appropriate. If a more general-purpose decomposition is needed, SVD might be more suitable. If blind source separation is desired, ICA would be the preferred technique. If interpretability and a parts-based representation are crucial, NMF is a good choice.

In some cases, it can be beneficial to use these techniques in combination. For example, PCA or SVD could be used for initial dimensionality reduction, and then ICA or NMF could be applied to extract interpretable features from the reduced data.

0.7 8. Extensions of Non-negative Matrix Factorization (NMF)

While the standard NMF algorithm is widely used, various extensions and variations have been developed to address specific needs and improve performance. Here are a few notable ones:

0.7.1 8.1 Sparse NMF

Sparse non-negative matrix factorization (NMF) is a variation of the standard NMF algorithm that enforces sparsity on the factor matrices W and H . This means that it encourages many elements of these matrices to be zero.

Reasoning: Sparsity is often desirable in NMF for several reasons: - Interpretability: Sparse factor matrices are easier to interpret because they highlight the most important features and reduce the complexity of the model. - Feature Selection: Sparsity can act as a form of feature selection, identifying the most relevant features for representing the data. - Noise Reduction: By focusing on a smaller set of features, sparse NMF can help reduce the influence of noise in the data.

Methods: Sparsity in NMF is typically achieved by adding sparsity-inducing regularization terms to the NMF objective function. These terms penalize non-zero elements in the factor matrices, encouraging them to become zero. Common regularization techniques include: - L1 regularization: Adds a penalty proportional to the sum of the absolute values of the elements in the factor matrices. - L2 regularization: Adds a penalty proportional to the sum of the squared values of the elements in the factor matrices. - Other sparsity constraints: Various other constraints can be used to enforce sparsity, such as limiting the number of non-zero elements in each row or column of the factor matrices.

Benefits: Sparse NMF offers several benefits compared to standard NMF: - Improved interpretability: Sparse factor matrices are easier to understand and relate to the original features. - Better feature selection: Sparsity helps identify the most relevant features for representing the data. - Reduced noise: By focusing on a smaller set of features, sparse NMF can help reduce the influence of noise. - Enhanced generalization: Sparse models often generalize better to new data because they are less prone to overfitting.

0.7.2 8.2 Constrained NMF

Constrained non-negative matrix factorization (NMF) is a variation of standard NMF where additional constraints are imposed on the factor matrices (W and H) during the factorization process. These constraints are often based on prior knowledge about the data or specific properties desired in the resulting factors.

Reasoning: Incorporating constraints allows for tailoring NMF to specific applications and data characteristics. This can improve the quality of the factorization, enhance interpretability, and ensure the results are more meaningful in the context of the problem.

Examples of Constraints:

- **Sum-to-one Constraint:** This constraint enforces that the elements in each row of H (or each column of W) sum to one. It's commonly used in topic modeling, where each row of H represents a document and the elements indicate the proportion of each topic in that document.
- **Orthogonality Constraint:** This constraint requires that the columns of W (or rows of H) be orthogonal to each other. It encourages the factors to be independent and represent distinct aspects of the data.
- **Sparsity Constraint:** Similar to sparse NMF, this constraint promotes sparsity in the factor matrices, leading to more interpretable and noise-resistant results.
- **Domain-Specific Constraints:** Constraints based on domain knowledge can be incorporated to guide the factorization. For example, in image processing, constraints could be used to ensure that the factors represent specific image features like edges or textures.

Benefits: Constrained NMF offers several advantages:

- **Improved Factorization:** Constraints can guide the factorization process toward more meaningful and relevant solutions.
- **Enhanced Interpretability:** Constraints can make the resulting factors easier to interpret and relate to the original data.
- **Tailored Solutions:** Constraints allow for adapting NMF to specific applications and data characteristics.
- **Control over Factor Properties:** Constraints can be used to enforce desired properties in the factors, such as sparsity, orthogonality, or specific patterns.

In essence, constrained NMF allows you to customize the NMF algorithm to your specific needs by incorporating prior knowledge or desired properties into the factorization process. This can lead to more insightful and relevant results compared to standard NMF.

0.7.3 8.3 Semi-NMF

Semi-NMF, or semi non-negative matrix factorization, is a variation of NMF where the non-negativity constraint is relaxed on one of the factor matrices (either W or H). This means that one of the matrices can contain both positive and negative values while the other remains non-negative.

Reasoning: The standard NMF algorithm requires all elements of the factor matrices to be non-negative. However, in some applications, negative values can be meaningful and provide valuable insights. For instance, in financial data analysis, negative returns are possible and should be considered in the factorization. Semi-NMF allows for the decomposition of such mixed-sign data while still preserving the interpretability and parts-based representation benefits of NMF.

How it Works: Semi-NMF modifies the standard NMF objective function and update rules to accommodate the relaxation of the non-negativity constraint. The algorithm still aims to find two matrices (W and H) whose product approximates the original data matrix (V), but one of the matrices can now have negative elements.

Benefits: - Handling Mixed-Sign Data: Semi-NMF enables the decomposition of data with both positive and negative values, which is crucial for applications where negative values are meaningful. - Preserving Interpretability: While allowing negative values, semi-NMF still retains the interpretability and parts-based representation benefits of NMF. - Flexibility: It offers more flexibility compared to standard NMF by accommodating a wider range of data types.

Considerations:

- When using semi-NMF, it's crucial to choose which factor matrix (W or H) should be allowed to have negative values based on the specific application and the interpretation of the factors.
- The interpretation of the factors may differ slightly from standard NMF due to the presence of negative values. Careful consideration should be given to the meaning of negative values in the context of the problem.

In summary, semi-NMF is a valuable extension of NMF that allows for the decomposition of mixed-sign data while still retaining many of the benefits of standard NMF. It offers more flexibility and can provide insights into data where negative values are meaningful.

0.7.4 8.4 Convex NMF

Convex non-negative matrix factorization (convex NMF) is a variant of NMF where convexity constraints are imposed on the factorization. This means that the factor matrices (W and H) are restricted to lie within a convex set.

Reasoning: Incorporating convexity constraints in NMF can lead to several benefits:

- Uniqueness: Convex NMF can help address the non-uniqueness issue inherent in standard NMF. By restricting the solution space to a convex set, it can reduce the number of possible solutions and potentially lead to a unique or more stable solution.
- Improved Convergence: Convexity constraints can improve the convergence properties of the NMF algorithm, making it more likely to find a good solution efficiently.
- Better Interpretability: In some cases, convexity constraints can lead to more interpretable factors by encouraging them to represent specific features or patterns in the data.

Methods: Convexity constraints in NMF are typically enforced by restricting the factor matrices to lie within a convex set. This can be achieved through various methods, such as:

- Convex Hull: The factor matrices are constrained to lie within the convex hull of a set of data points or features.
- Polytope Constraints: The factor matrices are restricted to lie within a specific polytope defined by a set of linear inequalities.
- Other Convex Constraints: Various other convex constraints can be used, such as restricting the factor matrices to be positive semidefinite or to have specific sparsity patterns.

Benefits: Convex NMF offers several advantages compared to standard NMF:

- Potential Uniqueness: It can help address the non-uniqueness issue of standard NMF, leading to more stable and reliable solutions.

- **Improved Convergence:** Convexity constraints can improve the convergence properties of the NMF algorithm.
- **Enhanced Interpretability:** In some cases, convexity constraints can lead to more interpretable factors.

Considerations:

- Choosing appropriate convex constraints is crucial for the success of convex NMF. The constraints should be relevant to the specific application and data characteristics.
- Enforcing convexity constraints can increase the computational complexity of the NMF algorithm. However, efficient algorithms have been developed to address this challenge.

In summary, convex NMF is a valuable variant of NMF that incorporates convexity constraints to improve the uniqueness, convergence, and interpretability of the factorization. It offers a powerful tool for various data analysis and machine learning tasks.

0.7.5 8.5 Online-NMF:

Online non-negative matrix factorization (Online NMF) is a variant of NMF designed to handle streaming data, where new data points arrive continuously over time.

Reasoning: Standard NMF assumes that the entire data matrix is available at once. However, in many real-world scenarios, data arrives in a streaming fashion, and it's not feasible to store and process the entire dataset at once. Online NMF addresses this challenge by updating the factorization incrementally as new data points become available.

How it Works: Online NMF algorithms typically employ incremental update rules to incorporate new data points without recomputing the entire factorization. These update rules adjust the factor matrices (W and H) based on the new data, gradually refining the factorization over time.

Benefits:

- **Handling Streaming Data:** Online NMF is well-suited for analyzing streaming data, where new data points arrive continuously.
- **Adaptability:** It can adapt to changes in the data distribution over time, making it suitable for dynamic environments.
- **Efficiency:** It avoids the need to store and process the entire dataset, making it more memory-efficient and computationally tractable for large datasets.
- **Real-time Analysis:** Online NMF enables real-time analysis of streaming data, providing timely insights as new data becomes available.

Key Considerations:

- Choosing an appropriate update rule is crucial for the performance of online NMF. Different update rules have different properties and may be more suitable for specific types of data or applications.
- The learning rate, which controls the step size of the updates, needs to be carefully tuned to balance stability and adaptability.
- Online NMF may require more iterations or data points to converge compared to standard NMF, as the factorization is updated incrementally.

In summary, online NMF is a valuable variant of NMF designed for handling streaming data. It offers adaptability, efficiency, and real-time analysis capabilities, making it suitable for various

dynamic data environments.

0.8 9. Application of NMF for Portfolio Diversification

In this section, we will explore a simplified example of portfolio diversification using the NMF technique. We will approach this by leveraging NMF's ability to extract meaningful features from financial data and providing a systematic way to diversify a portfolio based on underlying factors. This can be done with the following steps:

- **Correlation Matrix:** We start with the correlation matrix of asset returns, which captures the relationships between different assets.
- **NMF Decomposition:** NMF decomposes the correlation matrix into factor loadings (W) and factor scores (H). The factor loadings represent the contribution of each asset to each factor, while the factor scores represent the importance of each factor over time.
- **Factor Interpretation:** By analyzing the factor loadings, we can identify groups of assets that exhibit similar behavior and understand the underlying factors that drive asset co-movements.
- **Diversification:** We can diversify our portfolio by selecting assets that load highly on different factors. This reduces the overall portfolio risk by spreading investments across different sources of risk.
- **Portfolio Construction:** Based on the factor analysis, we can determine the weights for each asset in our portfolio, ensuring non-negativity and summing to 1.

In the following code snippet, we construct a simplified portfolio of data for five stocks: AAPL, MSFT, GOOG, AMZN, and TSLA. Then, we construct a `returns_df` DataFrame, containing daily returns for the selected assets.

```
[3]: # Download historical data for the tickers
tickers = ['AAPL', 'MSFT', 'GOOG', 'AMZN', 'TSLA', 'NVDA', 'META', 'JPM']
data = yf.download(tickers, start='2022-01-01', end='2023-01-01')['Close']

# 1. Load asset returns data
returns_df = data.pct_change().dropna()
returns_df
```

```
[*****100%*****] 8 of 8 completed
```

```
[3]: Ticker      AAPL      AMZN      GOOG      JPM      META      MSFT
Date
2022-01-04 -0.012691 -0.016916 -0.004535  0.037910 -0.005937 -0.017147 \
2022-01-05 -0.026600 -0.018893 -0.046830 -0.018282 -0.036728 -0.038388
2022-01-06 -0.016693 -0.006711 -0.000745  0.010624  0.025573 -0.007902
2022-01-07  0.000988 -0.004288 -0.003973  0.009908 -0.002015  0.000510
2022-01-10  0.000116 -0.006570  0.011456  0.000957 -0.011212  0.000733
...
2022-12-23 -0.002798  0.017425  0.017562  0.004745  0.007855  0.002267
2022-12-27 -0.013878 -0.025924 -0.020933  0.003504 -0.009827 -0.007414
2022-12-28 -0.030685 -0.014692 -0.016718  0.005465 -0.010780 -0.010255
2022-12-29  0.028325  0.028844  0.028799  0.005737  0.040132  0.027630
```

```
2022-12-30  0.002468 -0.002138 -0.002473  0.006606  0.000665 -0.004938
```

```
Ticker      NVDA      TSLA
Date
2022-01-04 -0.027589 -0.041833
2022-01-05 -0.057562 -0.053471
2022-01-06  0.020794 -0.021523
2022-01-07 -0.033040 -0.035447
2022-01-10  0.005616  0.030342
...
2022-12-23 -0.008671 -0.017551
2022-12-27 -0.071354 -0.114089
2022-12-28 -0.006019  0.033089
2022-12-29  0.040396  0.080827
2022-12-30  0.000753  0.011164
```

```
[250 rows x 8 columns]
```

Now that we have the `returns_df` DataFrame, containing daily returns for the selected assets, we can use it in the NMF portfolio diversification code:

```
[4]: # 2. Calculate the correlation matrix
correlation_matrix = returns_df.corr()
correlation_matrix
```

```
[4]: Ticker      AAPL      AMZN      GOOG      JPM      META      MSFT      NVDA
Ticker
AAPL      1.000000  0.695904  0.790572  0.547907  0.592900  0.824901  0.763022 \
AMZN      0.695904  1.000000  0.724023  0.501689  0.605782  0.741196  0.709078
GOOG      0.790572  0.724023  1.000000  0.511193  0.681990  0.845282  0.767571
JPM       0.547907  0.501689  0.511193  1.000000  0.393248  0.532506  0.528096
META      0.592900  0.605782  0.681990  0.393248  1.000000  0.625860  0.607685
MSFT      0.824901  0.741196  0.845282  0.532506  0.625860  1.000000  0.787883
NVDA      0.763022  0.709078  0.767571  0.528096  0.607685  0.787883  1.000000
TSLA      0.637218  0.591533  0.556651  0.364936  0.398646  0.563946  0.680243
```

```
Ticker      TSLA
Ticker
AAPL      0.637218
AMZN      0.591533
GOOG      0.556651
JPM       0.364936
META      0.398646
MSFT      0.563946
NVDA      0.680243
TSLA      1.000000
```

Non-negative matrix factorization (NMF) is fundamentally designed to work with non-negative

matrices. This means that the input matrix (V) and the resulting factor matrices (W and H) are expected to have only non-negative elements. However, in the specific context of portfolio diversification using NMF, the input matrix is often the correlation matrix of asset returns, which can contain negative values representing inverse relationships between assets.

In our example, the correlation matrix has only positive values. But if required, we would consider the following adjustments in certain cases:

- **Shifting:** we can shift the correlation matrix by adding 1 to all elements, ensuring all values are between 0 and 2. This might slightly alter the interpretation of the factor loadings but won't significantly affect the overall diversification strategy.
- **Absolute Values:** Another option is to use the absolute values of the correlation coefficients, focusing on the strength of the relationships rather than their direction. This might be useful when we want to identify groups of assets with strong co-movements regardless of whether they are positive or negative.

Next, we apply NMF to break down the correlation matrix into two smaller matrices:

```
[5]: # 3. Apply NMF
n_components = 5 # Number of factors to extract
model = NMF(n_components=n_components, init='random', random_state=0)
W = model.fit_transform(correlation_matrix) # Factor loadings
H = model.components_ # Factor scores
```

Here, we use the `NMF()` function from the `sklearn.decomposition` module in the scikit-learn library. - `n_components` specifies the number of (or components) to extract from the data. In the context of portfolio diversification, these factors represent underlying drivers of asset co-movements. In this specific case, setting `n_components = 5` assumes that there are likely around 5 major underlying factors driving the co-movements of the assets in the portfolio. This is a reasonable starting point, but we should experiment with different values to find the optimal number for our specific data and investment goals. Choosing the right number of components is often an iterative process involving domain expertise, experimentation, and model evaluation. - `init='random'` specifies the initialization method for the factor matrices (W and H). 'random' initializes them with random non-negative values. - `random_state=0` sets a random seed for reproducibility.

The next step involves analyzing the factor loadings (W) obtained from the NMF decomposition to understand the underlying factors driving asset co-movements and using this information to diversify the portfolio. We can access the factor loadings (W) using the variable `W` obtained in this step:

```
[6]: # Convert W to a pandas DataFrame for easier analysis
W_df = pd.DataFrame(W, index=returns_df.columns)
W_df
```

```
[6]:
```

	0	1	2	3	4
Ticker					
AAPL	0.464385	0.268748	2.308230e-01	0.780432	0.083933
AMZN	0.667705	0.000000	1.201913e-01	0.276371	0.499709
GOOG	0.458127	0.178795	1.562598e-01	0.803557	0.246558
JPM	0.251893	0.864869	1.184635e-07	0.396024	0.208779

META	0.003509	0.265374	2.902256e-01	0.680850	0.639629
MSFT	0.550906	0.165175	1.144755e-01	0.780762	0.174299
NVDA	0.406845	0.266955	3.439417e-01	0.672619	0.173927
TSLA	0.310215	0.169140	6.282628e-01	0.296560	0.046810

The factor loadings matrix (W) provides insights into how much each asset contributes to each factor.

- Rows: Represent the assets in your portfolio.
- Columns: Represent the extracted factors.
- Values: Indicate the strength of the relationship between an asset and a factor. Higher values suggest a stronger contribution of the asset to that factor.

To interpret the factors, we need to examine the assets that load highly on each factor. For each factor, we need to identify the assets with the highest loadings. These assets are most strongly associated with that factor.

```
[7]: # Identify top contributing assets for each factor
n_top_assets = 3
for factor_num in range(W_df.shape[1]):
    print(f"\nFactor {factor_num + 1}:")
    top_assets = W_df.iloc[:, factor_num].nlargest(n_top_assets)
    print(top_assets)
```

```
Factor 1:
Ticker
AMZN    0.667705
MSFT    0.550906
AAPL    0.464385
Name: 0, dtype: float64
```

```
Factor 2:
Ticker
JPM     0.864869
AAPL    0.268748
NVDA    0.266955
Name: 1, dtype: float64
```

```
Factor 3:
Ticker
TSLA    0.628263
NVDA    0.343942
META    0.290226
Name: 2, dtype: float64
```

```
Factor 4:
Ticker
GOOG    0.803557
```

```
MSFT    0.780762
AAPL    0.780432
Name: 3, dtype: float64
```

Factor 5:

```
Ticker
META    0.639629
AMZN    0.499709
GOOG    0.246558
Name: 4, dtype: float64
```

This output shows the top 3 contributing assets for each factor and their loadings. Remember that we can adjust `n_top_assets` and the interpretation of factors based on our specific data and investment goals. For now, we keep the top 3 contributing assets. We can now use this information to interpret the factors and develop a diversification strategy. Based on the top contributing assets for each factor, we can attempt to interpret their economic or financial meaning:

- **Factor 1:** Dominated by AMZN, MSFT, and AAPL. This factor could be interpreted as representing **Large-Cap Tech or Growth Stocks**. These companies are often associated with innovation, technological advancements, and high growth potential.
 - **Factor 2:** Primarily driven by JPM, with moderate contributions from AAPL and NVDA. This factor might represent **Financials or Value Stocks**. JPM is a major financial institution, and this factor could reflect the performance of the financial sector or companies with more established value characteristics.
 - **Factor 3:** Led by TSLA and NVDA, with some influence from META. This factor could be interpreted as **Electric Vehicles or Disruptive Technology**. TSLA is a leading electric vehicle manufacturer, and NVDA is a major semiconductor company, both associated with disruptive technologies.
 - **Factor 4:** Strongly influenced by GOOG, MSFT, and AAPL. This factor seems to represent **Big Tech or Market Leaders**. These companies are dominant players in the technology sector and have a significant impact on the overall market. It might overlap somewhat with Factor 1, but with a broader focus on market leadership.
 - **Factor 5:** Predominantly driven by META, AMZN, and GOOG. This factor could be interpreted as **Social Media and E-commerce**. META is a major social media company, AMZN is a dominant e-commerce player, and GOOG has a strong presence in both areas.
- Diversification Strategy

Based on this factor interpretation, a diversified portfolio could be constructed by selecting assets that load highly on different factors. For example, we could consider:

- **Large-Cap Tech/Growth:** AMZN or MSFT (Factor 1)
- **Financials/Value:** JPM (Factor 2)
- **Electric Vehicles/Disruptive Tech:** TSLA or NVDA (Factor 3)
- **Big Tech/Market Leaders:** GOOG or AAPL (Factor 4)
- **Social Media/E-commerce:** META (Factor 5)

By diversifying across these factors, we would be spreading investments across different sectors and risk profiles, potentially reducing overall portfolio risk and enhancing returns.

This was just a simplified interpretation for this demonstration. Further analysis might be needed to refine the factor definitions. And we also need to remember that the optimal diversification

strategy depends on individual investment goals, risk tolerance, and time horizon.

0.9 10. Conclusion

In this lesson, we learned about Non-negative Matrix Factorization (NMF), a dimensionality reduction technique with applications in financial engineering. We discussed the definition, properties, and variations of NMF, comparing it to other methods like PCA, SVD, and ICA. The lesson then focused on NMF's applications in finance, including portfolio diversification, risk management, and sentiment analysis. It demonstrated a simplified but practical example of using NMF for portfolio diversification with stock market data, showing how to extract and interpret factors to build a diversified portfolio. Overall, the lesson highlighted NMF as a powerful and interpretable tool for analyzing financial data and making informed investment decisions.

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Required Readings

Required Readings for this program are open access, which means you should be able to access them at no cost. The link provided in the citation will take you directly to the reading or to a page where you can download it.

1. Araci, Dogu. *FinBERT: Financial Sentiment Analysis with Pre-trained Language Models*. 2019. University of Amsterdam, master's thesis. *arXiv*, <https://arxiv.org/abs/1908.10063>.

- **Required pages:** All 11 pages
- **Estimated reading time:** 25 minutes

2. Yang, Yi, et al. "FinBERT: A Pretrained Language Model for Financial Communications." *arXiv* 2006.08097, 2020, <https://arxiv.org/abs/2006.08097>.

- **Required pages:** All 5 pages
- **Estimated reading time:** 20 minutes

Lesson 2 - News Data

December 25, 2025

MODULE 5 | LESSON 2

1 News Data

Reading Time	45 minutes
Prior Knowledge	Familiarity with Python programming: Basic Python syntax, data manipulation, DataFrames, yfinance, sklearn; Fundamental financial concepts: stocks, market sentiment, risk management, and trading strategies, sentiment analysis
Keywords	Non-negative Matrix Factorization (NMF), Feature Extraction, Parts-based Representation, Sentiment Analysis, Natural Language Processing (NLP), Sentiment Analysis, Topic Modeling, FinBERT, NMF, TF-IDF

In this lesson, we demonstrate how to use news data and natural language processing (NLP) techniques for financial analysis, specifically for Microsoft. We will explore different news sources, perform sentiment analysis using FinBERT, and use topic modeling (NMF) to uncover hidden themes in the news. The goal is to gain insights that could potentially lead to better-informed financial decisions.

```
[1]: !pip install pip --upgrade
      !pip install yfinance --upgrade
```

Requirement already satisfied: pip in /usr/local/lib/python3.11/site-packages (25.3)

WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manager, possibly rendering your system unusable. It is recommended to use a virtual environment instead: <https://pip.pypa.io/warnings/venv>. Use the --root-user-action option if you know what you are doing and want to suppress this warning.

Requirement already satisfied: yfinance in /usr/local/lib/python3.11/site-packages (1.0)
Requirement already satisfied: pandas>=1.3.0 in /usr/local/lib/python3.11/site-packages (from yfinance) (2.0.0)
Requirement already satisfied: numpy>=1.16.5 in /usr/local/lib/python3.11/site-packages (from yfinance) (1.26.2)
Requirement already satisfied: requests>=2.31 in /usr/local/lib/python3.11/site-packages (from yfinance) (2.32.3)
Requirement already satisfied: multitasking>=0.0.7 in /usr/local/lib/python3.11/site-packages (from yfinance) (0.0.11)
Requirement already satisfied: platformdirs>=2.0.0 in /usr/local/lib/python3.11/site-packages (from yfinance) (3.4.0)
Requirement already satisfied: pytz>=2022.5 in /usr/local/lib/python3.11/site-packages (from yfinance) (2023.3.post1)
Requirement already satisfied: frozendict>=2.3.4 in /usr/local/lib/python3.11/site-packages (from yfinance) (2.4.6)
Requirement already satisfied: peewee>=3.16.2 in /usr/local/lib/python3.11/site-packages (from yfinance) (3.17.8)
Requirement already satisfied: beautifulsoup4>=4.11.1 in /usr/local/lib/python3.11/site-packages (from yfinance) (4.12.2)
Requirement already satisfied: curl_cffi<0.14,>=0.7 in /usr/local/lib/python3.11/site-packages (from yfinance) (0.13.0)
Requirement already satisfied: protobuf>=3.19.0 in /usr/local/lib/python3.11/site-packages (from yfinance) (6.33.2)
Requirement already satisfied: websockets>=13.0 in /usr/local/lib/python3.11/site-packages (from yfinance) (15.0.1)
Requirement already satisfied: cffi>=1.12.0 in /usr/local/lib/python3.11/site-packages (from curl_cffi<0.14,>=0.7->yfinance) (1.15.1)
Requirement already satisfied: certifi>=2024.2.2 in /usr/local/lib/python3.11/site-packages (from curl_cffi<0.14,>=0.7->yfinance) (2025.11.12)
Requirement already satisfied: soupsieve>1.2 in /usr/local/lib/python3.11/site-packages (from beautifulsoup4>=4.11.1->yfinance) (2.4.1)
Requirement already satisfied: pycparser in /usr/local/lib/python3.11/site-packages (from cffi>=1.12.0->curl_cffi<0.14,>=0.7->yfinance) (2.21)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.11/site-packages (from pandas>=1.3.0->yfinance) (2.8.2)
Requirement already satisfied: tzdata>=2022.1 in /usr/local/lib/python3.11/site-packages (from pandas>=1.3.0->yfinance) (2023.3)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.11/site-

```
packages (from python-dateutil>=2.8.2->pandas>=1.3.0->yfinance) (1.16.0)
Requirement already satisfied: charset-normalizer<4,>=2 in
/usr/local/lib/python3.11/site-packages (from requests>=2.31->yfinance) (3.1.0)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.11/site-
packages (from requests>=2.31->yfinance) (3.4)
Requirement already satisfied: urllib3<3,>=1.21.1 in
/usr/local/lib/python3.11/site-packages (from requests>=2.31->yfinance)
(1.26.15)
WARNING: Running pip as the 'root' user can result in broken permissions
and conflicting behaviour with the system package manager, possibly rendering
your system unusable. It is recommended to use a virtual environment instead:
https://pip.pypa.io/warnings/venv. Use the --root-user-action option if you know
what you are doing and want to suppress this warning.
```

```
[2]: # Loading libraries
import torch
import datetime
import feedparser
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import requests
import yfinance as yf

from datetime import datetime, timedelta
from scipy.special import softmax
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.decomposition import NMF
from sklearn.preprocessing import MinMaxScaler
from transformers import AutoTokenizer, AutoModelForSequenceClassification

api_key = "API_KEY" # Replace with your actual News API key
```

2 1. Purpose of News Data

Exploring news data is crucial for informed financial and investment decision-making. Timely and accurate news analysis empowers investors to navigate complex markets effectively. In the dynamic world of finance, where markets fluctuate based on a multitude of factors, news data plays a pivotal role. Here are several reasons why exploring news data is crucial for informed financial and investment decisions:

Market Sentiment Analysis:

- **Emotional Drivers:** News articles capture investor emotions, expectations, and reactions. Positive news can boost confidence, leading to increased investments, while negative news

may trigger caution or even sell-offs.

- **Behavioral Insights:** Understanding market sentiment helps investors anticipate trends and align their strategies accordingly.

Event Impact Assessment: - **News-Driven Volatility:** Events such as earnings reports, geopolitical developments, or regulatory changes significantly impact stock prices. Analyzing news allows us to assess the potential effects of these events on specific assets or entire sectors. - **Timeliness Matters:** Investors who react swiftly to breaking news gain a competitive edge.

Risk Management: - **Early Warnings:** News provides early warnings about risks. Staying informed helps to mitigate risks, whether it's a sudden economic downturn, a corporate scandal, or a regulatory shift. - **Portfolio Adjustments:** By monitoring news, investors can adjust their portfolios proactively, minimizing exposure to adverse events.

Sector Insights: - **Industry Trends:** News highlights emerging trends, mergers, acquisitions, and disruptions within specific sectors. Investors can allocate funds based on sector-specific news. - **Sector Rotation:** Understanding which sectors are gaining momentum or facing challenges informs strategic asset allocation.

Algorithmic Trading: - **News-Driven Algorithms:** Algorithmic trading models rely on real-time news feeds. These algorithms react swiftly to breaking news, executing trades based on pre-defined rules. - **Quantitative Strategies:** Incorporating news sentiment scores into quantitative models enhances their predictive power.

Behavioral Finance: - **Herding Behavior:** News influences investor behavior. Herding (following the crowd) or panic selling during market downturns can be better understood through news analysis. - **Fear of Missing Out (FOMO):** News-driven biases impact decision-making. Recognizing these biases helps investors make more rational choices.

In summary, news data isn't just information; it's a compass guiding financial decisions. By analyzing news comprehensively, investors gain insights, manage risks, and adapt to the ever-evolving landscape of global markets.

#2. Sources of News Data

There's no single news data API that's universally used by all financial engineers. The choice of API often depends on specific needs, budget, and the type of analysis being performed. However, here are some popular and reputable news data APIs commonly used in the financial industry:

1. Refinitiv (formerly Thomson Reuters):

- Widely used by institutional investors and professional traders.
- Offers comprehensive news coverage: Global markets, equities, fixed income, commodities, FX, etc.
- Advanced features: Sentiment analysis, entity recognition, real-time alerts.
- Integration with Refinitiv Eikon platform: Provides access to a vast range of financial data and analytics tools.

2. Bloomberg:

- Another industry-standard for financial professionals.
- Extensive news coverage: Global markets, business news, economic data, company information, worldwide televised broadcasts

- Powerful analytics and data visualization tools. Integrated with Bloomberg Terminal: Provides a comprehensive suite of financial data and tools.

3. FactSet:

- Popular among investment professionals and asset managers.
- Strong focus on fundamental data and company research.
- Offers news coverage: Market news, company announcements, economic data.
- Integrated with FactSet workstation: Provides access to a wide range of financial data and analytics.

4. Dow Jones (Factiva):

- Well-known for its business and financial news coverage.
- Extensive historical archive: Access to news articles dating back decades.
- Global coverage: News from various regions and languages.
- Often used for sentiment analysis and text mining.

When choosing an API, the most important considerations are: - **Cost:** These APIs typically involve subscription fees, which can vary significantly based on data access, features, and usage volume. - **Data Coverage:** Each API has its strengths in terms of the types of news and markets it covers. Choose one that aligns with your specific needs. - **Integration:** Consider how easily the API integrates with your existing tools and workflows.

Unfortunately, the APIs used by financial engineers listed above (Refinitiv, Bloomberg, FactSet, Dow Jones) are typically not free to use. They are commercial products targeted toward professional investors and institutions, and they usually involve subscription fees that can be quite substantial. However, there are a few other alternative options available for free or limited free access to get news data, some of which we will consider in detail in this lesson.

Important Note: When using free options, be mindful of usage limits and terms of service to avoid exceeding quotas or violating any restrictions.

If you have specific budget constraints or are just starting to explore news data analysis, the free options can be a good starting point. However, for professional-grade financial analysis, investing in a commercial API will likely be necessary to access the depth and breadth of data required.

2.1 Using yfinance for News Data

While `yfinance` is primarily known for historical stock price data, it also provides access to financial news related to specific stocks.

For example, we can use the `yfinance` library to get stock information for Microsoft. We can leverage the fact that it's typically a list of dictionaries. Each dictionary represents a news article and contains keys like `title`, `publisher`, `link`, `published`, etc.

Once we have all the interesting details, we can print the news data, and we can also create a `DataFrame` from specific pieces of information to keep our data for further manipulations and exploratory data analysis.

```
[3]: # Create a Ticker object for Microsoft and access the news data
msft = yf.Ticker("MSFT")
news_data = msft.news
```

```

# Print the desired information for each article
for article in news_data:
    content = article.get('content')
    if content:
        # Assuming title, publisher, link are now within 'content'
        print("Published Time:", content.get('pubDate'))
        print("Title:", content.get('title'))
        print("Publisher:", content.get('provider').get('displayName'))
        print("Link:", content.get('canonicalUrl').get('url'))
        print("Content Type:", content.get('contentType'))
        print("-" * 30)

```

Published Time: 2025-12-23T21:15:25Z

Title: Apple spent 2025 setting itself up for the future - and its biggest moves weren't about AI

Publisher: Yahoo Finance

Link: <https://finance.yahoo.com/news/apple-spent-2025-setting-itself-up-for-the-future--and-its-biggest-moves-werent-about-ai-211525196.html>

Content Type: STORY

Published Time: 2025-12-25T05:10:47Z

Title: Nebius Group (NBIS) Is Up 20.8% After Blackwell AI Cloud Deal Wins With Microsoft And Meta - Has The Bull Case Changed?

Publisher: Simply Wall St.

Link: <https://finance.yahoo.com/news/nebius-group-nbis-20-8-051047431.html>

Content Type: STORY

Published Time: 2025-12-25T00:30:02Z

Title: This Dividend-Yielding Tech Stock Looks a 'Compelling Buy' for 2026 After 2 Years of Underperformance

Publisher: Barchart

Link: <https://www.barchart.com/story/news/36784733/this-dividend-yielding-tech-stock-looks-a-compelling-buy-for-2026-after-2-years-of-underperformance>

Content Type: STORY

Published Time: 2025-12-24T22:45:00Z

Title: Ranking the Best "Magnificent Seven" Stocks to Buy for 2026. Here's My No. 4

Publisher: Motley Fool

Link: <https://www.fool.com/investing/2025/12/24/ranking-the-best-magnificent-seven-stocks-to-buy-4/>

Content Type: STORY

Published Time: 2025-12-24T19:15:16Z

Title: Microsoft Will 'Prove Doubters Wrong' In 2026 With AI-Fueled Growth

Publisher: Investor's Business Daily

Link: <https://www.investors.com/news/technology/microsoft-stock-msft-core->

winner-in-ai/?src=A00220&yptr=yahoo

Content Type: STORY

Published Time: 2025-12-24T18:25:01Z

Title: Meta Ordered By Italian Regulators To Keep WhatsApp Open For Rival AI Chatbots

Publisher: Investor's Business Daily

Link: <https://www.investors.com/news/technology/meta-stock-whatsapp-ai-italy/?src=A00220&yptr=yahoo>

Content Type: STORY

Published Time: 2025-12-24T17:18:26Z

Title: Sector Update: Tech Stocks Advance Wednesday Afternoon

Publisher: MT Newswires

Link: <https://finance.yahoo.com/news/sector-tech-stocks-advance-wednesday-171826809.html>

Content Type: STORY

Published Time: 2025-12-24T16:51:09Z

Title: Top Midday Stories: Apple's Cook Buys \$3 Million in Nike Stock; OpenAI Explores Ads in ChatGPT; Sanofi-Dynavax \$2.2 Billion Deal

Publisher: MT Newswires

Link: <https://finance.yahoo.com/news/top-midday-stories-apple-apos-165109264.html>

Content Type: STORY

Published Time: 2025-12-24T15:49:00Z

Title: AI & Security Push Aids Google Cloud: What's Ahead for GOOGL Stock?

Publisher: Zacks

Link: <https://finance.yahoo.com/news/ai-security-push-aids-google-154900963.html>

Content Type: STORY

Published Time: 2025-12-24T15:20:00Z

Title: The Tech Stock Warren Buffett Wishes He'd Bought

Publisher: Motley Fool

Link: <https://www.fool.com/investing/2025/12/24/the-tech-stock-warren-buffett-wishes-hed-bought/>

Content Type: STORY

As you notice, here we only have 8 articles. `yfinance` does have some limitations when it comes to retrieving news articles.

- **Limited Number of Articles:** `yfinance` doesn't provide a direct way to control the number of news articles it fetches. The number of articles returned can vary and seems to be capped, often resulting in a smaller dataset than you might expect. The exact limit is not explicitly documented and might depend on factors like the stock ticker, news availability, and the underlying data source used by `yfinance`.

- **Reliance on External APIs:** `yfinance` doesn't have its own dedicated news database. It relies on aggregating news from various external sources and APIs. This means the availability and quantity of news data can be influenced by the limitations and potential changes in those external sources.
- **No Fine-grained Control:** `yfinance` doesn't offer options to filter news articles by specific criteria like date range, news provider, or keywords. You get a general set of recent news articles related to the stock ticker, but you can't customize the query further.

In summary: While `yfinance` is a convenient tool for getting basic stock information and a quick overview of recent news, it's not ideal for comprehensive news analysis or if you need a large and customizable news dataset. But despite limitations, `yfinance` still has variable use cases, especially when you need quick and easy access to financial data.

- **Simple News Monitoring:** Getting a snapshot of recent news related to a stock. While limited, the news feature in `yfinance` can be helpful for staying updated on major developments or headlines affecting a company.
- **Prototyping or educational purposes:** It's a great tool for learning about financial data analysis or for quickly testing ideas without the overhead of more complex data sources.

2.2 RSS Feeds: Google News RSS

An alternative way of getting news updates is to subscribe to an RSS feed. RSS stands for Really Simple Syndication or Rich Site Summary. It's a standardized web feed format that allows users to subscribe to updates from websites or blogs. These updates are typically delivered as news headlines, article summaries, or other content changes. Many well-known financial news sources provide RSS feeds. Some examples include the *Wall Street Journal*, Bloomberg, Reuters, *Financial Times*, Seeking Alpha, The Motley Fool, Benzinga, etc.

In this lesson, we explore using Google News RSS to access news articles from Google News in a structured format that can be easily processed by machines. While not strictly an API, we can parse Google News RSS feeds for free using libraries like `feedparser` Python library designed to parse syndicated feeds, most commonly RSS and Atom feeds. It can handle various feed formats and variations, making it a versatile tool for extracting information from different sources.

In the following code, we define the RSS feed URL and extract article information (title, link, publication date, etc.) for the query term "Microsoft":

```
[4]: # Define RSS feed URL and retrieve feed
query = "Microsoft"
rss_url = f"https://news.google.com/rss/search?q={query}&t&hl=en-US&gl=US&ceid=US:en"
feed = feedparser.parse(rss_url)

for entry in feed.entries:
    print("Published:", entry.published)
    print("Title:", entry.title)
    print("Link:", entry.link)
    print("-" * 30)
```

Published: Mon, 01 Dec 2025 08:00:00 GMT

Title: Tech Moves: Expedia names first AI chief; Textio founder joins Microsoft; T-Mobile exec departs - GeekWire

Link: <https://news.google.com/rss/articles/CBMiwAFBVV95cUxNdnhialhu0VhIZzVGQWVnb2d1UmVNTEFTbWw3Z1RwVGNENnhCMThxc085Yl8zYVNmT20tWnRFb1pTTW5rV21LOTBJVnpXRvVrRk51RDJXeFVialFjeC01Mmt1NE45VlZGT3VlN1FFV3c1UHBZczd0LWZvSUxMajFpeUFNVmh6bjBVZUVKQkRvWUFmeC16ZC1IUz15V05iYX1FYThBRnlQWHBOWkplbHRlWDZvMHZLVdJ3Yk5QdGVkbW8?oc=5>

Published: Tue, 29 Apr 2025 07:00:00 GMT

Title: Health Catalyst Announces Strategic Partnership with Microsoft t - GuruFocus

Link: <https://news.google.com/rss/articles/CBMirwJBVV95cUxPMTBRVkJZuMWE5d25XX3dsVS1MN0xDLURJYnlrbm5WN0h3a1pLVEFJUvZZLWxtYWhRdzRPX21YNVBiWVFIMXRUUmQxQTRLNTN5R2pWd1VMSG9kWERrb3R6RkhselZRazZ0Q1lpYmhEc3U0RFpseWtObFNLY1ZETFJMd2IzNFNyczRqUUNjZ1NsX1FzTEswclBPSWxvX1QxWkdCQ0pjOG8wdjN3UFhBbkVILVY2aFRWQ25jRTZORmpSRENLa1BSZFo1aXfkUzVsbDNFWUZxa0hT22IwSF1mRlk2Qm5MbDdbmVRwVC13SmlvUX1rbjEwU29peVozWVA3bDNSZWRYWjB4V21fWk9abDBIZmlCUUVoUDVHYnhCQVFMTTIwVVL1SaWlSWnlXZXNxVmc?oc=5>

Published: Wed, 18 Jan 2023 08:00:00 GMT

Title: What's up with... e&, Vodafone, Microsoft, T-Mobile US - telecomtv.com

Link: <https://news.google.com/rss/articles/CBMitgFBVV95cUxPazBGdkR1ZzZhZTloMEhpR2pMN1UwZVNPTmhZMXRhcEEzNklJQjRqTkFMeGhuZ1JiZG56b1VxX2tIaUUyNm9NZDVtaTZuTG9pRTdIQ1hjaVdDM2oyUkFGR1BSZ1JCaTdaQV9hN3JicGxpVy1IaTU5TDkwVkl3U29RY0V2VEItUTdJY1B6NGh6aDNSZzF1eV9RV2dMqkY0dVZBRWV1c3NpWW9KWV9PU2JQRldKY2ZNUdIBuwFBVV95cUxQeEk1d21RenU2Xy10T3BzeEFLaHl1TbVVxSzBISTZEcnhDLV12NjZhQkdiS19PcHNjWnR4VEhvSEE5SzJwWj1hNU5HR2ZKUWh4RG0oYVRma3pkQzVPdWMwbV9yRHVWdURiS1Ffm1pGNm9GdlZFeU83TFZWUTV6V3FNv1V5Tm9veHp6eURGM290Sk94MmUzTjB4OXRkMFh6Z11LbTBjBUZxdnlGeTJMcDVXeDNlT3ZTQ1doeWNR?oc=5>

Published: Tue, 13 Aug 2024 07:00:00 GMT

Title: Kudo adds AI speech translation for Microsoft Teams - inavateonthenet.net

Link: <https://news.google.com/rss/articles/CBMiogFBVV95cUxPNXpORnZlU0c5d08x0FBxQzFDRHRzeDVZqjVJTEdaX3BoVDhNMVBnZ0p1cnJRR1lydU9uVmhCcORtbG51bHVWmZJoX051eHh6WHVNCkMwa3lxZC1qdZBUb0xzBghJS1NLSEtnbEw2bDRUNlp3VmFyQ3RtdlRmanBtVFZiY010ZFVVTjZ5OTRXLWNqLTduWUxKNks4bGJpN0x00EE?oc=5>

Published: Wed, 06 May 2015 07:00:00 GMT

Title: Secret T-shirt message explains why Microsoft skipped Windows 9 - Business Insider

Link: <https://news.google.com/rss/articles/CBMihAFBVV95cUxPMHJGREdockM1d3J0R1hoMnByMDM5MmJGWM1MWV9MXzVJc1Y5QjBqRU1Vb3BrNH1Hbk0xUjUxM3I2U31RS1JfccktrbjZ3bXJZVU9LNkc2UkZlTkhNbG1NZWZNSTntQm45bVd5SzhJWVJkaFFwBFFrWFRzSXRUFUHZyQ2ZWNS?oc=5>

Published: Sat, 02 Mar 2024 08:00:00 GMT

Title: How Microsoft, T-Mobile, and Amazon use big bets to build durable businesses in turbulent times - GeekWire

Link: <https://news.google.com/rss/articles/CBMiwgFBVV95cUxQTjJEWDkxX1QzeVhCaWxhRFZ2N2w0RG5JVEVRcGFcd184dkdWa0pWTjVkr1puTEZjNWhPbzVtSnBvMktjQXd0MwP1SUcweHF4UXR1ZW52djBvV2o2bUNTaeC1aDNKX310NktiTOxWR3NPWUhfEDNCZ01hTnRPeEYxRkxLUGI2SE5kYVQ2NGJyb>

As you can see here, we have more news titles than we get from `yfinance`. But again, Google News RSS also has its advantages and limitations:

Advantages of Google News RSS: - Free Access: You can access and parse Google News RSS feeds without any API keys or subscription fees. - Wide Range of Topics: Google News covers a vast array of news categories and topics. - Fresh Content: RSS feeds are updated frequently, so you get access to the latest news articles.

Limitations: - No Fine-grained Control: You can't filter articles by specific criteria like date range or news source within the RSS feed itself. - Potential Rate Limiting: Google might impose rate limits on how frequently you can access their RSS feeds. - No Historical Data: RSS feeds typically provide only recent articles, not a historical archive.

Use Cases: - Staying Updated on Specific Topics: Subscribe to RSS feeds for topics relevant to your interests or investments. - Building Simple News Aggregators: Create a basic news aggregator that displays articles from various Google News RSS feeds. - Sentiment Analysis and Text Mining: Extract text from news articles for sentiment analysis or other text-based research.

Overall, Google News RSS is a valuable resource for accessing free news data, especially for smaller projects, personal use, or when you need a quick overview of recent news on specific topics.

2.3 2.3 News API

Although not a Python package, News API provides a straightforward way to fetch news data. It offers a free tier with limited requests per day. This might be sufficient for small-scale projects or learning purposes. You will need to sign up for a free API key on the NewsAPI website before using it.

News API is a cloud-based REST API that provides programmatic access to a vast collection of news articles from thousands of sources around the world. It aggregates news from reputable publishers, news agencies, blogs, and other online media outlets.

Compared to `yfinance` and Google News RSS: - More Articles: You can potentially retrieve a much larger number of news articles with News API. - Customization: You have much finer control over the news data you fetch through search and filtering options. - Additional Features: Sentiment analysis and various API endpoints add more analytical capabilities.

Limitations: - Cost: While it offers a free tier with limited requests, you'll need a paid subscription for larger-scale usage or access to all features. - Rate Limits: Even with paid plans, there are rate limits on how many requests you can make within a given time period.

Use Cases: - News Monitoring and Analysis: Track news trends, identify emerging topics, and analyze media coverage for specific companies, industries, or events. - Sentiment Analysis: Gauge public sentiment toward brands, products, or political figures. - Market Research: Gather insights on consumer behavior, competitor activity, and industry trends. - Content Curation: Build news aggregators or personalized news feeds. - Algorithmic Trading: Incorporate news sentiment and events into trading strategies.

Overall, News API is a more powerful and versatile tool for news data analysis compared to `yfinance` and Google News RSS. It's a great option if you need a larger dataset, more customization, and additional features like sentiment analysis. However, it does come with a cost for more extensive usage.

For the remainder of this lesson, we will explore some of the News API features. First, we need to sign up and get an API key.

- Sign Up: Create a free account on the News API website: <https://newsapi.org/pricing>, selecting the free Developer pricing plan. Sign up as an individual. The free tier offers limited requests per day and allows you to search news articles (with a 24-hour delay) up to a month old, but this should suffice for learning purposes.
- Get Your API Key: As soon as you sign up, you'll be greeted with your unique API key that you'll need to include in your requests.
- On the greeting page, you will also find a link to the "Getting Started Guide." Please take some time to explore the API documentation for detailed information on endpoints, parameters, and response formats.

In this lesson, we will explore some basic functionality and experiment with different queries and filters to tailor the results to our specific needs.

The following is the example code that fetches news articles related to "Microsoft" from News API, saves them in DataFrame, and handles potential errors during the process. Do not forget to get your own API key. Then, please navigate to the top of this lesson and replace `API_KEY` with your actual API key in the code cell at the top of this notebook. Please rerun that code cell and then proceed with the following code cell below:

```
[ ]: # Define News API variables
query = "Microsoft"
url = f"https://newsapi.org/v2/everything?
      q={query}&apiKey={api_key}&language=en"
response = requests.get(url)

# Get news data
results = []
if response.status_code == 200:
    news_data = response.json()
    for article in news_data['articles']:
        results.append({
            'Date': article['publishedAt'][:10], # Extract date
            'URL': article['url'],
            'Source': article['source']['name'],
            'Author': article['author'],
            'Title': article['title'],
            'Description': article['description'],
            'Content': article['content']
        })
else:
    print("Error fetching news:", response.status_code)

# Create DataFrame
df = pd.DataFrame(results)
df
```

We can also save the dataset locally in a `news_data.csv` file. Saving News API data to a CSV file is beneficial because of the limitations on data availability for certain time periods, particularly with the free plan. It is highly probable that the remainder code in this lesson will output different results when you run it. Thus, referring to `news_data.csv` might be beneficial.

```
[ ]: # Save News dataset
df.to_csv('news_data.csv', index=False)
```

`if response.status_code == 200:` This line checks if the value of `response.status_code` is equal to 200. In the context of HTTP requests, a status code of 200 typically indicates a successful request.

If the condition in the if statement is true (i.e., the status code is 200), this line executes and the code execution will move to next line. `response.json()` in the following line is a method that attempts to parse the response content as JSON (JavaScript Object Notation). JSON is a common data format used for data exchange on the web. The parsed JSON data is then assigned to the variable `news_data`.

The code output saves article metadata for each news item: information like publication date (`publishedAt`), source (`source`), author (`author`), title (`title`), description (`description`), content (`content`), and URL (`url`) provide context for the sentiment analysis.

Note the “[Removed]” in the Content column of dataframe. This is likely due to content restrictions imposed by the news source or by News API itself. The reasoning behind this is as follows:

- **Publisher Restrictions:** Some news publishers may choose to limit the distribution of their full article content through APIs. They may only provide headlines, summaries, or partial content. In such cases, News API might replace the unavailable content with “[Removed]”.
- **News API Policies:** News API itself might have policies in place to prevent the scraping or redistribution of full articles. They may intentionally remove or truncate content to comply with copyright regulations or publisher agreements.
- **Content Filtering:** In some cases, News API might apply content filtering to remove sensitive or inappropriate content. This could also result in “[Removed]” appearing in place of the filtered content.

Unfortunately, there’s no direct way to retrieve the removed content through News API. We would need to access the full article through the original source’s website if it is available.

The News API allows you to **filter news articles by category**. We can specify the category parameter in an API request to retrieve news from a specific category, such as business, entertainment, general, health, science, sports, and technology.

Important considerations: - **Category Availability:** The availability of certain categories might vary depending on News API plan and the geographic region we are targeting. - **Relevance:** While the business category is a good starting point for finance news, it might also include articles that are not strictly finance-related. We might need to further filter the results based on keywords or other criteria to refine the selection. - **Language and Country:** We can further refine search by specifying the language and country parameters to retrieve news from a specific region and in a particular language.

Unfortunately, the News API does not directly support searching for multiple categories simultaneously using the category parameter. We can only specify one category at a time. However, we

can achieve a similar result by making separate API requests for each category and then combining the results.

A significant limitation of searching by category is that NEWS API currently does not support category parameter on <https://newsapi.org/v2/everything> endpoint. Instead, we should use <https://newsapi.org/v2/top-headlines> endpoint. This endpoint allows filtering by category. However, this endpoint has some limitations compared to `/everything`: - Limited Articles: It returns a limited number of articles (usually the top 20-30 headlines) for a given query and category. It's not meant for comprehensive news retrieval. - Focus on Recent News: It primarily focuses on recent news and might not include older articles. - Source Restrictions: It sources articles only from a limited set of popular and well-known news sources, potentially missing out on smaller publications.

3 3. Sentiment Analysis

Financial traders can leverage news data and sentiment analysis in several ways to enhance their decision-making processes. News data and sentiment analysis empower traders to make informed decisions, manage risks, and adapt to market dynamics. By combining quantitative models with qualitative insights, traders gain a competitive edge.

How Financial Traders Might Use Sentiment Analysis: - Identify Market Sentiment: Analyze the overall sentiment of news articles related to a specific stock, sector, or the entire market to gauge investor sentiment and predict potential price movements. - Event-Driven Trading: Detect news events (like earnings announcements, product launches, regulatory changes) and analyze their sentiment to identify trading opportunities. - News-Based Sentiment Indicators: Create custom indicators based on news sentiment to supplement technical analysis and inform trading decisions. - Risk Management: Monitor news sentiment to assess potential risks and adjust portfolio positions accordingly.

Important Considerations: - Accuracy of Sentiment Analysis: Sentiment analysis is not always perfect, especially for financial news, which can contain complex language and jargon. It's crucial to validate the accuracy of your sentiment analysis models. - Timeliness: News sentiment can change rapidly. For trading purposes, real-time or near-real-time news feeds and sentiment analysis are often required. - Integration with Trading Systems: You'll need to integrate the news sentiment data with your trading platform or algorithms to automate trading decisions based on the analysis.

Remember that sentiment analysis is just one tool in a financial trader's toolkit. It should be used in conjunction with other forms of analysis and risk management strategies.

3.1 3.1 Sentiment analysis using FinBERT

Unfortunately, for us, News API's Sentiment Analysis is a paid feature. If you have access to such a plan, the API response would already include sentiment scores (positive, negative, neutral) for each article. You can then use these scores directly in your trading strategies.

In this lesson, we will use an alternative method by implementing the sensitive scoring function manually. For this, we will use **FinBERT**, a pre-trained NLP model specifically designed for financial text. It's based on the BERT architecture (Bidirectional Encoder Representations from Transformers), a type of Transformer model, and has been fine-tuned on a large corpus of financial data.

Transformers are a type of deep learning model that have been very successful in natural language processing (NLP) tasks. They are particularly good at understanding the relationships between different words in a sentence, which is important for tasks like sentiment analysis. FinBERT is a specific type of Transformer model that has been pre-trained on a large corpus of financial text. This means that it has already learned a lot about the language used in financial documents. This makes it particularly good at understanding the nuances of financial language, including jargon, sentiment, and specific financial concepts, and this makes it well suited for tasks like sentiment analysis of financial news articles.

In the following piece of code, we utilize the FinBERT model to apply a sentiment score to each news article. To do this, we first need to preprocess the content of each article using the `preprocess` function. Then, we apply `get_sentiment()` to perform sentiment analysis on a given preprocessed text using the FinBERT model. Here's how it works:

- **Tokenization:** The preprocessed text is tokenized using the tokenizer object. Tokenization is the process of breaking down the text into individual units (tokens) - words or subwords - that the model can understand. The `return_tensors='pt'` argument specifies that the tokens should be returned as PyTorch tensors. PyTorch tensors are multi-dimensional arrays that are a fundamental data structure in the PyTorch library. They are similar to NumPy arrays but with some key advantages. As an analogy, think of PyTorch tensors as containers that hold numerical data that can be of various dimensions (e.g., 1D for vectors, 2D for matrices, 3D or higher for more complex data).
- **Model Inference and Score Extraction using Softmax:** The tokenized input is passed to the FinBERT model for inference. The relevant scores are extracted from the model's output and converted to a NumPy array using `detach().numpy()`. The scores are then passed through the `softmax` function. Softmax converts the scores into probabilities, ensuring that they sum up to 1.

The `get_sentiment()` function returns a dictionary containing the probabilities for negative, neutral, and positive sentiment, with the highest probability first.

The following code snippet processes the response from the News API and performs sentiment analysis (by successive implementation of text preprocessing, tokenization, model inference, and getting a sentiment) on the 'Description' of each retrieved article. We are also dropping rows where 'Description' has been removed and the article is no longer available:

```
[5]: # Specify the FinBERT model
MODEL = f"ProsusAI/finbert"
tokenizer = AutoTokenizer.from_pretrained(MODEL)
model = AutoModelForSequenceClassification.from_pretrained(MODEL)

# Text processing function
def preprocess(text):
    if text is None: # Handle None values by returning an empty string if text
        ↪ is None
        return ""
    new_text = []
    for t in text.split(" "):
        t = '' if t.startswith('#') and len(t) > 1 else t # remove hashtags
```

```

t = '' if t.startswith('@') and len(t) > 1 else t # remove usernames
t = '' if t.startswith('http') else t # remove URLs
new_text.append(t)
return " ".join(new_text)

# Sentiment scoring function
def get_sentiment(text):
    text = preprocess(text)
    encoded_input = tokenizer(text, return_tensors='pt')
    output = model(**encoded_input)
    scores = output[0][0].detach().numpy()
    scores = softmax(scores)
    return {
        'positive': scores[0],
        'negative': scores[1],
        'neutral': scores[2]
    }

```

Now that we have initiated the FinBERT model and constructed helper functions, we can apply these techniques onto our `df` News data DataFrame. Please note that by the time you run the code in this lesson, availability of news data via the News API free plan would have changed since the plan allows one month of history. We saved the original News data when we wrote this lesson. You will need to load the 'WQU_FD_news_data.csv' file in order to get the same results in this lesson. Otherwise, please proceed to explore fresh data.

```

[6]: # Open saved DataFrame - OPTIONAL
df = pd.read_csv('WQU_FD_news_data.csv')
df.head()

```

```

[6]:
      Date                                URL      Source
0  2024-09-27          https://removed.com  [Removed] \
1  2024-10-09          https://removed.com  [Removed]
2  2024-10-10  https://www.theverge.com/2024/10/10/24266333/a...  The Verge
3  2024-10-01  https://www.theverge.com/2024/10/1/24258337/mi...  The Verge
4  2024-10-01  https://www.wired.com/story/mustafa-suleyman-i...    Wired

      Author                                Title
0         NaN          [Removed] \
1         NaN          [Removed]
2  Kylie Robison  Agents are the future AI companies promise - a...
3    Tom Warren  Microsoft is using AI to improve Windows search
4   Will Knight  Microsoft's AI Boss Wants to Bring 'Emotional ...

      Description
0          [Removed] \
1          [Removed]
2  OpenAI, Google, and Microsoft believe autonomo...
3  Microsoft is improving its Windows search acro...

```

4 Microsoft AI CEO Mustafa Suleyman is overseein...

	Content
0	[Removed]
1	[Removed]
2	Illustration by Cath Virginia / The Verge Ph...
3	Illustration by Alex Castro / The Verge\r\n\n ...
4	We don't save any of the material with Copilot...

```
[7]: # Remove rows with "[Removed]" or None in Description
df = df[df['Description'] != '[Removed]']
df = df.dropna(subset=['Description'])

# Apply sentiment scoring to the 'description' column and create new columns
df.loc[:, 'Sent_positive'] = df['Description'].apply(lambda x:
    get_sentiment(x)['positive'])
df.loc[:, 'Sent_negative'] = df['Description'].apply(lambda x:
    get_sentiment(x)['negative'])
df.loc[:, 'Sent_neutral'] = df['Description'].apply(lambda x:
    get_sentiment(x)['neutral'])
df
```

```
[7]:      Date      URL
2  2024-10-10  https://www.theverge.com/2024/10/10/24266333/a... \
3  2024-10-01  https://www.theverge.com/2024/10/1/24258337/mi...
4  2024-10-01  https://www.wired.com/story/mustafa-suleyman-i...
5  2024-10-01  https://www.theverge.com/2024/10/1/24259187/mi...
6  2024-10-02  https://www.theverge.com/2024/10/2/24260296/mi...
..      ...      ...
95 2024-10-12  https://gizmodo.com/forget-amazon-prime-day-th...
96 2024-10-04  https://www.fastcompany.com/91203365/microsoft...
97 2024-09-17  https://www.digitaltrends.com/computing/launch...
98 2024-10-09  https://finance.yahoo.com/news/microsoft-corp-...
99 2024-09-18  https://www.businessinsider.com/bill-gates-dau...
```

	Source	Author
2	The Verge	Kylie Robison \
3	The Verge	Tom Warren
4	Wired	Will Knight
5	The Verge	Tom Warren
6	The Verge	Tom Warren
..	...	...
95	Gizmodo.com	Gizmodo Deals
96	Fast Company	Associated Press
97	Digital Trends	Willow Roberts
98	Yahoo Entertainment	Neha Gupta
99	Business Insider	Nora Redmond

	Title	
2	Agents are the future AI companies promise - a...	\
3	Microsoft is using AI to improve Windows search	
4	Microsoft's AI Boss Wants to Bring 'Emotional ...	
5	Microsoft gives Copilot a voice and vision in ...	
6	Microsoft wants to know why some game develope...	
..	...	
95	Forget Amazon Prime Day, This Windows 11 Pro L...	
96	Microsoft and the DOJ intercept this Russian h...	
97	Launching Windows 11 apps could get up to 50% ...	
98	Microsoft Corp (MSFT): Stock That Will Make Yo...	
99	Bill Gates' daughter says friends have cut her...	

	Description	
2	OpenAI, Google, and Microsoft believe autonomo...	\
3	Microsoft is improving its Windows search acro...	
4	Microsoft AI CEO Mustafa Suleyman is overseein...	
5	Microsoft is overhauling its Copilot AI assist...	
6	Microsoft is looking for Xbox feedback from ga...	
..	...	
95	Right after Prime Day, StackSocial is offering...	
96	Russian hacking group Star Blizzard is targeti...	
97	has spotted a recent from Microsoft confirming...	
98	We recently compiled a list of the 10 stocks t...	
99	Phoebe Gates, a daughter of Bill and Melinda F...	

	Content	Sent_positive
2	Illustration by Cath Virginia / The Verge Ph...	0.582753 \
3	Illustration by Alex Castro / The Verge\r\n\n ...	0.771638
4	We don't save any of the material with Copilot...	0.335345
5	Image: Microsoft\r\n\n \n\n\n Copilot is trans...	0.281658
6	Image: Microsoft\r\n\n \n\n\n Microsoft wants to...	0.048003
..	...	...
95	Following the Prime Day event, a deal has emer...	0.080600
96	A hacking group tied to Russian intelligence t...	0.022818
97	Windows Latest has spotted a recent support do...	0.885826
98	We recently compiled a list of the 10 stocks t...	0.055370
99	Phoebe Gates told her dad, Bill Gates, about s...	0.016213

	Sent_negative	Sent_neutral
2	0.009541	0.407706
3	0.011051	0.217311
4	0.011903	0.652752
5	0.026102	0.692241
6	0.021824	0.930173
..	...	...

95	0.011433	0.907967
96	0.562415	0.414767
97	0.008787	0.105386
98	0.013606	0.931024
99	0.861046	0.122741

[89 rows x 10 columns]

The code output contains sentiment scores. This is how to interpret sentiment:

- Higher probability indicates stronger sentiment. For example, if **positive: 0.85**, it suggests a strong positive sentiment expressed in the article.
- Look for dominant sentiment. The category with the highest probability usually represents the overall sentiment of the article.
- Consider the context. Even with high sentiment scores, it's important to read the article content to understand the nuances and specific aspects driving the sentiment.

By analyzing the sentiment scores alongside the article content, we can gain insights into the overall sentiment toward Microsoft as reported in the news. This information can be valuable for understanding market perception, identifying potential trends, and making informed decisions.

3.2 Visualizing Sentiment Scores

We can now visualize sentiment scores. But before doing this, note that there are multiple articles on some days. Directly summing up the sentiment scores when there are multiple articles on the same day can lead to inflated values and spikes in the plot. To address this, we should normalize the sentiment scores by the number of articles on each day before plotting. By normalizing the sentiment scores, the plot should now show a more accurate representation of the overall sentiment trend without being affected by the number of articles on each day. The spikes should be reduced, and the plot will be more interpretable.

In the following code snippet we group articles by date and aggregate to get both the sum and count of articles for each date. Then, we divide the sum of each sentiment score by the count of articles for that date to get the average sentiment for the day. This normalizes the scores and prevents inflation due to varying numbers of articles. Then, we use the normalized sentiment scores for visualization.

```
[8]: # Group by date and calculate average sentiment scores
df_grouped = df.groupby('Date').agg(['sum', 'count']) # Get sum and count

# Create new columns for normalized sentiment scores
df_grouped['avg_positive'] = df_grouped['Sent_positive']['sum'] / \
    df_grouped['Sent_positive']['count'] # Normalize positive
df_grouped['avg_negative'] = df_grouped['Sent_negative']['sum'] / \
    df_grouped['Sent_negative']['count'] # Normalize negative
df_grouped['avg_neutral'] = df_grouped['Sent_neutral']['sum'] / \
    df_grouped['Sent_neutral']['count'] # Normalize neutral

# Convert df_grouped.index to DatetimeIndex
```

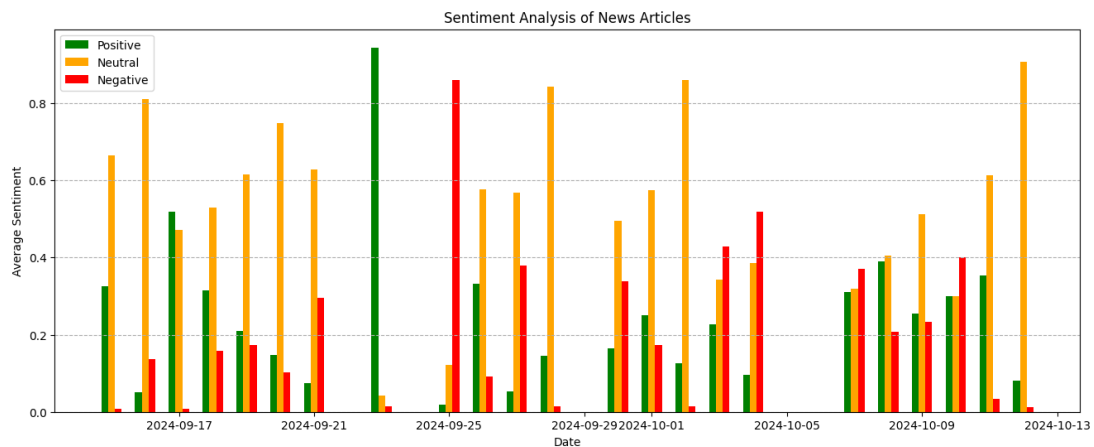
```

df_grouped.index = pd.to_datetime(df_grouped.index)

# Plot the sentiment data
fig, ax = plt.subplots(figsize=(16, 6))
width = 0.2
ax.bar(df_grouped.index - pd.DateOffset(days=width),
       df_grouped['avg_positive'], width=width, label='Positive', color='green')
ax.bar(df_grouped.index, df_grouped['avg_neutral'], width=width,
       label='Neutral', color='orange')
ax.bar(df_grouped.index + pd.DateOffset(days=width),
       df_grouped['avg_negative'], width=width, label='Negative', color='red')

# Set plot attributes (labels, ticks, title, legend, gridlines)
ax.set_xlabel('Date')
ax.set_ylabel('Average Sentiment')
ax.set_title('Sentiment Analysis of News Articles')
ax.legend()
ax.grid(True, axis='y', linestyle='--')
plt.show()

```



The resulting plot is a bar chart visualizing the average sentiment of news articles over time. There are three bars for each date, representing the three sentiment categories: positive (green), negative (red), and neutral (orange). The height of each bar indicates the magnitude of the average sentiment score for that category on the corresponding date. Taller bars indicate stronger sentiment.

The plot allows us to observe trends in sentiment over time. We can see how the average sentiment changes across dates, whether it's becoming more positive, negative, or staying neutral. By comparing the heights of the bars for each category, we can identify periods of stronger positive or negative sentiment. For example, if on a particular date, the red bar (negative) is much taller than the green bar (positive), it suggests that the news articles for that date were generally more negative in sentiment. Conversely, a taller green bar would indicate more positive sentiment.

3.3 Sentiment Analysis Scores in Conjunction with Stock Price

Let's consider the following example to see how we can use sentiment analysis scores in conjunction with Microsoft's stock price, keeping in mind the caveats about sentiment analysis accuracy and the need for other analysis tools.

In this example, we will visualize sentiment and price trends. We'll plot sentiment scores alongside historical stock prices. This can help to visually identify potential correlations between sentiment shifts and price movements.

```
[9]: # Fetch Microsoft stock data for the same date range
msft = yf.download("MSFT", start=df_grouped.index.min(), end=df_grouped.index.
    ↪max())

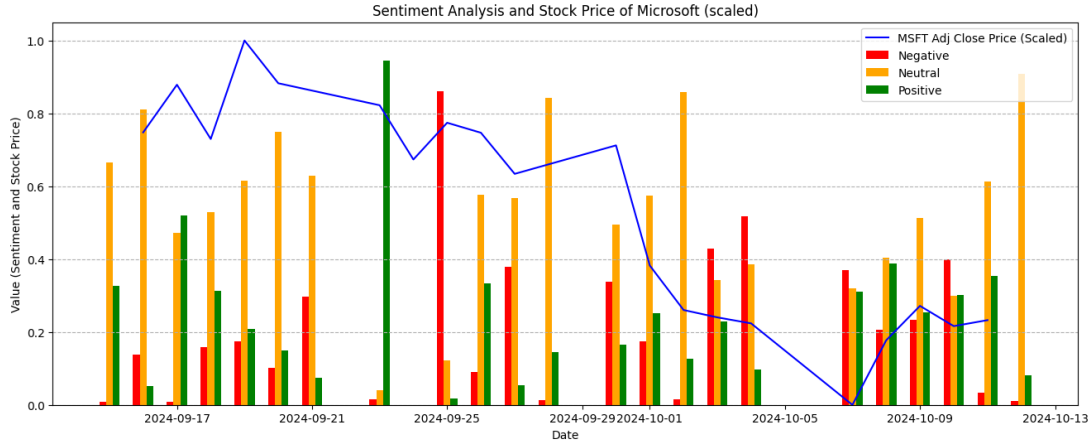
# Scale the Microsoft price data
scaler = MinMaxScaler()
scaled_stock_price = scaler.fit_transform(msft[['Close']])

# Plot scaled price plot
fig, ax = plt.subplots(figsize=(16, 6))
ax.plot(msft.index, scaled_stock_price, color='blue', label='MSFT Adj Close_
    ↪Price (Scaled)')

# Plot the sentiment data
width = 0.2 # Adjust the width as needed
ax.bar(df_grouped.index - pd.DateOffset(days=width),
    ↪df_grouped['avg_negative'], width=width, label='Negative', color='red')
ax.bar(df_grouped.index, df_grouped['avg_neutral'], width=width,
    ↪label='Neutral', color='orange')
ax.bar(df_grouped.index + pd.DateOffset(days=width),
    ↪df_grouped['avg_positive'], width=width, label='Positive', color='green')

# Set plot attributes (labels, ticks, title, legend, gridlines)
ax.set_xlabel('Date')
ax.set_title('Sentiment Analysis and Stock Price of Microsoft (scaled)')
ax.set_ylabel('Value (Sentiment and Stock Price)')
ax.legend()
ax.grid(True, axis='y', linestyle='--')
plt.show()
```

```
[*****100%*****] 1 of 1 completed
```



Now the plot also shows Microsoft’s stock price along with sentiment score bars. Please note that the stock price has been scaled, and we did this using `MinMaxScaler`. The `MinMaxScaler` from `sklearn.preprocessing` scales data to a specific range, typically between 0 and 1. It does this by applying the following formula:

$$X_{\text{scaled}} = \frac{(X - X_{\min})}{(X_{\max} - X_{\min})}$$

Where:

- X is the original data;
- X_{scaled} is the scaled data;
- $X_{\min}$ is the minimum value of each feature (column) in X ;
- $X_{\max}$ is the maximum value of each feature (column) in X ;

This means that the data point with the maximum value in the original data will be scaled to 1. The data point with the minimum value in the original data will be scaled to 0. And all other data points will be scaled proportionally between 0 and 1 based on their relative position within the original data range.

While the scaled stock price doesn’t show the actual price values, it still represents the stock price movement in a normalized way, allowing us to focus on the trends and patterns and compare them with other scaled features. When we plot the scaled stock price alongside the sentiment scores, we’re essentially comparing the trends and patterns of both features over time. We can observe whether positive sentiment tends to coincide with upward price movements, negative sentiment with downward movements, or if there are any other interesting relationships.

4 4. Scenario: Topic Modeling of Financial News

We will now use the News API data to discover underlying topics in financial news articles related to Microsoft and analyze sentiment associated with each topic.

Topic modeling is an unsupervised machine learning technique used to discover hidden thematic structures or topics within a collection of documents (also known as a **corpus**). It aims to automatically identify groups of words that frequently co-occur in documents, representing underlying themes or subjects. Topic modeling algorithms typically work by assuming that each document is a mixture of a small number of topics, and each topic is characterized by a distribution of words. The goal is to learn these topic distributions and the document-topic assignments.

Key terminology:

- **Document:** A single unit of text, such as an article, blog post, or tweet.
- **Corpus:** A collection of documents.
- **Topic:** A hidden thematic structure or theme within the corpus, represented by a distribution of words.
- **Document-Topic Distribution:** The probability or weight of each topic within each document.
- **Topic-Word Distribution:** The probability or weight of each word within each topic.

Non-negative matrix factorization (NMF) is a powerful technique for discovering hidden topics in text data. By applying it to financial news, we can uncover the main themes being discussed about Microsoft. Combining topic modeling with sentiment analysis provides a deeper understanding of the sentiment associated with different topics. This can help identify areas of positive or negative perception toward the company. The insights gained from this analysis can inform investment decisions, risk management, and public relations strategies.

Implementation steps:

- **Data Preparation:** We will use the existing `df` DataFrame containing News API data selecting the 'Description' column as the text data for analysis. We will preprocess the text data (remove stop words, punctuation, stemming/lemmatization).
- **Document-Term Matrix (TDM) Creation:** Then we will create a document-term matrix using TF-IDF (Term Frequency-Inverse Document Frequency). This matrix represents the frequency or importance of each word in each document.
- **NMF Application:** We apply NMF to the document-term matrix to decompose it into two matrices: W matrix representing the importance of each topic in each document and H matrix representing the importance of each word in each topic.
- **Topic Extraction and Interpretation:** Access the topic-word matrix (H) in order to examine the top words for each topic. Identify the words with the highest weights in each row of H to understand the theme of each topic. Based on the top words, we give meaningful names to the topics to make them interpretable.
- **Document-Topic Assignment:** Obtain the document-topic matrix (W) in order to assign dominant topics to documents. For each document, we find the topic with the highest weight in the corresponding row of W . This assigns the most prominent topic to each document.

4.1 4.1 Data Preparation

We start with data preparation. We preprocess the 'Description' column using the `preprocess()` function introduced earlier and save each processed description in a new column named 'Processed_Description':

```
[10]: # Preprocessing 'Description' column in df
df.loc[:, 'Processed_Description'] = df['Description'].apply(lambda x: preprocess(x))
```

Then, we are ready to apply the Term Frequency-Inverse Document Frequency (TF-IDF) technique.

4.2 4.2 Feature Extraction - Document-Term Matrix (TDM)

In the following code snippet, we fit the `TfidfVectorizer` vectorizer to our data in the 'Processed_Description' column. We apply the `TfidfVectorizer` vectorizer with the `max_features=250` parameter that limits the vocabulary size to the 250 most frequent words. This helps reduce the dimensionality of the document-term matrix (DTM) and can improve performance. We also use a `stop_words='english'` parameter that removes common English words (like "the," "a," "is") from the vocabulary, as they often don't carry much meaningful information. This will convert the text data in 'Processed_Description' into a document-term matrix (DTM) represented by the variable `dtm`:

```
[11]: # TF-IDF Vectorization
vectorizer = TfidfVectorizer(max_features=250, stop_words='english')
dtm = vectorizer.fit_transform(df['Processed_Description'])
dtm.shape
```

```
[11]: (89, 250)
```

The `dtm` variable now holds a sparse matrix where:

- Each row represents a document (a news article in our case).
- Each column represents a word (or feature) from the vocabulary.
- The values in the matrix represent the TF-IDF score of each word in each document.
- The `dtm` is typically stored as a sparse matrix to save memory because it often contains many zero values.

The typical range for `max_features` is usually between 1000 and 5000, but it can vary widely depending on the dataset and task depending on such factors as dataset size, task complexity, computation resources, experimentation, etc. For a dataset such as ours with only about 90 rows, a `max_features` value between 100 and 500 would likely be a good starting point for keeping balance between dimensionality and information.

4.3 4.3 NMF Application

After creating the `dtm`, we can now apply NMF to this matrix to discover underlying topics in the financial news articles. We are going to do this with `n_components=5` parameter that specifies the number of topics (or components) we want to extract from the data. In this case, we're aiming for five topics. `random_state=42` sets a random seed for reproducibility. Using the same seed ensures that we get the same results each time we run the code:

```
[12]: # NMF
nmf_model = NMF(n_components=5, random_state=42) # 5 topics
W = nmf_model.fit_transform(dtm)
H = nmf_model.components_
```

NMF is used to discover underlying topics in text data. It decomposes the document-term matrix into two lower-dimensional matrices that represent the relationships between words and topics and between topics and documents. Now we have obtained:

- W (Document-Topic Matrix) represents the distribution of topics within each document. Each row in W corresponds to a document in corpus, and each column corresponds to a topic discovered by NMF. The values in W indicate the weight or probability of each topic being present in each document.
- H (Topic-Word Matrix) represents the distribution of words within each topic. Each row in H corresponds to a topic, and each column corresponds to a word in vocabulary. The values in H indicate the weight or importance of each word within each topic.

The `n_components=5` parameter determines the number of topics (or components) we want to extract from data. In our case, we're aiming to discover five distinct topics within the financial news articles related to Microsoft. A smaller number of topics (like five) often leads to more interpretable results. It's easier to understand and assign meaningful labels to five topics compared to, say, 20 or 50 topics. When starting with topic modeling, it's often a good practice to begin with a smaller number of topics and then gradually increase it if needed. This makes it possible to get a general understanding of the main themes in the data before diving into more granular analysis. A smaller number of topics also reduces the computational cost of NMF, making the process faster, especially for larger datasets.

4.4 Topic Extraction and Interpretation

Let's now examine the top words in each topic to understand the theme of the topic. The goal is to understand the themes or subjects represented by each of the five topics extracted by NMF.

Recall that the H matrix stores the weight of each word in each topic. We want to identify the words with the highest weights for each topic, as these words are most representative of the topic's theme. For each topic, we'll display the top n words with the highest weights. This will give us a glimpse into the subject matter of the topic. Based on the top words, we'll try to assign a meaningful label or interpretation to each topic. This requires domain knowledge and careful consideration of the context of the financial news articles.

```
[13]: # Retrieves the names of the features (words)
feature_names = vectorizer.get_feature_names_out()

# Extract top words
n_top_words = 10 # Top 10 words
for topic_idx, topic in enumerate(H):
    print(f"\nTopic {topic_idx + 1}:")
    print(" ".join([feature_names[i]
                     for i in topic.argsort()[: -n_top_words - 1 : -1]]))
```

Topic 1:

copilot plus feature new pcs search features windows ai microsoft

Topic 2:

data energy centers power nuclear ai tech run microsoft carbon

Topic 3:

hololens microsoft windows headsets according update vr production uploadvr
macos

Topic 4:

ai openai google microsoft investments stay date way financial best

Topic 5:

flight game simulator 2024 microsoft test based office 2020 reboot

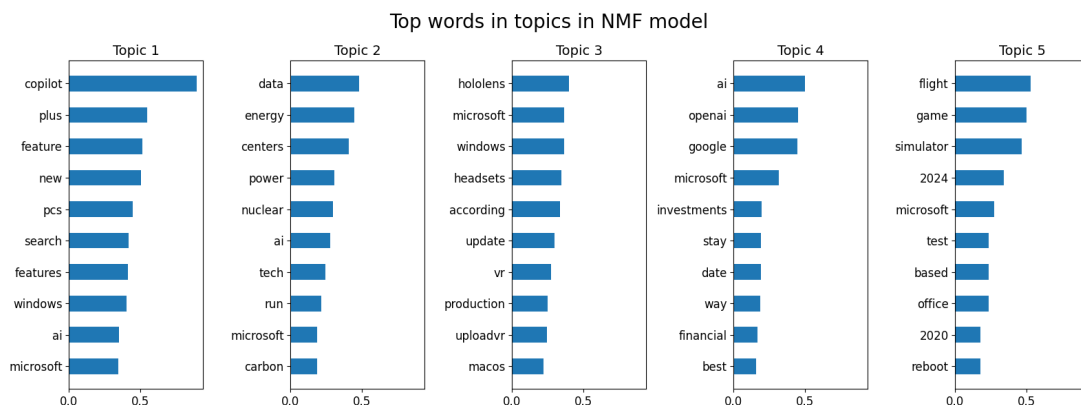
We can also plot top words for each topic for better visualization. In the following code snippet, we plot the top 10 words for each topic as bar subplots:

```
[14]: # Set figure size
fig, axes = plt.subplots(1, nmf_model.n_components, figsize=(16, 6),
    ↪sharex=True)
axes = axes.flatten() # Convert to 1D array for easier indexing

# Plot bar subplots for each topic
for topic_idx, topic in enumerate(H):
    top_features_ind = topic.argsort()[::-n_top_words - 1:-1]
    top_features = [feature_names[i] for i in top_features_ind]
    weights = topic[top_features_ind]

    ax = axes[topic_idx]
    ax.barh(top_features, weights, height=0.5, fill='blue')
    ax.set_title(f'Topic {topic_idx + 1}', fontdict={'fontsize': 14})
    ax.invert_yaxis()
    ax.tick_params(axis='both', which='major', labelsize=12)

# Set figure attributes (title)
fig.suptitle('Top words in topics in NMF model', fontsize=20, y=0.98)
fig.tight_layout(h_pad=2.0)
plt.show()
```



The code snippet's main purpose is to display the top words for each topic extracted by NMF. This helps in interpreting the topics and understanding the themes or subjects they represent.

As of this writing, get the following top words for each topic (you might get different results as News API will have another set of news articles by the time you run this code):

- **Topic 1: Microsoft Windows and AI Features**
 - Top Words: copilot, plus, feature, new, pcs, features, search, voice, ai, windows
 - Interpretation: This topic appears to be related to new features and enhancements in Microsoft Windows, particularly those involving AI and voice search capabilities. Copilot, a potential new AI assistant, is also a prominent theme.
- **Topic 2: Microsoft's Data Centers and Energy Initiatives**
 - Top Words: data, energy, centers, power, nuclear, ai, tech, run, microsoft, carbon
 - Interpretation: This topic seems to focus on Microsoft's data centers and their energy consumption. It might involve discussions about power sources, including nuclear energy, and initiatives to reduce carbon emissions. The use of AI technology in data centers is also a possible theme.
- **Topic 3: Microsoft's AR/VR Efforts (Hololens)**
 - Top Words: microsoft, windows, hololens, headsets, according, vr, update, production, uploadvr, 11
 - Interpretation: This topic likely revolves around Microsoft's augmented and virtual reality efforts, specifically focusing on the Hololens headset. It might involve updates on Hololens production, new features, or partnerships with VR-related platforms like UploadVR.
- **Topic 4: AI Competition (Microsoft, OpenAI, Google)**
 - Top Words: openai, google, ai, microsoft, stay, date, financial, way, competition, amid
 - Interpretation: This topic centers on the competition in the field of artificial intelligence, primarily involving Microsoft, OpenAI, and Google. It might discuss financial aspects, strategic partnerships, and the overall landscape of the AI race.
- **Topic 5: Microsoft Flight Simulator and Gaming**
 - Top Words: flight, game, simulator, microsoft, 2024, test, based, reboot, 2020, office
 - Interpretation: This topic seems to be related to Microsoft Flight Simulator, a popular flight simulation game. It might involve discussions about updates, new features, testing phases, or potential releases in the future. The inclusion of "office" is a bit unusual and might require further context to understand its relevance.

Based on these interpretations, here's a suggested set of labels for the topics:

1. Windows AI Enhancements

2. Data Center Sustainability
3. Hololens and AR/VR
4. AI Industry Competition
5. Flight Simulator and Gaming

These labels provide a concise and informative representation of the themes captured by each topic. Remember that topic interpretation can be subjective, so they can be adjusted based on your specific understanding of the data and context.

By carefully analyzing the top words and considering the broader context, we can gain valuable insights into the main themes being discussed in financial news articles related to Microsoft.

4.5 Document-Topic Assignment

Now, we'll use W to assign the most prominent topic to each document. We can do this by finding the topic (column) with the highest weight for each document (row) in W :

```
[15]: # Assigns the dominant topic for each document
df['Dominant_Topic'] = W.argmax(axis=1) + 1 # +1 to start topic numbering from 1
↳1
```

Here, $W.argmax(axis=1)$ finds the index of the maximum value (highest weight) in each row of W . This index corresponds to the dominant topic for that document. And $+ 1$ adds 1 to the topic index to start topic numbering from 1 instead of 0 (which is the default in Python indexing).

Bar charts can be used to visualize the distribution of topics within individual documents or across the entire corpus. The following bar plot demonstrates average topic distribution across all documents:

```
[16]: # Visualize average topic distribution across all documents
avg_topic_distribution = W.mean(axis=0)

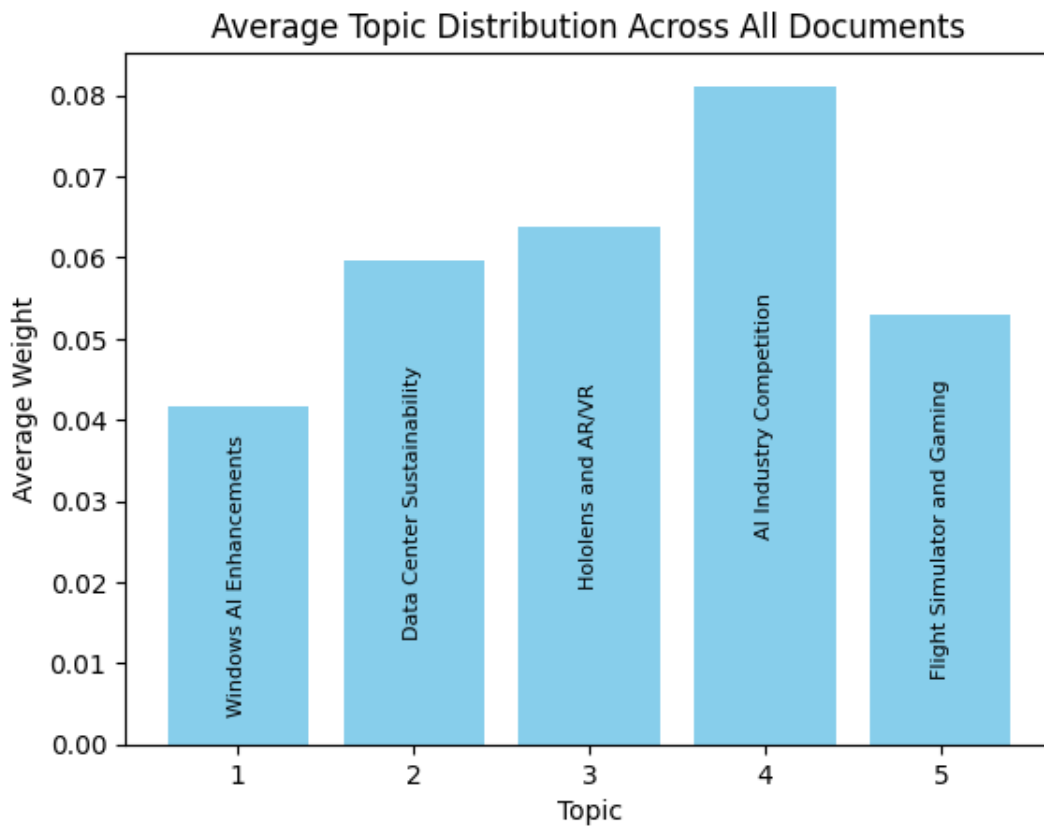
# Define topic labels
topic_labels = ["Windows AI Enhancements", "Data Center Sustainability",
               "Hololens and AR/VR", "AI Industry Competition",
               "Flight Simulator and Gaming"]

# Create bar plot
bars = plt.bar(np.arange(nmf_model.n_components) + 1, avg_topic_distribution,
↳color='skyblue')

# Add bar labels inside bars
for bar, label in zip(bars, topic_labels):
    plt.text(bar.get_x() + bar.get_width() / 2, bar.get_height() / 2, label,
             ha='center', va='center', rotation=90, color='black', fontsize=8)

plt.xticks(np.arange(nmf_model.n_components) + 1)
plt.xlabel("Topic")
plt.ylabel("Average Weight")
plt.title("Average Topic Distribution Across All Documents")
```

```
plt.show()
```



This bar plot illustrates the average prevalence of each topic across all documents in the dataset. The height of each bar represents the average weight or importance of that topic across all the documents analyzed. Higher bars indicate topics that are more prominent or discussed more frequently overall.

We can observe that “AI Industry Competition” is the most prevalent topic, followed by “Hololens and AR/VR,” based on the heights of their respective bars.

By examining this visualization, we can gain insights into the overall thematic distribution within the dataset. This information will guide further analysis, potentially focusing on documents related to “AI Industry Competition” and “Hololens and AR/VR” for a more in-depth understanding of these dominant themes.

4.6 5. Conclusion

In this lesson, we explored different news data sources. We compared and contrasted various sources like yfinance, Google News RSS, and News API. Ultimately, News API was selected for its flexibility and features.

Then, we performed sentiment analysis using FinBERT, a pre-trained NLP model for financial

text. Sentiment scores (positive, negative, and neutral) were calculated for news articles about Microsoft. The sentiment scores were plotted over time to observe trends and potential correlations with Microsoft's stock price.

Finally, we applied topic modeling, using NMF (non-negative matrix factorization) to discover underlying topics or themes within the news data. This involved preprocessing the text, creating a document-term matrix with TF-IDF, and applying the NMF algorithm.

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Required Readings

Required Readings for this program are open access, which means you should be able to access them at no cost. The link provided in the citation will take you directly to the reading or to a page where you can download it.

1. Bodas-Sagi, Diego, and José Labeaga. "Using GDELT Data to Evaluate the Confidence on the Spanish Government Energy Policy." *Big Data and AI*, special issue of *International Journal of Interactive Multimedia and Artificial Intelligence*, vol. 3, no. 6, 2016, pp. 38–43, <https://doi.org/10.9781/ijimai.2016.366>.

- **Required pages:** Entire article (pp. 38–43)
- **Estimated reading time:** 20 minutes

2. Consoli, Sergio, et al. "Information Extraction from the GDELT Database to Analyse EU Sovereign Bond Markets." *Mining Data for Financial Applications*, edited by Valerio Bitetta et al., Springer, 2021, pp. 55-67, https://doi.org/10.1007/978-3-030-66981-2_5.

- **Required pages:** Entire article (pp. 55–67)
- **Estimated reading time:** 35 minutes

3. Consoli, Sergio, et al. "Using the GDELT Dataset to Analyse the Italian Sovereign Bond Market." *Machine Learning, Optimization, and Data Science*, edited by Giuseppe Nicosia et al., Springer, 2020, pp. 190–202, https://doi.org/10.1007/978-3-030-64583-0_18.

- **Required pages:** Entire article (pp. 190–202)
- **Estimated reading time:** 40 minutes

Recommended Readings

- Elshendy, Mohammed, et al. "Using Four Different Online Media Sources to Forecast the Crude Oil Price." *Journal of Information Science*, vol. 44, no. 3, 2017, pp.408–421, <https://doi.org/10.1177/0165551517698298>.
- Alamro, Rawan, et al. "Predicting Saudi Stock Market Index by Incorporating GDELT Using Multivariate Time Series Modelling." *Advances in Data Science, Cyber Security and IT Applications*, edited by Auhood Alfaries et al., Springer, 2019, pp. 317–328, https://www.researchgate.net/publication/338097803_Predicting_Saudi_Stock_Market_Index_by_Incorporating_GDELT_Using_Multivariate_Time_Series_Modelling.

Lesson 3 - Large-Scale News Data Collections

December 25, 2025

MODULE 5 | LESSON 3

1 Large-Scale News Data Collections

Reading Time	60 minutes
Prior Knowledge	Fundamental Python Programming: Data Structures, Control Flow, Functions, File Handling, Regular Expressions, HTML Basics; Financial Engineering: A basic understanding of financial engineering concepts and applications for contextualizing the use of news data.
Keywords	WARC (Web ARChive), GDELT (Global Database of Events, Language, and Tone), GKG (Global Knowledge Graph), CSV, Warcio, Newspaper3k, Tqdm, Regular Expressions (Regex), Natural Language Processing (NLP), Large-Scale Data, Big Data

In this lesson, we learn how to leverage two large-scale news datasets, Common Crawl News Crawl and GDELT, for financial engineering applications. We learn how to access and process raw news articles from Common Crawl's WARC files, extracting key information like titles and content, and filtering by language and keywords. Additionally, we learn how to work with GDELT's GKG data to analyze news events, entities, and sentiment, including downloading, filtering, and analyzing GDELT's Tone scores for nuanced sentiment insights. We explore the code and techniques necessary to harness the power of these datasets for financial research and analysis.

```
[1]: !pip install pip --upgrade
      !pip install yfinance --upgrade
```

```
Requirement already satisfied: pip in /usr/local/lib/python3.11/site-packages
(22.3.1)
```

```
Collecting pip
```

```
  Downloading pip-25.3-py3-none-any.whl (1.8 MB)
```

```
1.8/1.8 MB
```

```

21.1 MB/s eta 0:00:0000:010:01
Installing collected packages: pip
  Attempting uninstall: pip
    Found existing installation: pip 22.3.1
    Uninstalling pip-22.3.1:
      Successfully uninstalled pip-22.3.1
Successfully installed pip-25.3
WARNING: Running pip as the 'root' user can result in broken permissions
and conflicting behaviour with the system package manager. It is recommended to
use a virtual environment instead: https://pip.pypa.io/warnings/venv

Requirement already satisfied: yfinance in /usr/local/lib/python3.11/site-
packages (0.2.51)
Collecting yfinance
  Downloading yfinance-1.0-py2.py3-none-any.whl.metadata (6.0 kB)
Requirement already satisfied: pandas>=1.3.0 in /usr/local/lib/python3.11/site-
packages (from yfinance) (2.0.0)
Requirement already satisfied: numpy>=1.16.5 in /usr/local/lib/python3.11/site-
packages (from yfinance) (1.26.2)
Requirement already satisfied: requests>=2.31 in /usr/local/lib/python3.11/site-
packages (from yfinance) (2.32.3)
Requirement already satisfied: multitasking>=0.0.7 in
/usr/local/lib/python3.11/site-packages (from yfinance) (0.0.11)
Requirement already satisfied: platformdirs>=2.0.0 in
/usr/local/lib/python3.11/site-packages (from yfinance) (3.4.0)
Requirement already satisfied: pytz>=2022.5 in /usr/local/lib/python3.11/site-
packages (from yfinance) (2023.3.post1)
Requirement already satisfied: frozendict>=2.3.4 in
/usr/local/lib/python3.11/site-packages (from yfinance) (2.4.6)
Requirement already satisfied: peewee>=3.16.2 in /usr/local/lib/python3.11/site-
packages (from yfinance) (3.17.8)
Requirement already satisfied: beautifulsoup4>=4.11.1 in
/usr/local/lib/python3.11/site-packages (from yfinance) (4.12.2)
Collecting curl_cffi<0.14,>=0.7 (from yfinance)
  Downloading curl_cffi-0.13.0-cp39-abi3-
manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (13 kB)
Collecting protobuf>=3.19.0 (from yfinance)
  Downloading protobuf-6.33.2-cp39-abi3-manylinux2014_x86_64.whl.metadata (593
bytes)
Collecting websockets>=13.0 (from yfinance)
  Downloading websockets-15.0.1-cp311-cp311-
manylinux_2_5_x86_64.manylinux1_x86_64.manylinux_2_17_x86_64.manylinux2014_x86_6
4.whl.metadata (6.8 kB)
Requirement already satisfied: cffi>=1.12.0 in /usr/local/lib/python3.11/site-
packages (from curl_cffi<0.14,>=0.7->yfinance) (1.15.1)
Collecting certifi>=2024.2.2 (from curl_cffi<0.14,>=0.7->yfinance)
  Downloading certifi-2025.11.12-py3-none-any.whl.metadata (2.5 kB)

```

Requirement already satisfied: soupsieve>1.2 in /usr/local/lib/python3.11/site-packages (from beautifulsoup4>=4.11.1->yfinance) (2.4.1)
 Requirement already satisfied: pycparser in /usr/local/lib/python3.11/site-packages (from cffi>=1.12.0->curl_cffi<0.14,>=0.7->yfinance) (2.21)
 Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.11/site-packages (from pandas>=1.3.0->yfinance) (2.8.2)
 Requirement already satisfied: tzdata>=2022.1 in /usr/local/lib/python3.11/site-packages (from pandas>=1.3.0->yfinance) (2023.3)
 Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.11/site-packages (from python-dateutil>=2.8.2->pandas>=1.3.0->yfinance) (1.16.0)
 Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.11/site-packages (from requests>=2.31->yfinance) (3.1.0)
 Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.11/site-packages (from requests>=2.31->yfinance) (3.4)
 Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.11/site-packages (from requests>=2.31->yfinance) (1.26.15)
 Downloading yfinance-1.0-py2.py3-none-any.whl (127 kB)
 Downloading
 curl_cffi-0.13.0-cp39-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (8.3 MB)

8.3/8.3 MB

88.2 MB/s 0:00:00

Downloading certifi-2025.11.12-py3-none-any.whl (159 kB)
 Downloading protobuf-6.33.2-cp39-abi3-manylinux2014_x86_64.whl (323 kB)
 Downloading websockets-15.0.1-cp311-cp311-manylinux_2_5_x86_64.manylinux1_x86_64.manylinux_2_17_x86_64.manylinux2014_x86_64.whl (182 kB)

Installing collected packages: websockets, protobuf, certifi, curl_cffi, yfinance

Attempting uninstall: certifi[90m

1/5 [protobuf]

Found existing installation: certifi 2022.12.7

1/5 [protobuf]

Uninstalling certifi-2022.12.7:

1/5 [protobuf]

Successfully uninstalled certifi-2022.12.7

1/5 [protobuf]

Attempting uninstall: yfinance

3/5 [curl_cffi]

Found existing installation: yfinance 0.2.51

3/5 [curl_cffi]

Uninstalling yfinance-0.2.51:[90m

3/5 [curl_cffi]

Successfully uninstalled yfinance-0.2.51m

3/5 [curl_cffi]

5/5

[yfinance]4/5 [yfinance]

Successfully installed certifi-2025.11.12 curl_cffi-0.13.0

protobuf-6.33.2 websockets-15.0.1 yfinance-1.0

WARNING: Running pip as the 'root' user can result in broken permissions

and conflicting behaviour with the system package manager, possibly rendering your system unusable. It is recommended to use a virtual environment instead:

<https://pip.pypa.io/warnings/venv>. Use the --root-user-action option if you know what you are doing and want to suppress this warning.

```
[2]: # Loading libraries
import gc
import gzip
import io
import nltk
import pandas as pd
import plotly.graph_objects as go
import re
import requests
import time
import tqdm
import warcio
import yfinance as yf
import zipfile

from datetime import datetime, timedelta
from newspaper import Article
from nltk.corpus import stopwords
from sklearn.decomposition import NMF
from sklearn.feature_extraction.text import TfidfVectorizer
```

Large-scale data collection refers to the process of gathering and storing massive amounts of data from various sources. In this lesson, we consider two: Common Crawl and GDELT. Both initiatives involve large-scale data collection from a vast number of online news sources. They utilize web crawling techniques and other methods to gather news articles from diverse websites and publications worldwide. This extensive data collection allows for comprehensive analysis and insights into global news coverage. While there are challenges associated with large-scale data collection, the benefits in terms of comprehensive analysis, improved accuracy, and the discovery of new knowledge make it a valuable approach for news analysis and research, including applications in financial engineering.

2 1. Common Crawl

Common Crawl is a non-profit organization that crawls the web and provides an open-source dataset of web pages. Common Crawl uses web crawlers to collect web pages and store them in a publicly accessible repository called the Common Crawl corpus.

They also have a separate dataset called News Crawl, which uses RSS feeds and sitemaps to collect news articles specifically. This dataset has around 600,000 news articles a day in multiple languages, going back a few years, and includes the full HTML of the pages.

Common Crawl use different techniques like following links, using sitemaps, and respecting robots.txt rules to find new pages and update existing ones. The data is stored in a format called WARC (Web ARChive) files. These files contain not only the HTML content of web pages but also metadata like timestamps, URLs, and content type. This data can be accessed and analyzed by anyone using various tools and techniques.

The Common Crawl corpus is estimated to be over 250 TiB of raw data, compressed. This data includes petabytes of text data, images, and metadata. Based on its massive volume, high velocity, variety of formats, and potential value, the Common Crawl data is unequivocally considered big data. It requires specialized tools and infrastructure, such as distributed computing frameworks and cloud-based storage, to handle the data effectively and extract meaningful insights. While challenges related to veracity and data quality need to be addressed, the scale and scope of the Common Crawl dataset firmly place it within the realm of big data for news and information analysis.

Interestingly, Common Crawl is a crucial resource for building large language models. This allows the models to benefit from the vast and diverse information available on the web, improving their ability to generate human-like text and understand a wide range of topics. The Common Crawl corpus, or a filtered version of it, played a role in training GPT models. The GPT-3, or third-generation Generative Pre-trained Transformer, a neural network machine learning model developed by OpenAI, was trained on a mixture of filtered Common Crawl (60% of GPT-3's data) and other corpuses such as WEBTEXT2, BOOKS1 and BOOKS2, and English-language Wikipedia (Brown et al.).

2.1 1.1 Getting News Crawl Data

It is essential to understand the structure of News Crawl data and how to access it. Common Crawl data files are primarily stored in the **WARC (Web ARChive) format**. WARC is a widely used standard for archiving web content, and it's well suited for storing large amounts of data, including web pages, metadata, and associated assets. These files contain the raw crawled web pages, along with metadata such as the URL, timestamp, content type, and HTTP headers. They are the primary source of data in Common Crawl.

The Common Crawl also provides its main datasets in **WAT files (Web Annotation Text)**, where it stores important metadata about the records providing additional context and information, and in **WET files (Web Extracted Text)** with the extracted plain text content of the web pages, making it easier to search and analyze the text without parsing HTML. You can see examples of each file format on Common Crawl's website here: <https://commoncrawl.org/get-started>.

Unlike Common Crawl's main datasets, News Crawl datasets are provided only in WARC format. Common Crawl's News datasets are organized into monthly segments, **WARC File Path Listings**, and the data within each segment is stored in WARC files. These files are further grouped into daily crawls. WARC File Path Listings are available on AWS (Amazon Web Services) and can be accessed via the following pattern `s3://commoncrawl/CC-NEWS/yyyy/mm/warc.paths.gz` (access to AWS cloud can be subject to an inter-region data transfer fee depending on your location). Alternatively, data is also accessible via the URL scheme at

<https://data.commoncrawl.org/CC-NEWS/yyyy/mm/warc.paths.gz>. We just need to replace yyyy and mm with the numerical year and month values, respectively.

Inside each WARC paths listing, there is a list of all crawls that were saved in this segment's WARC files during the month. The WARC files are released on a daily basis, and WARC file names follow the pattern: `crawl-data/CC-NEWS/yyyy/mm/CC-NEWS-yyyymmddHHMMSS-nnnnn.warc.gz`. The timestamp (yyyymmddHHMMSS) indicates the time the first record in the WARC file was created with:

yyyy	year
mm	month (01..12)
dd	day of month (01, etc.)
HH	hour (00..23)
MM	minute (00..59)
SS	second (00..59)
nnnnn	serial WARC file number. The serial number is reset when the crawl process is resumed.

Now let's try to get News Crawl data using the Common Crawl's WARC File Path Listings. The following code snippet aims to download and display the list of WARC file paths for the Common Crawl News dataset for September 2024. This code downloads a compressed file containing a list of WARC file paths for the Common Crawl News dataset, decompresses it, and then prints each path to the console:

```
[3]: # Get September 2024 WARC paths list
url = "https://data.commoncrawl.org/crawl-data/CC-NEWS/2024/09/warc.paths.gz"

# Fetch the file
response = requests.get(url, stream=True)
response.raise_for_status() # Raise an exception for bad responses (4xx or 5xx)

# Decompress and print content
with gzip.GzipFile(fileobj=io.BytesIO(response.content)) as gz:
    for line in gz:
        print(line.decode('utf-8').strip()) # Print each line, removing leading/
        ↪trailing spaces
```

```
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240901011006-05863.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240901040313-05864.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240901061815-05865.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240901082058-05866.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240901100849-05867.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240901115147-05868.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240901132402-05869.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240901145127-05870.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240901161309-05871.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240901174029-05872.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240901191613-05873.warc.gz
```

crawl-data/CC-NEWS/2024/09/CC-NEWS-20240901210159-05874.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240901230057-05875.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240902013240-05876.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240902035909-05877.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240902060627-05878.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240902073934-05879.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240902090600-05880.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240902102119-05881.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240902113551-05882.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240902124845-05883.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240902140154-05884.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240902151007-05885.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240902161834-05886.warc.gz
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crawl-data/CC-NEWS/2024/09/CC-NEWS-20240902184143-05888.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240902200726-05889.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240902214146-05890.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240902231517-05891.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240903010221-05892.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240903031847-05893.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240903052128-05894.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240903065755-05895.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240903082335-05896.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240903093529-05897.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240903104052-05898.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240903114456-05899.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240903125242-05900.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240903135605-05901.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240903145856-05902.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240903155837-05903.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240903170339-05904.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240903180601-05905.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240903191349-05906.warc.gz
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crawl-data/CC-NEWS/2024/09/CC-NEWS-20240904032710-05911.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240904051857-05912.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240904065038-05913.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240904081044-05914.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240904092642-05915.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240904103523-05916.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240904113931-05917.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240904123755-05918.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240904133441-05919.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240904143625-05920.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240904153623-05921.warc.gz

crawl-data/CC-NEWS/2024/09/CC-NEWS-20240904163717-05922.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240904174006-05923.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240904184511-05924.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240904195249-05925.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240904210940-05926.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240904223752-05927.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240905002041-05928.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240905022237-05929.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240905042413-05930.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240905060522-05931.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240905073724-05932.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240905085710-05933.warc.gz
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crawl-data/CC-NEWS/2024/09/CC-NEWS-20240905112031-05935.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240905122602-05936.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240905133109-05937.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240905143153-05938.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240905153656-05939.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240905164601-05940.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240905175318-05941.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240905190722-05942.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240905202310-05943.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240905214353-05944.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240905231222-05945.warc.gz
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crawl-data/CC-NEWS/2024/09/CC-NEWS-20240919232308-06171.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240920011440-06172.warc.gz
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crawl-data/CC-NEWS/2024/09/CC-NEWS-20240920054029-06174.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240920072903-06175.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240920085438-06176.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240920101150-06177.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240920112400-06178.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240920122830-06179.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240920133425-06180.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240920144308-06181.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240920154658-06182.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240920165443-06183.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240920181129-06184.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240920193432-06185.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240920210356-06186.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240920224850-06187.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240921010627-06188.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240921034714-06189.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240921055059-06190.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240921075106-06191.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240921094007-06192.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240921111859-06193.warc.gz
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crawl-data/CC-NEWS/2024/09/CC-NEWS-20240921142152-06195.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240921155205-06196.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240921173232-06197.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240921192053-06198.warc.gz
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crawl-data/CC-NEWS/2024/09/CC-NEWS-20240922102909-06205.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240922121527-06206.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240922135525-06207.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240922152907-06208.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240922170439-06209.warc.gz

crawl-data/CC-NEWS/2024/09/CC-NEWS-20240922184410-06210.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240922202709-06211.warc.gz
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crawl-data/CC-NEWS/2024/09/CC-NEWS-20240923035139-06214.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240923060350-06215.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240923074837-06216.warc.gz
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crawl-data/CC-NEWS/2024/09/CC-NEWS-20240923102329-06218.warc.gz
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crawl-data/CC-NEWS/2024/09/CC-NEWS-20240923123101-06220.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240923133940-06221.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240923143733-06222.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240923154058-06223.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240923164018-06224.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240923174201-06225.warc.gz
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crawl-data/CC-NEWS/2024/09/CC-NEWS-20240923202615-06227.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240923220036-06228.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240923235105-06229.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240924015653-06230.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240924041710-06231.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240924055946-06232.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240924073735-06233.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240924090019-06234.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240924101831-06235.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240924112743-06236.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240924123324-06237.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240924134506-06238.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240924144607-06239.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240924154631-06240.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240924164911-06241.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240924180107-06242.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240924191818-06243.warc.gz
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crawl-data/CC-NEWS/2024/09/CC-NEWS-20240924222818-06245.warc.gz
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crawl-data/CC-NEWS/2024/09/CC-NEWS-20240925020605-06247.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240925041112-06248.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240925054624-06249.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240925071505-06250.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240925083918-06251.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240925095325-06252.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240925110445-06253.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240925121530-06254.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240925132202-06255.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240925142612-06256.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240925153828-06257.warc.gz

crawl-data/CC-NEWS/2024/09/CC-NEWS-20240925164632-06258.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240925175152-06259.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240925190432-06260.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240925203004-06261.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240925220746-06262.warc.gz
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crawl-data/CC-NEWS/2024/09/CC-NEWS-20240926021249-06264.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240926042219-06265.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240926060949-06266.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240926074542-06267.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240926091134-06268.warc.gz
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crawl-data/CC-NEWS/2024/09/CC-NEWS-20240926113551-06270.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240926124341-06271.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240926135053-06272.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240926144744-06273.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240926155114-06274.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240926164717-06275.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240926174829-06276.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240926190018-06277.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240926201130-06278.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240926212823-06279.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240926225648-06280.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240927004505-06281.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240927024051-06282.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240927044057-06283.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240927063703-06284.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240927081602-06285.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240927093514-06286.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240927105445-06287.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240927120902-06288.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240927132212-06289.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240927143355-06290.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240927154732-06291.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240927165808-06292.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240927181445-06293.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240927193612-06294.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240927211106-06295.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240927225219-06296.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240928011132-06297.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240928040334-06298.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240928061306-06299.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240928081118-06300.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240928100319-06301.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240928115144-06302.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240928133641-06303.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240928152459-06304.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240928170532-06305.warc.gz

```

crawl-data/CC-NEWS/2024/09/CC-NEWS-20240928184547-06306.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240928203900-06307.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240928225518-06308.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240929011801-06309.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240929035219-06310.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240929061044-06311.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240929082310-06312.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240929102111-06313.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240929120854-06314.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240929135520-06315.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240929154117-06316.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240929171839-06317.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240929190454-06318.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240929204551-06319.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240929225111-06320.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240930012849-06321.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240930041208-06322.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240930061854-06323.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240930074933-06324.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240930091334-06325.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240930103308-06326.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240930114624-06327.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240930125257-06328.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240930135959-06329.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240930151334-06330.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240930162313-06331.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240930173229-06332.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240930183901-06333.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240930195256-06334.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240930212123-06335.warc.gz
crawl-data/CC-NEWS/2024/09/CC-NEWS-20240930230449-06336.warc.gz

```

Each line represents the path to a single WARC file. These files are compressed archives containing web pages and their associated metadata, specifically news articles in this case. These paths are essential for accessing the actual news data within the Common Crawl dataset. We would typically use these paths in conjunction with tools or libraries that can read and process WARC files.

Let's now determine the size of a specific WARC file within the Common Crawl News dataset. Let's choose the `crawl-data/CC-NEWS/2024/09/CC-NEWS-20240923074837-06216.warc.gz` file corresponding to data crawled on 23 September 2024 at about 7:48 AM:

```

[4]: # Specify WARC file path
warc_file_url = "https://data.commoncrawl.org/crawl-data/CC-NEWS/2024/09/
↳CC-NEWS-20240923074837-06216.warc.gz"

# Use HEAD request to get headers only
response = requests.get(warc_file_url, stream=True)
response.raise_for_status()

```

```
# Get file size from headers and print
file_size = int(response.headers.get('content-length', 0))
print(f"File size: {file_size} bytes")
print(f"File size: {file_size / (1024 * 1024):.2f} MB") # Convert to MB
```

File size: 1072759398 bytes

File size: 1023.06 MB

This code efficiently retrieves the size of a specific WARC file from the Common Crawl News dataset without downloading the entire file. As we see this specific WARC file has a size of approximately 1.02 GB (1023.06 MB). By understanding the size of the WARC files, we can better plan data acquisition and processing workflows. In our case, we should note that a 1 GB file can take a significant amount of time to download, especially on slower internet connections.

2.2 1.2 Processing WARC file records

For large WARC files like this, we can process them in smaller chunks to avoid memory issues. The `requests.get(..., stream=True)` line in the above code snippet initiates a streaming download, meaning it doesn't load the entire WARC file into memory at once. Instead, it downloads chunks of data as needed. This helps to minimize memory usage, especially for large WARC files. The `requests.get()` function returns a `Response` object, which is assigned to the variable `response`. This object contains information about the server's response, including the status code, headers, and the content of the response (in this case, the WARC file data). However, it is important to understand that the `response` object does not hold an entire WARC file. Think of `response` as a container with multiple compartments. One compartment holds information about the response (headers, status code), and another compartment is a pipeline connected to the server where the WARC file data is flowing through in chunks. As an analogy think of it like a water pipe connected to a reservoir. The pipe (the `response` object) exists, but the entire reservoir (the WARC file data) is not flowing through the pipe at once. Instead, water flows through in controlled amounts as needed.

Later, we can use this streaming feature efficiently to keep the memory footprint relatively low. Luckily for us, libraries like `warcio`, a Python library for reading and writing WARC files, allow us to iterate through records in a WARC file without loading the entire file into memory. When we access the content using `response.raw`, we're not getting a complete file in memory; we're accessing the pipeline through which the data is streaming. `warcio.ArchiveIterator` is designed to work with this pipeline. It reads chunks of data as they come through the pipeline and processes them one by one. For example, the following code snippet iterates over all records and creates a `DataFrame` with URL, date, and content length for each record in our chosen WARC file. It focuses only on "response" records, which typically contain the actual web page content:

```
[5]: # Function to process the WARC file and extract data
def process_warc_file(warc_file_url):
    data = []
    with requests.get(warc_file_url, stream=True) as response:
        response.raise_for_status()

    # Process WARC records
```

```

for record in warcio.ArchiveIterator(response.raw):
    if record.rec_type == 'response':

        # Extract URL, date and content length
        url = record.rec_headers.get_header('WARC-Target-URI')
        date = record.rec_headers.get_header('WARC-Date')
        content_length = record.rec_headers.get_header('Content-Length')

        # Store extracted info in 'data' list
        data.append([url, date, content_length])

# Create DataFrame
df = pd.DataFrame(data, columns=['URL', 'Date', 'Content-Length'])
return df

# Process the WARC file and get the DataFrame
df = process_warc_file(warc_file_url)
df

```

```

[5]:
                                URL
0      https://www.indiatvnews.com/sports/cricket/roh... \
1      https://ilsaronno.it/2024/09/23/la-maternita-u...
2      https://www.dailymail.co.uk/wires/reuters/arti...
3      https://aspicts.substack.com/p/tech-giants-pus...
4      https://topick.hket.com/article/3831279/%E5%B0...
...
25493  https://webcatalog.io/pt/apps/tag/api-generati...
25494  https://www.parkiet.com/technologie/art4117331...
25495  https://www.sussexexpress.co.uk/sport/boxing/c...
25496  https://www.etvbharat.com/mr/!health-and-lifes...
25497  https://www.bta.bg/bg/news/world/747292-kabine...

                                Date Content-Length
0      2024-09-23T07:48:37Z          218429
1      2024-09-23T07:48:37Z          271976
2      2024-09-23T07:48:37Z          527487
3      2024-09-23T07:48:37Z          214833
4      2024-09-23T07:48:37Z          137772
...
25493  2024-09-23T09:13:39Z          128452
25494  2024-09-23T09:13:39Z          506580
25495  2024-09-23T09:13:39Z          269726
25496  2024-09-23T09:13:40Z          360289
25497  2024-09-23T09:13:39Z          374772

```

[25498 rows x 3 columns]

Here, we see that there are 25,498 news records in this chosen WARC file that was crawled at about

7:48AM on September 23, 2024. This amount is large, but the code took less than a minute to iterate over all records. Processing of the above code was also relatively fast because all the record attributes (URL, crawling date, and content length) are saved in record headers.

However, we do not yet have any information about the contents of the news articles. To work with the news article content, we need to access the HTML content from the body of the news record and then use an HTML parsing library to extract specific elements from the HTML, such as the article title, body text, author, and/or publication date. Processing HTML, especially large HTML files or a large number of HTML files, can be time-consuming depending on the complexity of the HTML and the operations being performed. There are advanced techniques like parallel processing or distributed computing to further optimize the processing time. However, for simplicity, we'll just limit the number of articles to process to just the first couple thousand records. This should suffice to understand how extraction of news record attributes from News Crawl WARC files is done.

Let's now add more functionality to the `process_warc_file()` function in the above code snippet. This time, we will implement the reusable function that is designed to extract news articles related to Netflix from a WARC file. This function handles language filtering, keyword filtering, data extraction, error handling, and data organization, providing a streamlined way to process WARC files for specific information retrieval:

```
[6]: # Helper function to process WARC file
def process_warc_file(warc_file_url, limit=1000):
    data = []
    count = 0

    with requests.get(warc_file_url, stream=True) as response:
        response.raise_for_status()

        # Wrap the iterator with tqdm to process records with a progress bar up
        # to the limit
        for record in tqdm.tqdm(warcio.ArchiveIterator(response.raw),
                                total=limit, desc="Processing records"):
            if record.rec_type == 'response':

                # Proceed with data extraction and filtering
                url = record.rec_headers.get_header('WARC-Target-URI')
                date = record.rec_headers.get_header('WARC-Date')
                content_length = record.rec_headers.get_header('Content-Length')

                try:
                    html_content = record.content_stream().read().
                    decode('utf-8', 'ignore')

                    # Check for lang="en" before <head> (handling line breaks)
                    if re.search(r'lang\s*=\s*["']?en["']?\s*[s\S]*?<head>',
                                html_content, re.IGNORECASE):

                        # Extract title and article content using newspaper3k
```

```

        article = Article(url, language='en')
        article.download(input_html=html_content)
        article.parse()
        title = article.title
        news_article = article.text

        # Filter news article texts containing "netflix"
↪ (case-insensitive)
        if news_article and re.search(r'netflix', news_article,
↪ re.IGNORECASE):
            data.append([url, date, content_length, title,
↪ news_article])

        # Error handling
        except UnicodeDecodeError as e:
            print(f"Error decoding HTML content from {url}: {e}")
        except Exception as e:
            print(f"Error extracting article from {url}: {e}")

        # Increment count and check limit after processing each record
        count += 1
        if count > limit:
            break # Exit the loop if the limit is reached

        # Create DataFrame
        df = pd.DataFrame(data, columns=['URL', 'Date', 'Content-Length', 'Title',
↪ 'News_Article'])
        return df

```

Here we use `lang\s*=\s*["']?en["']?[\s\S]*?<head>` regex (regular expression). This regex is looking for the `lang` attribute in the HTML content, specifically checking if it's set to "en", and then it captures everything from that point up to the `<head>` tag, even if there are line breaks in between. This expression allows for flexibility in how the attribute is written, e.g. `lang="en"`, `lang = 'en'`, `lang=en`.

Then, we proceed with extracting news article details. The code does this with `newspaper3k` library to extract the title and main body text of news article from its HTML content. `newspaper3k` is a Python library used for extracting and analyzing articles from news websites and blogs. It is designed to make web scraping of news articles easier and more efficient. Then, the code filters news article texts containing "netflix" and appends extracted information to a `data` list, which is used to create the final `df` pandas DataFrame returned from function.

Please also note that, this time around, `warcio.ArchiveIterator` is wrapped within `tqdm.tqdm(...)` to create a progress bar that updates as the loop iterates through the WARC records. The progress bar will indicate the current record number, percentage of completion, and estimated remaining time, giving you a clear visual representation of the processing progress.

Now that we have a more functional `process_warc_file()` function, let's call it:

```
[7]: # Record the start time
start_time = time.time()

# Process the first 5000 English records with "netflix" in the title
df = process_warc_file(warc_file_url, limit=5000)

# Calculate and print total processing time
end_time = time.time()
processing_time = end_time - start_time
print(f"\nTotal processing time: {processing_time:.2f} seconds")

# Display DataFrame
df
```

```
Processing records: 82%|          | 4086/5000 [00:45<00:07,
127.38it/s]/usr/local/lib/python3.11/site-
packages/dateutil/parser/_parser.py:1207: UnknownTimezoneWarning: tzname PST
identified but not understood. Pass `tzinfos` argument in order to correctly
return a timezone-aware datetime. In a future version, this will raise an
exception.
  warnings.warn("tzname {tzname} identified but not understood. "
Processing records: 100%|          | 5000/5000 [00:53<00:00, 93.68it/s]
```

Total processing time: 53.42 seconds

```
[7]:
```

	URL	Date
0	https://www.indiatvnews.com/technology/news/ip...	2024-09-23T07:48:43Z \
1	https://www.express.co.uk/news/royal/1951925/p...	2024-09-23T07:53:11Z

	Content-Length	Title
0	225423	iPhone, Mac and iPad must be updated soon, as ... \
1	201548	Prince Harry and Meghan Markle 'feeling wrath ...

	News_Article
0	Follow us on Image Source : FILE iPhone, Mac a...
1	The pair were absent from a post-Emmy party th...

This was just a simplified example to demonstrate how to get a news dataset from a News Crawl WARC file. As a result, we now have filtered news data in `df` DataFrame, which in our case has only a handful of articles concerning Netflix. We can use this DataFrame in a similar manner as we could with other news datasets obtained by other means to process it further.

3 2. GDELT (Global Database of Events, Language, and Tone)

3.1 2.1 Introduction to GDELT

GDELT (Global Database of Events, Language, and Tone) is another massive, open-source database that monitors the world’s news media in over 65 languages. It uses sophisticated techniques to identify and categorize events, analyze the tone and sentiment of news coverage, and provide insights into global trends and patterns. GDELT’s data is freely available to researchers, journalists, and the public, enabling a deeper understanding of human society and global events.

GDELT data is available in various formats including CSV, JSON, and a specialized format called GKG (Global Knowledge Graph). The database is updated in real time or near real time, with new data being added continuously. The GDELT Project claims to have “the largest, most comprehensive, and highest resolution open database of human society ever created.” They process 100 million news articles and social media posts daily with about 65 languages represented in their dataset. Data is available from 1979 to present.

While not all aspects of big data might be equally pronounced in GDELT, the combination of its volume, velocity, variety, and potential value places it firmly within the realm of big data. The sheer scale and complexity of GDELT data necessitate specialized tools and techniques for efficient processing, analysis, and storage. This often involves using cloud-based platforms, distributed computing frameworks, and in-memory processing methods to handle the data effectively. However, it’s worth noting that the “big data” definition can be subjective and context dependent. While GDELT undoubtedly presents challenges associated with big data, the dataset might be considered relatively small compared to datasets from social media platforms or web-scale applications. Nonetheless, the volume, velocity, and variety of GDELT data are significant enough to warrant considering it as big data in the context of global event and news analysis.

3.2 2.2 GDELT vs. News Crawl

Both GDELT and News Crawl are open-source datasets focused on news analysis. Both provide a large amount of news data from various sources. Both allow for analysis of news trends and patterns. Both are publicly accessible and can be used for research and analysis. But they have some key differences:

Feature	News Crawl	GDELT
Data Source	Primarily news articles collected through RSS feeds and sitemaps.	News articles, social media posts, and other online media.
Data Format	Raw HTML of web pages (WARC files)	CSV, JSON, GKG (Global Knowledge Graph)
Focus	Provides the full content of news articles for analysis.	Extracts events, entities, and sentiment from news data.
Analysis Supported	Text mining, natural language processing (NLP), topic modeling.	Event detection, sentiment analysis, geospatial analysis, and network analysis.
Update Frequency	Updated multiple times per day.	Updated in real-time or near real-time (every 15 minutes for GKG).

Feature	News Crawl	GDELT
Data Size	Large (around 600,000 articles per day).	Massive (processes 100 million articles and posts daily).
Timeframe	Data available from the past few years.	Data available from 1979 to present.
Languages	Multiple languages supported.	Over 65 languages supported.
Strengths	Access to full article content, flexible for custom analysis.	Pre-processed data, efficient for specific analysis tasks, comprehensive analysis of events and sentiment.
Weaknesses	Requires more preprocessing, less structured data.	Limited access to raw article content, potential biases in data collection.

Key differences between News Crawl and GDELT data can be summarized as:

- **Data Format:** News Crawl primarily provides the full HTML content of news articles. This means you get the complete webpage as it appears online. GDELT provides data in various formats like CSV, JSON, and GKG. The choice of format depends on specific needs and the tools to use for analysis. GDELT's multiple formats provide flexibility for different types of analysis, while News Crawl's raw HTML requires more processing but gives access to the complete web page content.
- **Analysis:** GDELT provides analysis of events, tone, and sentiment. News Crawl provides the raw data for you to perform your own analysis.
- **Update Frequency:** GDELT is updated in real time or near real time. News Crawl updates less frequently, just multiple times a day.

GDELT datasets are publicly available and primarily stored on Google Cloud Storage. The specific location and access methods might vary depending on the dataset and format. GDELT offers different datasets. Global Knowledge Graph (GKG), Events, and Mentions are the main datasets of GDELT. The additional datasets and tools can be used in conjunction with the main datasets to provide a more comprehensive understanding of global events, news coverage, and public discourse. It's worth exploring the full range of GDELT's offerings on their website to see which datasets and tools are most relevant to our needs <https://www.gdeltproject.org/data.html#rawdatafiles>. GDELT's website provides detailed instructions and documentation on accessing their data.

3.3 2.3 Datasets Available from GDELT

Here's a summary of some of the GDELT datasets and their potential use in financial engineering:

- **GKG 2.0:** The Global Knowledge Graph, which extracts a wide range of information from online news articles and other online media sources globally including entities, themes, events, locations, sentiment, and more.
 - Financial Engineering Use:
 - * **Sentiment Analysis:** Analyze the tone and sentiment of news articles related to specific companies, industries, or economic indicators to gauge market sentiment and predict potential price movements.

- * Event Detection: Identify relevant events that could impact financial markets, such as company mergers, regulatory changes, or geopolitical developments.
 - * Theme Extraction: Extract key themes and topics from news articles to understand emerging trends and their potential influence on investment decisions.
- Granularity: Hourly updates, providing a near real-time view of global news and information. Data is available from February 2015 to the present.
- **Events 2.0:** Focuses on events and their relationships, providing details like actor roles, event codes, and geographic locations.
 - Financial Engineering Use:
 - * Risk Assessment: Assess potential risks associated with specific investments by analyzing events related to companies, countries, or industries.
 - * Portfolio Optimization: Use event data to optimize investment portfolios by identifying potential opportunities or threats based on global events and their potential impact.
 - * Quantitative Trading: Develop quantitative trading strategies based on signals extracted from event data, such as changes in relationships between companies or countries.
 - Granularity: Event-level data, capturing individual occurrences with specific details. Data is available from January 1, 1979, to the present.
- **Mentions 2.0:** Tracks mentions of people, organizations, and locations across the GDELT dataset.
 - Financial Engineering Use:
 - * Track news coverage and identify related entities for specific companies.
 - * Gauge public sentiment and perception toward companies.
 - Granularity: Mention-level data, providing insights into the frequency and context of entity mentions. Data is available from February 2015 to the present.
- **Global Frontpage Graph (GFG):** Captures the top headlines from news websites around the world, providing insights into the most prominent news stories.
 - Financial Engineering Use: Identify breaking news and potential market-moving events quickly.
 - Granularity: Hourly snapshots of the most prominent news stories. Data is available from April 2017 to the present.
- **Television Explorer:** Analyzes television news broadcasts and extracts information about the topics covered, people mentioned, and visual content.
 - Financial Engineering Use: Track sentiment and narratives on financial news programs.
 - Granularity: Data linked to specific television broadcasts and segments. Data availability depends on the specific television news channels and programs analyzed. Check the GDELT Television Explorer documentation for details.
- **Geo 2.0:** A geospatial dataset that links GDELT events and mentions to specific geographic locations and administrative regions.
 - Financial Engineering Use: Analyze the geographical impact of events on investments and markets.
 - Granularity: Location-specific data associated with events and mentions. Data is available from January 1, 1979, to the present (aligned with Events 2.0).
- **WEB NGrams 3.0:** Tracks the frequency of phrases and terms across a massive collection of online news articles and web pages.
 - Financial Engineering Use: Identify emerging trends and topics in financial discourse.
 - Granularity: Data on the frequency of word and phrase usage over time. Data is available

from January 1, 2010, to the present.

By understanding the strengths and granularity of each dataset, financial engineers can leverage GDELT's vast information to gain a competitive edge in their analyses and decision-making.

4 3. GDELT Data

4.1 3.1 Importing GDELT GKG Data

In this lesson, we focus on using GDELT to analyze data. Before running the code, we need to understand a few important details:

- **Naming Convention:** GKG data is updated every 15 minutes and data is stored in separate compressed files. The file names follow a specific pattern: `YYYYMMDDHHMMSS.gkg.csv.zip`, where:
 - YYYY: Year
 - MM: Month
 - DD: Day
 - HH: Hour
 - MM: Minute
 - SS: Second
- **No Header Row:** GKG files typically do not have a header row with column names. You need to refer to the GDELT GKG Codebook to identify the meaning of each column. GKG has 27 columns representing different aspects of news articles. Please refer to the [GDELT 2.0 Global Knowledge Graph Codebook \(V2.1\)](#) for full guidance.
- **Complex Data:** The data within each column can be complex, with multiple values separated by delimiters (e.g., semicolons, commas) or encoded information.

By understanding the organization of GDELT GKG data, we can effectively access, process, and analyze the information to gain insights from global news and events. Remember to refer to the GDELT GKG Codebook for detailed information on the data format and column descriptions.

Let's now run the code. The following code downloads the specified GDELT GKG data - `20240930081500.gkg.csv.zip` - corresponding to the GDELT data captured at 8:15 AM on September 30, 2024. We will then unzip it, read it into a pandas DataFrame, add column names, and filter the DataFrame to only include rows where the "Organizations" column contains the word "netflix":

```
[8]: # URL for GDELT GKG data
url = "http://data.gdeltproject.org/gdeltv2/20240930081500.gkg.csv.zip"

# Download and save the zipped file
response = requests.get(url, stream=True)
with open("gdeltdata.csv.zip", "wb") as file:
    for chunk in response.iter_content(chunk_size=1024):
        if chunk:
            file.write(chunk)

# Unzip the file
!unzip gdeltdata.csv.zip -d gdeltdata
```

```

# Read the CSV file into a pandas DataFrame
file_path = "gdelt_data/20240930081500.gkg.csv"
df = pd.read_csv(file_path, sep='\t', header=None)

# Add column names (replace with actual column names from GDELT documentation)
gkg_columns = [
    "GKGRECORDID", "DATE", "SourceCollectionIdentifier", "SourceCommonName",
    "DocumentIdentifier", "Counts", "V2Counts", "Themes", "V2Themes",
    "Locations", "V2Locations", "Persons", "V2Persons", "Organizations",
    "V2Organizations", "V2Tone", "Dates", "GCAM", "SharingImage",
    "RelatedImages", "SocialImageEmbeds", "SocialVideoEmbeds", "Quotations",
    "AllNames", "Amounts", "TranslationInfo", "Extras"
]
df.columns = gkg_columns

# Filter for news related to netflix (adjust column and keyword as needed)
netflix_news = df[df['Organizations'].str.contains("netflix", na=False)].copy()
netflix_news

```

Archive: gdelt_data.csv.zip
inflatng: gdelt_data/20240930081500.gkg.csv

```

[8]:
      GKGRECORDID      DATE SourceCollectionIdentifier
3      20240930081500-3  20240930081500              1 \
208  20240930081500-208  20240930081500              1
280  20240930081500-280  20240930081500              1
530  20240930081500-530  20240930081500              1
552  20240930081500-552  20240930081500              1
943  20240930081500-943  20240930081500              1

      SourceCommonName      DocumentIdentifier
3      malvern gazette.co.uk  https://www.malvern gazette.co.uk/news/24617666... \
208      campaignasia.com  https://www.campaignasia.com:443/article/how-c...
280      tomsguide.com  https://www.tomsguide.com/entertainment/netfli...
530  thenorthernecho.co.uk  https://www.thenorthernecho.co.uk/news/2461766...
552      tomsguide.com  https://www.tomsguide.com/entertainment/stream...
943  thenorthernecho.co.uk  https://www.thenorthernecho.co.uk/news/2461762...

      Counts V2Counts      Themes
3      NaN      NaN  UNREST_CRACKDOWN;TAX_ECON_PRICE;USPEC_POLICY1;... \
208      NaN      NaN  EPU_ECONOMY_HISTORIC;TAX_FNCACT;TAX_FNCACT_FOU...
280      NaN      NaN      TAX_FNCACT;TAX_FNCACT_GUIDE;
530      NaN      NaN  UNREST_CRACKDOWN;TAX_ECON_PRICE;USPEC_POLICY1;...
552      NaN      NaN  MANMADE_DISASTER_IMPLIED;MEDIA_MSM;WB_678_DIGI...
943      NaN      NaN  TAX_FNCACT;TAX_FNCACT_SECRETARY;WB_696_PUBLIC_...

```

V2Themes

3 USPEC_POLICY1,1700;EPU_POLICY_POLICY,1700;UNGP... \

208 EPU_ECONOMY_HISTORIC,267;EPU_ECONOMY_HISTORIC,...

280 TAX_FNCACT_GUIDE,814;TAX_FNCACT_GUIDE,1805;TAX...

530 UNGP_FORESTS_RIVERS_OCEANS,1721;USPEC_POLICY1,...

552 ALLIANCE,1766;MANMADE_DISASTER_IMPLIED,44;MEDI...

943 SOC_GENERALCRIME,388;EPU_CATS_MIGRATION_FEAR_F...

Locations ...

3 NaN ... \

208 1#China#CH#CH#35#105#CH;4#To Cheng, Guangdong,...

280 1#United States#US#US#39.828175#-98.5795#US;1#...

530 NaN ...

552 3#Hollywood, California, United States#US#USCA...

943 1#United Kingdom#UK#UK#54#-4#UK ...

GCAM

3 wc:304,c1.3:1,c12.1:26,c12.10:20,c12.12:10,c12... \

208 wc:573,c12.1:45,c12.10:78,c12.12:16,c12.13:24,...

280 wc:1455,c1.3:2,c12.1:128,c12.10:167,c12.12:63,...

530 wc:304,c1.3:1,c12.1:26,c12.10:20,c12.12:10,c12...

552 wc:654,c12.1:74,c12.10:71,c12.12:20,c12.13:19,...

943 wc:264,c12.1:19,c12.10:29,c12.12:11,c12.13:10,...

SharingImage

3 <https://www.malvern gazette.co.uk/resources/ima...> \

208 <https://cdn.i.haymarketmedia.asia/?n=campaign-...>

280 <https://cdn.mos.cms.futurecdn.net/5hRcycNRCD2d...>

530 <https://www.thenorthernecho.co.uk/resources/im...>

552 <https://cdn.mos.cms.futurecdn.net/PYKEwCu69pnq...>

943 <https://www.thenorthernecho.co.uk/resources/im...>

	RelatedImages	SocialImageEmbeds	
3	NaN	NaN	\
208	NaN	NaN	
280	https://img.youtube.com/vi/UKFMYWNatQM/maxresd...	NaN	
530	NaN	NaN	
552	https://img.youtube.com/vi/3HraByA23eA/maxresd...	NaN	
943	NaN	NaN	

SocialVideoEmbeds

3 NaN \

208 <https://youtube.com/user/CampaignAsia;>

280 <https://youtube.com/vi/PRZ1ELeGepo/maxresdefau...>

530 NaN

552 <https://youtube.com/vi/FRoEjAlhmsw/maxresdefau...>

943 NaN

	Quotations	
3	NaN	\
208	NaN	
280	1477 34 Tim Dillon : This is Your Country#234...	
530	NaN	
552	NaN	
943	NaN	

	AllNames
3	Martin Lewis,559;Money Saving Expert,763;Extra... \
208	New Marketing Paradigm',103;Charlene Ree,338;F...
280	Tim Dillon,994;Official Trailer,1074;Official ...
530	Martin Lewis,559;Money Saving Expert,763;Extra...
552	Love Is Blind',550;Blind Season,597;Official T...
943	Culture Secretary Lisa Nandy,30;Justice Secret...

	Amounts	TranslationInfo
3	99,a month if they,183;99,for either,1217;	NaN \
208	70,different languages,1398;5,different labels...	NaN
280	12,singles for a dating,165;7,premiere,2300;7,...	NaN
530	99,a month if they,183;99,for either,1217;	NaN
552	7, Official Trailer ,464;7, Official Traile...	NaN
943	NaN	NaN

	Extras
3	<PAGE_LINKS>https://t.co/KOW1ZZxoF3;https://tw...
208	<PAGE_LINKS>https://www.iabhongkong.com/joinC2...
280	<PAGE_LINKS>https://www.anrdoezrs.net/links/89...
530	<PAGE_LINKS>https://t.co/KOW1ZZxoF3;https://tw...
552	<PAGE_LINKS>https://www.tomsguide.com/entertai...
943	<PAGE_LINKS>https://www.mirror.co.uk/news/poli...

[6 rows x 27 columns]

First, please note that GDELT GKG 2.0 (Global Knowledge Graph) files are typically in the range of 15-25 MB. This means that we do not really need to worry about how to plan data acquisition and analysis workflows as this is relatively small size that won't pose a major memory concern.

GDELT GKG 2.0 data has 27 columns. The columns contain information about:

- Article Metadata: Unique ID, date/time, source, URL.
- Entities: Mentions of themes, locations, people, and organizations with counts and relevance.
- Sentiment: Overall tone, positive/negative scores, polarity, etc.
- Events & Context: Dates mentioned, content analysis, related images/videos, quotes.
- Additional: All names, amounts, translation info, and extra metadata.

Essentially, these columns provide a comprehensive view of a news article's content, context, and sentiment.

Please note that we only filtered a small GDELT dataset since its GKG data is updated every 15 minutes. Such small timeframe datasets or a sequence of consecutive datasets could potentially be useful in some financial engineering scenarios such as market microstructure analysis, real-time risk management, algorithmic execution, and news sentiment analysis.

Important Considerations: While small timeframe datasets can be useful in these scenarios, it's important to be aware of their limitations. The results obtained from small timeframe datasets might not be as robust or statistically significant as those obtained from larger timeframe datasets. It's essential to carefully validate the results and ensure that the chosen time window is appropriate for the specific application. Combining data from multiple sources and using robust statistical techniques can help improve the reliability of analysis based on small timeframe datasets.

4.2 3.2 Tone in GDELT Data

GDELT employs a sophisticated process to determine the emotional tone and sentiment expressed in news articles and online media. GDELT uses a mix of linguistic analysis (looking at words and grammar) and contextual information (events, entities, location, time) to score the tone of news articles. GDELT leverages a sophisticated blend of NLP, machine learning, statistical methods, contextual analysis, and rule-based systems to calculate its Tone scores. GDELT combines a deep understanding of language with a broad awareness of context to derive nuanced sentiment scores for news articles. This provides valuable insights into the emotional landscape of global events and online discourse.

Advantages of GDELT's Tone score are:

- **Vast Scale and Scope:** GDELT processes massive amounts of global news data in multiple languages, providing a broad perspective on sentiment and tone across diverse sources. This scale is unmatched by most other sentiment analysis tools.
- **Real-time and Historical Analysis:** GDELT provides both real-time updates and a historical archive, allowing you to analyze sentiment trends over time and identify emerging patterns. This historical context is invaluable for understanding the evolution of sentiment toward specific topics or entities.
- **Granular Sentiment Breakdown:** The `V2Tone` column offers a detailed breakdown of sentiment components, including Tone, Positive Score, Negative Score, Polarity, Activity, Self Direction, and Word Count. This granularity allows for a more nuanced understanding of sentiment beyond simple positive/negative classifications.
- **Open and Accessible:** GDELT is a free and open-source resource, making it readily available for researchers, analysts, and developers. This accessibility removes barriers to entry and promotes wider use of sentiment analysis for various applications.
- **Integration with Other GDELT Data:** GDELT's Tone score can be combined with other GDELT datasets, such as event data or geographic information, to gain a more holistic understanding of events and their emotional context. This integrated analysis can provide deeper insights into global trends and patterns.

Limitations: While sophisticated, the accuracy of GDELT's Tone scoring can be influenced by different factors like language ambiguity, cultural nuances, and the complexity of news content. It's also important to understand the limitations of automated sentiment analysis:

- **Accuracy Challenges:** Automated sentiment analysis is inherently complex, and GDELT's Tone score is not immune to errors. Language ambiguity, cultural nuances, and sarcasm can

sometimes lead to misinterpretations of sentiment.

- **Contextual Dependence:** Sentiment can be highly context dependent, and GDELT's analysis might not always capture the full nuances of the text's meaning. It's crucial to consider the surrounding context and potential biases when interpreting Tone scores.
- **Limited to News and Online Media:** GDELT primarily focuses on news articles and online media sources. Sentiment expressed in other forms of communication, such as social media posts or personal conversations, might not be accurately reflected in the Tone score.
- **Potential for Bias:** The sources and content included in GDELT's dataset can introduce biases into the Tone score. It's essential to be aware of these potential biases and consider their implications when interpreting results.

Overall, GDELT's Tone score is a valuable tool for understanding sentiment and tone in global news and online media. It offers vast scale, historical context, and granular analysis. However, it's crucial to be aware of its limitations, particularly regarding accuracy and contextual dependence. By carefully considering the advantages and limitations, we can effectively leverage GDELT's Tone score to gain valuable insights into global events and online discourse.

4.3 3.3 Getting Tone Components in GDELT data

We can use the **V2Tone** column to filter articles based on their sentiment, identify trends in news tone over time, or analyze the emotional context surrounding specific events or entities. This column contains a set of values that represent the sentiment and emotional tone of the associated news article or online media source. The values within the **V2Tone** column are typically separated by commas and include:

- **Tone:** Represents the overall tone, with higher values indicating positivity. Represented as a floating-point number. This is calculated as Positive Score minus Negative Score.
- **Positive Score:** This provides more granular sentiment details. Can be used to identify articles with strong positive sentiment. A measure of the positive sentiment expressed by a floating-point number ranging from 0 to 100. A score of 0 indicates the absence of positive sentiment. Higher values indicate stronger positive sentiment with 100 indicating the strongest possible positive sentiment in the article.
- **Negative Score:** This provides more granular sentiment details. Can be used to identify articles with strong negative sentiment. A measure of the negative sentiment expressed by a floating-point number ranging from 0 to 100. A score of 0 indicates the absence of negative sentiment in the article. Higher values indicate stronger negative sentiment with 100 indicating the strongest negative sentiment in the article.
- **Polarity:** An indicator of how emotionally polarized or charged the text is. Derived from a complex algorithm that considers various linguistic features and contextual factors.
- **Activity Reference Density:** A measure of the density of words related to activity or action in the text.
- **Self Direction:** A measure of the degree to which the text focuses on the author or the subject.
- **WordCount:** The number of words analyzed for tone in the article (integer value).

The following code snippet transforms the GDELT data into a more usable format for sentiment analysis. We take the **V2Tone** column and then separate and convert the relevant sentiment components from it. This allows you to easily work with and analyze the Tone data in subsequent steps.

```
[9]: # Split V2Tone into separate columns
netflix_news[['Tone', 'Positive Score', 'Negative Score', 'Polarity',
↳ 'Activity', 'Self Direction', 'WordCount']] = netflix_news['V2Tone'].str.
↳ split(',', expand=True)

# Convert numeric columns to appropriate data types
numeric_columns = ['Tone', 'Positive Score', 'Negative Score', 'Polarity',
↳ 'Activity', 'Self Direction']
netflix_news[numeric_columns] = netflix_news[numeric_columns].astype(float).
↳ round(3)
netflix_news[['Tone', 'Positive Score', 'Negative Score', 'Polarity',
↳ 'Activity', 'Self Direction', 'WordCount']]
```

```
[9]:      Tone  Positive Score  Negative Score  Polarity  Activity  \
3   -2.102              0.601            2.703      3.303    28.228  \
208  3.086              3.858            0.772      4.630    22.685
280 -0.493              4.310            4.803      9.113    20.135
530 -2.102              0.601            2.703      3.303    28.228
552  0.682              4.502            3.820      8.322    23.602
943 -3.460              2.076            5.536      7.612    26.644

      Self Direction  WordCount
3              3.303         304
208             0.463         573
280             1.293        1455
530             3.303         304
552             0.955         654
943             1.730         264
```

From this code output we can draw the following observations:

- Negative Tone Prevails: More articles have a negative “Tone” value, suggesting a generally negative or critical slant in the text.
- Subtle Sentiment: The “Positive Score” and “Negative Score” values remain relatively low, indicating that the sentiment expressed might be nuanced or indirect rather than strongly positive or negative.
- Relationship Between Polarity and Tone: It seems that in this particular dataset, even articles with a negative Tone can still have an overall positive sentiment direction as captured by Polarity. This highlights the importance of considering both Tone and Polarity together to get a complete picture of the sentiment expressed in the text. While Tone might indicate a generally negative or critical tone, the positive Polarity suggests that there might be underlying positive elements or a more nuanced sentiment conveyed within those articles.
- Higher Activity in Shorter Articles: There seems to be a trend of higher “Activity Reference Density” in articles with a small “WordCount”. This could indicate that shorter articles tend to have more action-oriented language or describe more events.
- Varied Self Direction: The “Self Direction” values vary, suggesting that some articles focus more on the author’s perspective while others are more objective or focused on external events.

How this information could help: By carefully analyzing this output in conjunction with other information and domain expertise, financial engineers can gain insights into market sentiment toward Netflix and develop strategies for investment and risk management. Possible interpretations

of the above results are:

- **Active Change and Adaptation:** The results might reflect a period of significant change or transition for Netflix. News coverage could be critical of specific decisions or challenges (negative Tone) but also acknowledge the company's active efforts to adapt, innovate, or respond to market dynamics (positive Polarity and Activity).
- **Competitive Landscape:** The mixed sentiment might reflect the competitive landscape of the streaming industry. Articles could be discussing Netflix in relation to its competitors, highlighting both challenges and opportunities.
- **Investor Caution:** The negative Tone might indicate a degree of caution or skepticism among investors, even amidst positive developments. This could suggest a need for Netflix to address concerns or improve transparency to maintain investor confidence.

Focusing on the Tone and Polarity aspects, here are some potential conclusions from a financial engineering perspective:

- **Sentiment-Based Trading Strategies:** The Tone and Polarity data can be used to develop trading strategies that capitalize on shifts in market sentiment toward Netflix. For example, a strategy might involve buying Netflix stock when Polarity is strongly positive, even if Tone is slightly negative, indicating potential undervaluation.
- **Risk Management:** The Tone data can be used to assess potential risks associated with investing in Netflix. Negative Tone might indicate increased uncertainty or negative news coverage, prompting further investigation or risk mitigation strategies.
- **Predictive Modeling:** The Tone and Polarity data can be incorporated into predictive models for Netflix's stock price or other financial metrics. The combination of Tone and Polarity might provide valuable insights into market sentiment that can improve the accuracy of these models.

Overall, the data suggests a dynamic and potentially volatile period for Netflix, with news coverage reflecting both challenges and opportunities. By carefully considering all the V2Tone components and combining them with other GDELT data and fundamental analysis, financial engineers can gain a more comprehensive understanding of market sentiment and its potential impact on Netflix's financial performance.

However, we also need to note the important considerations:

- **Limited Data:** This analysis is based on a small subset of data. A larger dataset and more comprehensive analysis are needed for more robust conclusions.
- **External Factors:** Consider other factors that might be influencing market sentiment toward Netflix, such as industry trends, competitor actions, or overall economic conditions.
- **News Content:** Analyze the actual content of the news articles to understand the specific factors driving the Tone and Polarity scores. This qualitative analysis can provide valuable context for the numerical data.

4.4 3.4 How to Enhance Sentiment Analysis

Besides the V2Tone column, there are other columns in the GDELT GKG 2.0 dataset that we could potentially use to enhance sentiment analysis:

Themes and V2Themes: These columns contain a list of themes identified in the article. We can use this to identify themes related to positive or negative sentiment (e.g., "ECON_INFLATION" might be associated with negative sentiment). We can create a dictionary mapping themes to sentiment scores and use it to analyze the sentiment associated with the themes present in the articles.

We can use these columns to create a sentiment dictionary or lexicon based on the co-occurrence of specific themes with positive or negative keywords. We can then use this lexicon to score the sentiment of articles based on the presence of these themes or names.

Quotations: This column contains direct quotations from the article. Analyzing the sentiment of these quotations can provide a more nuanced understanding of the sentiment expressed in the article. We can apply sentiment analysis techniques directly to the text of the quotations to get a more granular understanding of the sentiment expressed.

GCAM: GCAM stands for Global Content Analysis Measures. This column in the GKG dataset provides a comprehensive set of content analysis measures derived from applying Google’s Cloud Vision API to the images and videos associated with a news article. This column contains the output of Google’s Cloud Vision API, which can include labels and descriptions of images associated with the article. These labels and descriptions can sometimes provide clues about the sentiment of the article. We can apply sentiment analysis to the text extracted from the GCAM data to infer sentiment related to the visual content of the article.

We can extract text from the GCAM data and apply sentiment analysis models to this text to infer the sentiment associated with visual content.

AllNames: This column contains a list of all names mentioned in the article. This could be used to identify individuals or entities that are frequently associated with positive or negative sentiment. By analyzing the sentiment associated with the mentioned names, we can potentially identify key influencers or stakeholders driving the overall sentiment.

Important Considerations: These columns might require more pre-processing and analysis compared to the **V2Tone** column. Also, the accuracy of sentiment analysis based on these columns might be lower compared to using **V2Tone**, which is specifically designed for sentiment analysis. Combining the insights from these columns with the **V2Tone** data can provide a more comprehensive understanding of the overall sentiment.

5 4. Dynamics of Change in Tone

5.1 4.1. Considering Two Methods

Let’s now consider methods for observing the dynamics of change in tone toward Netflix. There could be many approaches to this, but let’s delve deeper into the two very basic simple methods for observing the dynamics of change in tone:

Method 1: Averaging Tone for Each 15-Minute File and Comparing Average Scores

This method prioritizes observing the overall sentiment trend towards Netflix over time by analyzing the average Tone score within consecutive 15-minute intervals. Here’s a detailed breakdown of the process:

- **Data Acquisition and Filtering:** The process begins by acquiring the relevant GDELT GKG data files for the desired time period, typically encompassing a series of 15-minute intervals. Each file contains information about various news articles and events captured during that specific 15-minute window. The next step is to filter these files to isolate news articles specifically related to Netflix, using keywords, entity mentions, or other relevant criteria.
- **Tone Extraction and Aggregation:** Once the Netflix-related articles are identified within each

file, the “V2Tone” column is extracted. This column contains a comma-separated string of values representing different sentiment components, including the overall Tone score. The “V2Tone” string is then parsed to isolate the Tone value, which is typically a numerical score indicating the sentiment expressed in the article. Tone scores are often normalized to a specific range, such as -100 to +100, with higher values indicating more positive sentiment. For each 15-minute file, the Tone scores of all the Netflix-related articles are aggregated by calculating their average. This average Tone score represents the overall sentiment toward Netflix during that specific 15-minute interval.

- **Time Series Construction and Analysis:** The average Tone scores, along with their corresponding timestamps (usually the start time of each 15-minute interval), are then used to construct a time series. This time series provides a chronological view of how sentiment toward Netflix has evolved over the extended period. The time series can be visually inspected for patterns, trends, and significant shifts in sentiment. Additionally, various analytical techniques can be applied, including: moving averages, to smooth out short-term fluctuations and identify longer-term trends; volatility analysis, to quantify the variability of sentiment over time; and correlation with other data, such as stock prices or news events, to explore potential relationships.
- **Advantages:** This method’s primary advantage lies in its clear focus on observing the overall change in sentiment and its relative simplicity for implementation and interpretation. It’s also computationally efficient, making it suitable for real-time applications or large datasets. However, it’s worth noting that this method sacrifices some granularity by averaging individual article sentiments, and it might be sensitive to outliers. Additionally, it provides limited insight into the specific news stories or events that drive sentiment changes.

Method 2: Combining All Samples and Analyzing Tone Elements for Individual Articles

This method focuses on analyzing the sentiment expressed in individual news articles and its potential relationship to specific events or news stories. It provides a more granular and event-centric perspective compared to Method 1. Here’s a detailed description:

- **Data Consolidation and Tone Extraction:** All the GDELT GKG data files for the desired time period are merged into a single dataset. This consolidated dataset includes all Netflix-related articles and their associated information, including timestamps, Tone scores, and other relevant features. The “V2Tone” column is extracted for each article, and the Tone score is parsed and isolated, similar to Method 1.
- **Article-Level Sentiment Analysis:** The Tone scores of individual articles are then analyzed, considering their timestamps and other relevant features such as themes, entities mentioned, or quotations. This analysis aims to understand the detailed sentiment expressed within specific news stories and how it might relate to the context or content of the article.
- **Event Detection and Correlation:** This method allows for identifying specific events or news stories that coincide with significant changes in sentiment for individual articles. By examining the timestamps of articles with notable Tone scores, researchers can pinpoint potential triggers for sentiment shifts. Furthermore, the Tone scores can be correlated with other data sources, such as stock prices or news event databases, to explore relationships and understand the impact of specific events on sentiment toward Netflix.
- **Advantage:** The advantage of this method lies in its detailed sentiment information and the

potential for deeper analysis, including exploring the relationship between Tone and other article features. It also allows for understanding the impact of specific events on sentiment. However, this approach can be more computationally demanding and complex, potentially losing clear temporal information as it focuses on individual articles rather than aggregated trends. The interpretation of results might also require more effort due to the large volume of individual article sentiments.

5.2 4.2. Application of Method 1: Averaging Tone for Each 15-Minute File and Comparing Average Scores

In this section, we will utilize Method 1. The below code downloads GDELT data for specific 15-minute intervals from 8 AM to 11:45 PM on September 23, 2024, filters for news related to Netflix, extracts sentiment information (Tone scores), and calculates the average sentiment for each interval. It does this efficiently by processing each data file in memory without saving it to disk, making it suitable for analyzing larger datasets:

```
[10]: # Generate timestamps from 8 AM to 11:45 PM on September 23, 2024
start_time = datetime(2024, 9, 23, 8, 0, 0)
end_time = datetime(2024, 9, 23, 23, 45, 0)
timestamps = []
current_time = start_time
while current_time <= end_time:
    timestamps.append(current_time.strftime("%Y%m%d%H%M%S"))
    current_time += timedelta(minutes=15)

# Create a dictionary to store average Tone for each timestamp
avg_tones = {}

# Loop through timestamps, download, process, and discard each file
for timestamp in timestamps:
    url = f"http://data.gdeltproject.org/gdeltv2/{timestamp}.gkg.csv.zip"
    response = requests.get(url, stream=True)

    # Process the zip file in memory without extracting to disk
    with zipfile.ZipFile(io.BytesIO(response.content)) as zip_ref:
        for file_name in zip_ref.namelist():
            if file_name.endswith(".gkg.csv"):
                with zip_ref.open(file_name) as file:
                    # Specify the encoding as 'latin-1' to handle potential
↪ encoding issues
                    df = pd.read_csv(file, sep='\t', header=None,
                                   on_bad_lines='skip', # Skip lines with
↪ errors
                                   engine='python', # Use Python engine to
↪ handle large files
                                   encoding='latin-1') # Explicitly set
↪ encoding to 'latin-1'
```

```

        # Add column names (refer to GDELT documentation)
        gkg_columns = [
            "GKGRECORDID", "DATE", "SourceCollectionIdentifier",
↪ "SourceCommonName",
            "DocumentIdentifier", "Counts", "V2Counts", "Themes",
↪ "V2Themes",
            "Locations", "V2Locations", "Persons", "V2Persons",
↪ "Organizations",
            "V2Organizations", "V2Tone", "Dates", "GCAM",
↪ "SharingImage",
            "RelatedImages", "SocialImageEmbeds", "SocialVideoEmbeds",
↪ "Quotations",
            "AllNames", "Amounts", "TranslationInfo", "Extras"
        ]
        df.columns = gkg_columns

        # Filter for news related to Netflix
        netflix_news = df[df['Organizations'].str.contains("netflix",
↪ na=False)].copy()

        # Extract Tone components (similar to previous code)
        netflix_news[['Tone', 'Positive Score', 'Negative Score',
↪ 'Polarity', 'Activity', 'Self Direction', 'WordCount']] =
↪ netflix_news['V2Tone'].str.split(',', expand=True)
        numeric_columns = ['Tone', 'Positive Score', 'Negative Score',
↪ 'Polarity', 'Activity', 'Self Direction']
        netflix_news[numeric_columns] = netflix_news[numeric_columns].
↪ astype(float, errors='ignore').round(3) #ignore errors

        # Calculate and store average Tone
        avg_tone = netflix_news['Tone'].mean()
        avg_tones[timestamp] = avg_tone

        # File is automatically discarded when exiting the 'with' block

# Create a DataFrame from the avg_tones dictionary converting 'Timestamp'
↪ column to datetime objects
tone_df = pd.DataFrame(list(avg_tones.items()), columns=['Timestamp',
↪ 'AvgTone'])
tone_df['Timestamp'] = pd.to_datetime(tone_df['Timestamp'],
↪ format='%Y%m%d%H%M%S')
tone_df

```

```

[10]:          Timestamp    AvgTone
0  2024-09-23 08:00:00 -0.531833

```

```

1 2024-09-23 08:15:00 -3.236250
2 2024-09-23 08:30:00 -0.025667
3 2024-09-23 08:45:00 1.060857
4 2024-09-23 09:00:00 -2.863167
..
59 2024-09-23 22:45:00 0.120400
60 2024-09-23 23:00:00 0.489000
61 2024-09-23 23:15:00 -0.892667
62 2024-09-23 23:30:00 1.537625
63 2024-09-23 23:45:00 0.226905

```

[64 rows x 2 columns]

Here, we first create a list of timestamps (timestamps) representing 15-minute intervals between 4 AM and 8 PM on September 23, 2024. These timestamps will be used to access the GDELT data files.

Then, we iterate through each timestamp to download GDELT Data, process Data, calculate Average Tone, and discard File. This code utilizes in-memory processing. In-memory processing is essential for GDELT data analysis due to its ability to handle large data volumes efficiently, enhance speed and responsiveness, improve scalability, optimize resource usage, and provide greater flexibility for analysis. By processing each file individually and discarding it before moving to the next, in-memory approaches enable researchers to extract meaningful insights from GDELT data without being constrained by storage limitations or performance bottlenecks.

Finally, we create a pandas DataFrame named the `avg_tones`, which contains timestamps and their corresponding average Tone scores for Netflix. This allows us to further analyze and visualize the sentiment dynamics over time.

5.2.1 4.2.1 Get Netflix Intraday Data

Let's now get intraday stock price data for Netflix (NFLX) for September 23, 2024, using the `yfinance` library. The following code snippet downloads Netflix's 15-minute interval intraday stock data for September 23, 2024 (including pre- and post-market), saves it to a CSV file, and then displays it:

```

[11]: # Download the intraday data using yfinance
nflx_intraday = yf.download(
    tickers='NFLX',
    start='2024-09-23',
    end='2024-09-24',
    interval='15m',
    prepost=True)

# Save the data to a CSV file
nflx_intraday.to_csv("netflix_intraday_20240923.csv")

# Display DataFrame
nflx_intraday

```

```
$NFLX: possibly delisted; no price data found (15m 2024-09-23 -> 2024-09-24)
(Yahoo error = "15m data not available for startTime=1727064000 and
endTime=1727150400. The requested range must be within the last 60 days.")
[*****100%*****] 1 of 1 completed
```

1 Failed download:

```
['NFLX']: possibly delisted; no price data found (15m 2024-09-23 -> 2024-09-24)
(Yahoo error = "15m data not available for startTime=1727064000 and
endTime=1727150400. The requested range must be within the last 60 days.")
```

[11]: Empty DataFrame

```
Columns: [(Adj Close, NFLX), (Close, NFLX), (High, NFLX), (Low, NFLX), (Open,
NFLX), (Volume, NFLX)]
Index: []
```

Important consideration for data availability: Intraday data is typically available from Yahoo Finance for the past 60 days. As of this writing, available data was also downloaded and saved as 'WQUnetflix_intraday_20240923.csv' for later retrieval. In case you run this notebook when intraday data is no longer available, please load data from the locally saved 'WQUnetflix_intraday_20240923.csv' file for the remainder of this lesson.

[12]: *# Open saved DataFrame - OPTIONAL*

```
nflx_intraday = pd.read_csv('WQU netflix_intraday_20240923.csv')
nflx_intraday.set_index('Datetime', inplace=True)
nflx_intraday
```

[12]:

		Open	High	Low	Close	
Datetime						
2024-09-23	04:00:00	701.030000	701.030000	700.980000	700.980000	\
2024-09-23	04:15:00	700.980000	700.980000	700.900000	700.960000	
2024-09-23	04:30:00	699.780000	699.780000	699.780000	699.780000	
2024-09-23	05:15:00	700.570000	700.990000	700.570000	700.790000	
2024-09-23	05:45:00	700.700000	701.500000	700.700000	701.500000	
2024-09-23	06:00:00	701.400000	701.500000	701.400000	701.500000	
2024-09-23	06:30:00	701.910000	701.910000	701.910000	701.910000	
2024-09-23	07:00:00	702.210000	702.210000	702.210000	702.210000	
2024-09-23	07:15:00	702.210000	702.210000	701.250000	701.250000	
2024-09-23	07:30:00	701.470000	701.610000	701.250000	701.250000	
2024-09-23	07:45:00	701.310000	701.320000	701.250000	701.250000	
2024-09-23	08:00:00	703.048800	703.048800	697.561200	701.950000	
2024-09-23	08:15:00	702.000000	702.500000	701.040000	701.350000	
2024-09-23	08:30:00	701.670000	702.250000	700.974000	702.190000	
2024-09-23	08:45:00	701.625000	702.250000	701.000000	702.187500	
2024-09-23	09:00:00	701.625000	702.470000	701.000000	702.470000	
2024-09-23	09:15:00	702.350000	703.100000	700.940000	702.130000	
2024-09-23	09:30:00	702.390015	704.099976	699.159973	703.799988	
2024-09-23	09:45:00	704.010010	712.176514	703.534973	708.840027	
2024-09-23	10:00:00	708.840027	711.960022	708.523926	709.650024	

2024-09-23 10:15:00	709.510010	711.000000	708.799988	709.684998
2024-09-23 10:30:00	710.130005	710.500000	707.710022	710.499878
2024-09-23 10:45:00	710.320007	711.519897	708.309998	709.010010
2024-09-23 11:00:00	709.239929	709.239929	705.859985	707.599976
2024-09-23 11:15:00	707.380005	707.380005	705.549988	706.570007
2024-09-23 11:30:00	706.640015	708.460022	706.640015	707.280029
2024-09-23 11:45:00	707.364990	708.900024	706.690002	707.337280
2024-09-23 12:00:00	707.030029	707.534973	703.599976	704.179993
2024-09-23 12:15:00	703.700012	705.642517	703.340027	704.849976
2024-09-23 12:30:00	704.875000	705.299988	703.809998	705.299988
2024-09-23 12:45:00	704.830017	705.599976	704.434998	705.268982
2024-09-23 13:00:00	705.010010	705.700012	703.979980	705.590027
2024-09-23 13:15:00	705.000000	707.200012	705.000000	706.979980
2024-09-23 13:30:00	707.190002	707.809387	706.263184	707.289978
2024-09-23 13:45:00	707.309998	708.190002	707.164978	708.150024
2024-09-23 14:00:00	708.109985	709.150024	707.190125	708.294983
2024-09-23 14:15:00	708.270020	708.299988	706.440002	706.592407
2024-09-23 14:30:00	706.599976	707.330017	706.369995	706.905029
2024-09-23 14:45:00	706.849976	707.715027	705.760010	707.255005
2024-09-23 15:00:00	706.979980	707.539978	706.809998	707.080017
2024-09-23 15:15:00	706.940002	707.656982	705.450012	705.450012
2024-09-23 15:30:00	705.400024	705.690002	704.002991	704.739990
2024-09-23 15:45:00	705.000000	706.539978	704.505005	705.349976
2024-09-23 16:00:00	705.370000	706.460000	705.370000	705.875000
2024-09-23 16:15:00	705.875000	706.637500	705.370000	705.577500
2024-09-23 16:30:00	705.550000	706.650000	705.370000	705.370000
2024-09-23 16:45:00	706.095000	706.640000	705.550000	706.095000
2024-09-23 17:00:00	706.095000	706.700000	705.550000	706.125000
2024-09-23 17:15:00	706.125000	706.700000	705.578700	705.775000
2024-09-23 17:30:00	705.775000	706.297100	705.550000	706.125000
2024-09-23 17:45:00	706.100000	706.700000	705.550000	706.125000
2024-09-23 18:00:00	706.075000	706.075000	705.770000	706.000000
2024-09-23 18:15:00	705.460000	706.200000	705.460000	705.750000
2024-09-23 18:30:00	705.800000	706.671200	705.000000	705.800000
2024-09-23 18:45:00	705.750000	706.000000	705.578700	705.578700
2024-09-23 19:00:00	705.770000	706.050000	705.325000	705.325000
2024-09-23 19:15:00	705.700000	705.700000	705.000000	705.500000
2024-09-23 19:30:00	705.640000	705.760500	705.405000	705.760500
2024-09-23 19:45:00	705.360000	706.671200	705.010100	705.400000

	Adj	Close	Volume
Datetime			
2024-09-23 04:00:00	700.980000		0
2024-09-23 04:15:00	700.960000		0
2024-09-23 04:30:00	699.780000		0
2024-09-23 05:15:00	700.790000		0
2024-09-23 05:45:00	701.500000		0

2024-09-23 06:00:00	701.500000	0
2024-09-23 06:30:00	701.910000	0
2024-09-23 07:00:00	702.210000	0
2024-09-23 07:15:00	701.250000	0
2024-09-23 07:30:00	701.250000	0
2024-09-23 07:45:00	701.250000	0
2024-09-23 08:00:00	701.950000	0
2024-09-23 08:15:00	701.350000	0
2024-09-23 08:30:00	702.190000	0
2024-09-23 08:45:00	702.187500	0
2024-09-23 09:00:00	702.470000	0
2024-09-23 09:15:00	702.130000	0
2024-09-23 09:30:00	703.799988	565971
2024-09-23 09:45:00	708.840027	218229
2024-09-23 10:00:00	709.650024	88469
2024-09-23 10:15:00	709.684998	78553
2024-09-23 10:30:00	710.499878	63312
2024-09-23 10:45:00	709.010010	91744
2024-09-23 11:00:00	707.599976	74905
2024-09-23 11:15:00	706.570007	41229
2024-09-23 11:30:00	707.280029	51673
2024-09-23 11:45:00	707.337280	88883
2024-09-23 12:00:00	704.179993	33891
2024-09-23 12:15:00	704.849976	36095
2024-09-23 12:30:00	705.299988	34663
2024-09-23 12:45:00	705.268982	17953
2024-09-23 13:00:00	705.590027	17369
2024-09-23 13:15:00	706.979980	28843
2024-09-23 13:30:00	707.289978	126802
2024-09-23 13:45:00	708.150024	28786
2024-09-23 14:00:00	708.294983	39972
2024-09-23 14:15:00	706.592407	40875
2024-09-23 14:30:00	706.905029	69318
2024-09-23 14:45:00	707.255005	36271
2024-09-23 15:00:00	707.080017	44198
2024-09-23 15:15:00	705.450012	96335
2024-09-23 15:30:00	704.739990	59030
2024-09-23 15:45:00	705.349976	243493
2024-09-23 16:00:00	705.875000	0
2024-09-23 16:15:00	705.577500	0
2024-09-23 16:30:00	705.370000	0
2024-09-23 16:45:00	706.095000	0
2024-09-23 17:00:00	706.125000	0
2024-09-23 17:15:00	705.775000	0
2024-09-23 17:30:00	706.125000	0
2024-09-23 17:45:00	706.125000	0
2024-09-23 18:00:00	706.000000	0

2024-09-23	18:15:00	705.750000	0
2024-09-23	18:30:00	705.800000	0
2024-09-23	18:45:00	705.578700	0
2024-09-23	19:00:00	705.325000	0
2024-09-23	19:15:00	705.500000	0
2024-09-23	19:30:00	705.760500	0
2024-09-23	19:45:00	705.400000	0

5.2.2 4.2.2 Netflix Intraday Data in Conjunction with Average Tone in News

In previous code, we get intraday data for stock price data for Netflix. Please note the `prepost=True` parameter, which indicates whether to include pre-market and post-market data. In our case, we include pre- and post-market data. As you can see, the Volume column displays zeros before 9:30 AM (market open) and after 4:00 PM (market close). Yahoo uses the exchange's native time zone. For Netflix, it is NASDAQ located in New York City. On September 23 in New York, we observe Eastern Daylight Time (EDT), which is 4 hours behind the UTC timezone. Let's convert Datetime column values in the `nflx_intraday` DataFrame to UTC timezone:

```
[13]: # Convert Datetime to UTC timezone
nflx_intraday.index = pd.to_datetime(nflx_intraday.index).tz_localize('America/
↪New_York').tz_convert('UTC')
nflx_intraday
```

```
[13]:
```

		Open	High	Low	Close	
Datetime						
2024-09-23	08:00:00+00:00	701.030000	701.030000	700.980000	700.980000	\
2024-09-23	08:15:00+00:00	700.980000	700.980000	700.900000	700.960000	
2024-09-23	08:30:00+00:00	699.780000	699.780000	699.780000	699.780000	
2024-09-23	09:15:00+00:00	700.570000	700.990000	700.570000	700.790000	
2024-09-23	09:45:00+00:00	700.700000	701.500000	700.700000	701.500000	
2024-09-23	10:00:00+00:00	701.400000	701.500000	701.400000	701.500000	
2024-09-23	10:30:00+00:00	701.910000	701.910000	701.910000	701.910000	
2024-09-23	11:00:00+00:00	702.210000	702.210000	702.210000	702.210000	
2024-09-23	11:15:00+00:00	702.210000	702.210000	701.250000	701.250000	
2024-09-23	11:30:00+00:00	701.470000	701.610000	701.250000	701.250000	
2024-09-23	11:45:00+00:00	701.310000	701.320000	701.250000	701.250000	
2024-09-23	12:00:00+00:00	703.048800	703.048800	697.561200	701.950000	
2024-09-23	12:15:00+00:00	702.000000	702.500000	701.040000	701.350000	
2024-09-23	12:30:00+00:00	701.670000	702.250000	700.974000	702.190000	
2024-09-23	12:45:00+00:00	701.625000	702.250000	701.000000	702.187500	
2024-09-23	13:00:00+00:00	701.625000	702.470000	701.000000	702.470000	
2024-09-23	13:15:00+00:00	702.350000	703.100000	700.940000	702.130000	
2024-09-23	13:30:00+00:00	702.390015	704.099976	699.159973	703.799988	
2024-09-23	13:45:00+00:00	704.010010	712.176514	703.534973	708.840027	
2024-09-23	14:00:00+00:00	708.840027	711.960022	708.523926	709.650024	
2024-09-23	14:15:00+00:00	709.510010	711.000000	708.799988	709.684998	
2024-09-23	14:30:00+00:00	710.130005	710.500000	707.710022	710.499878	

2024-09-23	14:45:00+00:00	710.320007	711.519897	708.309998	709.010010
2024-09-23	15:00:00+00:00	709.239929	709.239929	705.859985	707.599976
2024-09-23	15:15:00+00:00	707.380005	707.380005	705.549988	706.570007
2024-09-23	15:30:00+00:00	706.640015	708.460022	706.640015	707.280029
2024-09-23	15:45:00+00:00	707.364990	708.900024	706.690002	707.337280
2024-09-23	16:00:00+00:00	707.030029	707.534973	703.599976	704.179993
2024-09-23	16:15:00+00:00	703.700012	705.642517	703.340027	704.849976
2024-09-23	16:30:00+00:00	704.875000	705.299988	703.809998	705.299988
2024-09-23	16:45:00+00:00	704.830017	705.599976	704.434998	705.268982
2024-09-23	17:00:00+00:00	705.010010	705.700012	703.979980	705.590027
2024-09-23	17:15:00+00:00	705.000000	707.200012	705.000000	706.979980
2024-09-23	17:30:00+00:00	707.190002	707.809387	706.263184	707.289978
2024-09-23	17:45:00+00:00	707.309998	708.190002	707.164978	708.150024
2024-09-23	18:00:00+00:00	708.109985	709.150024	707.190125	708.294983
2024-09-23	18:15:00+00:00	708.270020	708.299988	706.440002	706.592407
2024-09-23	18:30:00+00:00	706.599976	707.330017	706.369995	706.905029
2024-09-23	18:45:00+00:00	706.849976	707.715027	705.760010	707.255005
2024-09-23	19:00:00+00:00	706.979980	707.539978	706.809998	707.080017
2024-09-23	19:15:00+00:00	706.940002	707.656982	705.450012	705.450012
2024-09-23	19:30:00+00:00	705.400024	705.690002	704.002991	704.739990
2024-09-23	19:45:00+00:00	705.000000	706.539978	704.505005	705.349976
2024-09-23	20:00:00+00:00	705.370000	706.460000	705.370000	705.875000
2024-09-23	20:15:00+00:00	705.875000	706.637500	705.370000	705.577500
2024-09-23	20:30:00+00:00	705.550000	706.650000	705.370000	705.370000
2024-09-23	20:45:00+00:00	706.095000	706.640000	705.550000	706.095000
2024-09-23	21:00:00+00:00	706.095000	706.700000	705.550000	706.125000
2024-09-23	21:15:00+00:00	706.125000	706.700000	705.578700	705.775000
2024-09-23	21:30:00+00:00	705.775000	706.297100	705.550000	706.125000
2024-09-23	21:45:00+00:00	706.100000	706.700000	705.550000	706.125000
2024-09-23	22:00:00+00:00	706.075000	706.075000	705.770000	706.000000
2024-09-23	22:15:00+00:00	705.460000	706.200000	705.460000	705.750000
2024-09-23	22:30:00+00:00	705.800000	706.671200	705.000000	705.800000
2024-09-23	22:45:00+00:00	705.750000	706.000000	705.578700	705.578700
2024-09-23	23:00:00+00:00	705.770000	706.050000	705.325000	705.325000
2024-09-23	23:15:00+00:00	705.700000	705.700000	705.000000	705.500000
2024-09-23	23:30:00+00:00	705.640000	705.760500	705.405000	705.760500
2024-09-23	23:45:00+00:00	705.360000	706.671200	705.010100	705.400000

	Adj Close	Volume
Datetime		
2024-09-23 08:00:00+00:00	700.980000	0
2024-09-23 08:15:00+00:00	700.960000	0
2024-09-23 08:30:00+00:00	699.780000	0
2024-09-23 09:15:00+00:00	700.790000	0
2024-09-23 09:45:00+00:00	701.500000	0
2024-09-23 10:00:00+00:00	701.500000	0
2024-09-23 10:30:00+00:00	701.910000	0

2024-09-23	11:00:00+00:00	702.210000	0
2024-09-23	11:15:00+00:00	701.250000	0
2024-09-23	11:30:00+00:00	701.250000	0
2024-09-23	11:45:00+00:00	701.250000	0
2024-09-23	12:00:00+00:00	701.950000	0
2024-09-23	12:15:00+00:00	701.350000	0
2024-09-23	12:30:00+00:00	702.190000	0
2024-09-23	12:45:00+00:00	702.187500	0
2024-09-23	13:00:00+00:00	702.470000	0
2024-09-23	13:15:00+00:00	702.130000	0
2024-09-23	13:30:00+00:00	703.799988	565971
2024-09-23	13:45:00+00:00	708.840027	218229
2024-09-23	14:00:00+00:00	709.650024	88469
2024-09-23	14:15:00+00:00	709.684998	78553
2024-09-23	14:30:00+00:00	710.499878	63312
2024-09-23	14:45:00+00:00	709.010010	91744
2024-09-23	15:00:00+00:00	707.599976	74905
2024-09-23	15:15:00+00:00	706.570007	41229
2024-09-23	15:30:00+00:00	707.280029	51673
2024-09-23	15:45:00+00:00	707.337280	88883
2024-09-23	16:00:00+00:00	704.179993	33891
2024-09-23	16:15:00+00:00	704.849976	36095
2024-09-23	16:30:00+00:00	705.299988	34663
2024-09-23	16:45:00+00:00	705.268982	17953
2024-09-23	17:00:00+00:00	705.590027	17369
2024-09-23	17:15:00+00:00	706.979980	28843
2024-09-23	17:30:00+00:00	707.289978	126802
2024-09-23	17:45:00+00:00	708.150024	28786
2024-09-23	18:00:00+00:00	708.294983	39972
2024-09-23	18:15:00+00:00	706.592407	40875
2024-09-23	18:30:00+00:00	706.905029	69318
2024-09-23	18:45:00+00:00	707.255005	36271
2024-09-23	19:00:00+00:00	707.080017	44198
2024-09-23	19:15:00+00:00	705.450012	96335
2024-09-23	19:30:00+00:00	704.739990	59030
2024-09-23	19:45:00+00:00	705.349976	243493
2024-09-23	20:00:00+00:00	705.875000	0
2024-09-23	20:15:00+00:00	705.577500	0
2024-09-23	20:30:00+00:00	705.370000	0
2024-09-23	20:45:00+00:00	706.095000	0
2024-09-23	21:00:00+00:00	706.125000	0
2024-09-23	21:15:00+00:00	705.775000	0
2024-09-23	21:30:00+00:00	706.125000	0
2024-09-23	21:45:00+00:00	706.125000	0
2024-09-23	22:00:00+00:00	706.000000	0
2024-09-23	22:15:00+00:00	705.750000	0
2024-09-23	22:30:00+00:00	705.800000	0

2024-09-23	22:45:00+00:00	705.578700	0
2024-09-23	23:00:00+00:00	705.325000	0
2024-09-23	23:15:00+00:00	705.500000	0
2024-09-23	23:30:00+00:00	705.760500	0
2024-09-23	23:45:00+00:00	705.400000	0

Let's now plot a candlestick chart. The candlestick charts are considered a valuable tool in financial technical analysis for several reasons. Their visual clarity in illustrating price action, ability to signal reversals, and widespread acceptance make them a valuable tool for financial technical analysis and informed trading decisions. The following code generates both the candlestick chart for `nflx_intraday` and the `tone_df` data on the same interactive plot. The code uses Plotly's graphing library, which enables us to interactively explore the visualization. We can zoom into a section of the visualization or hover over individual candlesticks/bars to glean precise values of stock price/volume at a specific time:

```
[14]: # Create candlestick trace
candlestick_trace = go.Candlestick(x=nflx_intraday.index,
                                   open=nflx_intraday['Open'],
                                   high=nflx_intraday['High'],
                                   low=nflx_intraday['Low'],
                                   close=nflx_intraday['Close'],
                                   name='Netflix Price')

# Create tone trace
tone_trace = go.Scatter(x=tone_df['Timestamp'],
                        y=tone_df['AvgTone'],
                        mode='lines',
                        name='Average Tone',
                        line=dict(color='blue', width=1),
                        yaxis='y2') # Assign to secondary y-axis

# Create figure with both traces
fig = go.Figure(data=[candlestick_trace, tone_trace])

# Update layout with secondary y-axis
fig.update_layout(title_text='Netflix Price and Average Tone',
                  yaxis_title='Price (USD)',
                  yaxis2=dict(title='Average Tone',
                              overlaying='y',
                              side='right'))

fig.show()
```



This code generates a chart with both the Netflix intraday candlestick chart and the average tone data plotted together, allowing us to visually analyze the relationship between price movements and sentiment changes.

Each candlestick represents the price movement of Netflix stock during a specific time interval. The green candlestick indicates a price increase during that interval. The bottom of the green candlestick represents the opening price, the top represents the closing price, and the lines (wicks) extending from the body represent the high and low prices during that interval. The red candlestick indicates a price decrease during that interval. The top of the red candlestick represents the opening price, the bottom represents the closing price, and the wicks represent the high and low prices.

The layout is updated to include a secondary y-axis on the right side (`yaxis2`) and set titles for both y-axes. The primary y-axis on the left represents the price (USD) of Netflix stock, and the secondary y-axis represents average tone values. Using separate y-axes allows for a clear and meaningful representation of both datasets without distorting their respective values. Now we can visually compare the price movements of Netflix stock with the changes in average sentiment over time. This can help to identify potential relationships or correlations between price and sentiment.

By observing the patterns and trends in the candlestick prices and the tone line, we can explore potential correlations and insights:

- **Positive Tone and Price:** When the Tone line is high (positive sentiment) and the candlesticks are green (price increase), it could suggest that positive news coverage is having a favorable impact on Netflix's stock price. We can observe an abundance of news that is mostly positive in tone a few hours prior to market opening from about 10 AM. This could potentially explain green candlesticks at the opening time at 13:30.
- **Negative Tone and Price:** Conversely, a low Tone line (negative sentiment) coinciding with red candlesticks (price decrease) could indicate that negative news is influencing the stock negatively.
- **Divergence:** We can also observe that there are instances where the Tone line and candlestick patterns diverge. This might warrant further investigation. For example, positive news sentiment (high Tone) but a declining stock price (red candlesticks) could raise questions about other market factors at play.

Overall, the chart provides a visual tool to explore the potential link between sentiment (Tone) derived from GDELT data and Netflix's stock price movement. It enables analysts and traders to observe correlations, spot potential trends, and make more informed decisions. Remember that this is just one aspect of analysis, and many other factors influence stock prices. The chart shows

potential correlations, but further investigation is needed to establish any causal relationships between sentiment and price movement. Numerous other factors, such as market trends, economic news, and company-specific developments, can impact stock prices. We should also consider that the Tone score is just one measure of sentiment and may not capture all nuances of news coverage.

5.3 4.3. Possible Financial Engineering Scenarios

We explored an application of Method 1 in the previous example. By strategically selecting and applying these methods, financial engineers can leverage sentiment analysis to gain valuable insights, inform decision-making, and enhance their strategies across various applications. Here are some potential financial engineering scenarios where observing the dynamics of change in tone toward Netflix can be applied:

Method 1: Averaging Tone for Each 15-Minute File and Comparing Average Scores

This method, focusing on overall sentiment trends, is particularly useful for scenarios demanding real-time or near real-time analysis and decision-making due to its efficiency in capturing broad sentiment shifts. Here's a closer look at some key applications:

- **Real-Time Sentiment-Based Trading:** This scenario involves leveraging real-time sentiment data to inform trading decisions. By tracking the average Tone score for Netflix across consecutive 15-minute intervals, traders can identify emerging trends in market sentiment. For instance, if the average Tone score consistently exceeds a predefined positive threshold, it might signal an increasing positive sentiment toward Netflix, potentially leading to a price increase. This could trigger a buy signal for automated trading systems or inform a trader's decision to enter a long position. Conversely, a sustained drop in the average Tone score below a negative threshold could indicate growing negative sentiment and a potential price decline, prompting a sell signal or a short position.
- **Risk Management and Portfolio Allocation:** Risk management is crucial in financial engineering, and sentiment analysis can play a significant role in assessing and mitigating risk associated with investments. By monitoring the average Tone scores for Netflix, portfolio managers can gain insights into the level of uncertainty or negativity surrounding the company. A significant and sustained drop in sentiment could indicate heightened risk, prompting a reassessment of portfolio exposure. In such cases, risk managers might consider reducing their allocation to Netflix or implementing hedging strategies to protect against potential losses. Conversely, a period of consistently positive sentiment could signal lower risk and potentially justify increasing exposure to Netflix within the portfolio.
- **Market Microstructure Analysis:** This scenario delves into the intricate dynamics of how news and sentiment impact short-term price movements of Netflix stock. By correlating the average Tone scores with high-frequency price data, researchers can identify patterns and relationships between sentiment changes and price fluctuations. For example, a sudden spike in positive sentiment might be followed by a rapid increase in Netflix's stock price, indicating a strong correlation between sentiment and market reaction. Understanding these relationships can help traders anticipate price movements and potentially identify arbitrage opportunities.
- **Sentiment-Driven Event Detection:** This scenario involves using sentiment analysis to detect significant events or news stories that trigger notable shifts in sentiment toward Netflix. By analyzing the time series of average Tone scores for sudden spikes or drops, researchers can pinpoint potential triggers for sentiment change. For instance, a sudden drop

in sentiment might coincide with a negative news report or a regulatory announcement affecting Netflix. Identifying these events in real time enables proactive responses and informed decision-making, helping traders and investors adjust their strategies accordingly.

Method 2: Combining All Samples and Analyzing Tone Elements for Individual Articles

This method, providing granular sentiment information for individual articles, is more suitable for scenarios requiring in-depth analysis and understanding the context behind sentiment changes. Here's a closer look at some key applications:

- **Event-Driven Trading Strategies:** This scenario involves developing trading strategies based on the sentiment expressed in specific news articles or events related to Netflix. By analyzing the Tone scores of individual articles, traders can identify those with strong positive or negative sentiment. For instance, a highly positive article discussing a successful new Netflix original series might trigger a buy signal, while a negative article highlighting subscriber churn could prompt a sell signal. This approach allows traders to capitalize on market reactions to specific information and events.
- **Fundamental Analysis:** Fundamental analysis plays a crucial role in investment decision-making, and incorporating sentiment analysis can provide valuable insights. By analyzing the Tone scores of articles related to Netflix's earnings reports, product launches, or other significant events, analysts can gauge market perception and assess the potential impact on financial performance. For example, if articles discussing Netflix's latest earnings report express overwhelmingly positive sentiment, it might reinforce a positive outlook for the company's future prospects. Conversely, negative sentiment surrounding a new product launch could raise concerns about market acceptance and potential impact on revenue.
- **Competitive Analysis:** Understanding the competitive landscape is essential for strategic decision-making in financial engineering. Sentiment analysis can help track sentiment toward Netflix and its competitors, providing insights into market dynamics and potential opportunities or threats. By analyzing the Tone scores of articles mentioning Netflix and its competitors, analysts can identify shifts in competitive positioning and market perception. For example, a surge in positive sentiment toward a competitor's new streaming service might signal a potential threat to Netflix's market share, while a decline in negative sentiment toward Netflix could indicate an improving competitive outlook.

In essence, while both methods provide valuable insights for financial engineering applications, they cater to different needs and scenarios:

- Method 1 (averaging Tone scores) is ideal for real-time or near real-time analysis of overall sentiment trends, enabling rapid decision-making in trading, risk management, and market microstructure analysis. Its focus on aggregated sentiment makes it well-suited for capturing broad market sentiment shifts.
- Method 2 (analyzing individual Tone elements) is more appropriate for in-depth analysis of individual articles and their impact on sentiment, providing a more nuanced understanding of specific events and their potential consequences. This method is often used for fundamental analysis and competitive intelligence, where understanding the context behind sentiment changes is crucial.

5.4 5. Conclusion

In this lesson, we embarked on a practical exploration of two large-scale news datasets, Common Crawl News Crawl and GDELT, with a focus on their applications in financial engineering. We delved into accessing and processing raw news articles from Common Crawl's WARC files, learning how to extract key information, filter by language and keywords, and utilize the Python libraries for efficient article content extraction. Furthermore, we explored GDELT's GKG data, discovering techniques to analyze news events, entities, and sentiment. We learned how to download, filter, and analyze GDELT's Tone scores to gain valuable sentiment insights. Through this lesson, we learned how to leverage these datasets for financial research and analysis.

References

- Brown, Tom B., et al. "Language Models are Few-Shot Learners." *Proceedings of the 34th International Conference on Neural Information Processing Systems*, no. 159, 2020, pp. 1877-1901. <https://dl.acm.org/doi/abs/10.5555/3495724.3495883>.
- THE GDELT Project. *The GDELT Global Knowledge Graph (GKG) Data Format Codebook*. v2.1, 2015. http://data.gdeltproject.org/documentation/GDELT-Global_Knowledge_Graph_Codebook-V2.1.pdf.

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Required Readings

Required Readings for this program are open access, which means you should be able to access them at no cost. The link provided in the citation will take you directly to the reading or to a page where you can download it.

1. Alma Mater Studiorum. *Information Literacy Tutorial*. Università di Bologna, 2018.

<https://sba.unibo.it/en/services/training/tutorial-di-information-literacy-1>.

- **Required pages:** TU2: Identify the Required Resources and TU5: Evaluate the Resources
- **Estimated reading time:** 30 minutes

2. Ismail, W. Omar Ali Saifuddin Wan, et al. "The Essential of Engineering Education Involving Critical Thinking and Problems Solving Skills among Mechanical Engineer Employees." *IOP Conference Series: Materials Science Engineering*, vol. 697, 2019, <https://iopscience.iop.org/article/10.1088/1757-899X/697/1/012017>.

- **Required pages:** All
- **Estimated reading time:** 20 minutes

3. Hattingh, Marié, et al. "The Use of Critical Thinking to Identify Fake News: A Systematic Literature Review." *Responsible Design, Implementation and Use of Information and Communication Technology*, 2020, pp. 235–246, <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7134234/>.

- **Required pages:** All
- **Estimated reading time:** 25 minutes

CRITICAL THINKING TO ASSESS DATA VERACITY

Reading Time	15 minutes
Prior Knowledge	All Financial Data content, Financial Markets Module 7 content
Keywords	Information Literacy, Fake News

In this lesson, we examine the characteristics of critical thinking. We'll look at a systematic literature review that addresses how critical thinking skills were used to determine if news was "fake news."

1. Critical Thinking

Critical thinking is not just something we'll discuss once. As Ismail writes, "Critical thinking involves the purposeful, goal-directed thinking during the process of making decisions based on the evidence in the scientific problem-solving process" (1). Throughout this program, you'll be expected to utilize critical thinking skills for a variety of purposes, but here, we'll focus on using critical thinking to determine whether a source is reliable.

The article by Ismail addresses the importance of critical thinking skills for getting hired and being productive once employed. The article highlights a study that was done on employees who had five years of experience in an engineering job, particularly mechanical engineering. The period of five years is considered to be a starting point for building maturity in one's career.

In accordance with this study, the following questionnaire was given to mechanical engineers with 5 years of work experience. The questions relate to:

- Problem identification in a complex system
- Improving explanations, analysis, and evaluation skills
- Searching for alternative solutions
- Searching for innovative solutions
- Justifying decisions based on strong evidence
- Ability to persevere to accomplish responsibilities assigned
- Ability to understand and adapt to the culture of the community

Critical thinking skills are highly desirable in the workforce because they're necessary for working effectively on a team, analyzing data, crafting arguments, making decisions, and solving problems. As such, it's important to strengthen this skill as much as you can throughout the program.

Your critical thinking skills will be evaluated in a variety of ways—from forum posts to group work projects to your capstone project. One critical thinking skill we focus on here is information literacy, which is essential as you search for valid and credible sources to find the data you need, to support and develop your ideas, and more. These skills are incredibly important in financial engineering jobs as well as other STEM-related professions. Later in this lesson, we'll apply critical thinking skills to determine whether news is fake or credible.

2. Information Literacy

The American Library Association defines information literacy as "a set of abilities requiring individuals to recognize when information is needed and have the ability to locate, evaluate, and use effectively the needed information." As such, information literacy means that you first know how to research and find the sources you're looking for and how to evaluate whether a source is a credible and reliable source of information. While this may sound simple, it has become an increasingly difficult task with the advent of fake news, crowdsourced information, and anonymous internet authors.

As such, the first thing we'll have you do is complete two modules from the Information Literacy tutorial, as outlined in the Required Readings for this lesson. Please be sure to complete the following modules:

- TU2: Identify the Required Resources
- TU5: Evaluate the Resources

Given the vast amount and variety of information on the internet, evaluating online resources is more important than ever. Can you trust information from a blog? What about using information that was posted as an answer to someone's question?

As you prepare yourself to get data and research for your empirical studies, you may encounter news articles and headlines that you use in a sentiment analysis or text analysis. Be warned that there is plenty of fake news on the internet.

3. Unreliable Sources

Journalism has changed drastically over the last few decades with the rise of the internet, smartphones, and social media. The development of 24-hour news networks has encouraged cable news to embrace sensationalist headlines while promoting the partisan viewpoints of its owners and balancing the needs and demands of advertisers. TV and social media have overtaken print as sources of news, so newspapers and local journalism have struggled to survive. As a result, people must look to other sources for their news, and oftentimes, these sources may have conflicts of interest that must be considered when evaluating whether their reporting is reliable and credible. There's a lot of pressure in journalism to break the story first, regardless of whether reporters have all the facts. As a result, initial reports are often wrong, and networks or publications may find themselves issuing corrections or retractions later.

These are all important realities to keep in mind when finding sources on the internet. This isn't to say that all cable news is unreliable or that you should never trust a blog. Instead, the takeaway should be to do your research. When deciding whether or not a source is credible, ask a few key questions:

- Who wrote this text? Do they have a reputation as a credible scholar or expert in this field or on this topic? Is this a reliable organization that has demonstrated a commitment to getting the facts right? Is there any potential political bias or conflict of interest you should be aware of?

-When was this text written? Is the source outdated or irrelevant, considering the topic you're researching? How does the source respond to or incorporate modern developments pertaining to the topic?

-Does the text's argument, logic, and reasoning hold up? Can the information in the text be easily disproven? Can most facts or statistics be substantiated by another source? Does the author cite reliable sources as evidence?

-What is the style or tone of the text? Is it more academic and objective, or is it emotional and provocative? Is the text trying to persuade you to buy/use/read/watch/do something?

As you complete the information literacy tutorial, you'll learn the skills and strategies you need to evaluate a source and answer these questions.

Suppose you were building a trading strategy using machine learning algorithms that perform sentiment analysis. These methods require headlines, news articles, and perhaps even social media posts. The existence of fake news could turn these models into garbage in, garbage out boxes. Alternatively, perhaps in doing your capstone project, you decide to use a source that turns out to be fake. This could have disastrous implications for your research findings. In our age of misinformation, it is more important than ever to be able to identify (and avoid) unreliable sources.

4. Critical Thinking about Fake News

Please read Machete and Turpin's "The Use of Critical Thinking to Identify Fake News: A Systematic Literature Review." The article reviews the literature on how critical thinking can be used to identify fake news. The articles that explicitly discussed critical thinking found that most students do not have sufficient critical thinking skills to identify news that is fake.

Once you have read the article, apply the information you have learned from the information literacy course to determine whether something is fake news.

1. Is the information accurate, as determined by checking other sources?
2. Is the article written by an authority? Is the authority recognized by peers?
3. Is the information complete? Take care to note stories that omit key information.
4. Is the story timely? Have there been updates or other news that indicate the information may be outdated?
5. Was the information presented as clickbait, merely to attract an audience, or is the information provided as a genuine news story?

Suppose in your due diligence, you answer some of the above questions in the negative. The question to ask then is "has this story been repeated elsewhere, and has it been confirmed or verified by reliable sources?" Any doubts on your part could then prevent that article from becoming part of the data for your sentiment analysis.

5. Conclusion

Clearly, creating fake news would be a violation of good ethical practice. As an FE, you can become more information-literate by learning how to assess the quality, authority, comprehensiveness, and timeliness of information you find on the web. This information may become data that you use to build and train your models for FE. Building information literacy will actually help promote good ethical practice as well.

In concluding this module, let's remember to be honest and vigilant in avoiding the creation, spread, or inclusion of fake news in your studies. Even if you use AI to determine what is fake, be sure that you confirm what gets used. The saying "garbage in, garbage out" applies, but the situation can be worse if we assume fake stories to be true and make key decisions based on false information.

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